Sharp Identification for Regressions with Interval-Observed and Missing Covariates*

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Abstract

Researchers frequently estimate regressions when key variables are missing or observed only through intervals. Imputation or assumptions about the observation process can yield seemingly precise but misleading conclusions. We develop methods for identification and inference that remain robust to such incomplete data. For a broad class of conditional moment models, we characterize the sharp identified set, the set of all parameter values consistent with the observed data and maintained regression model, without restricting how incomplete observations arise. We establish Hausdorff-consistent estimation and uniformly valid confidence regions, and develop computationally tractable implementations for common regression models. In an application relating interval-elicited discount rates to on-time high-school graduation, conventional inference excludes zero, whereas our interval-data-robust confidence regions include it. Exploiting sharp identification nevertheless yields substantially tighter confidence regions than non-sharp alternatives.

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1 Introduction

Regression models are ubiquitous in applied economics, yet the variables entering them are often not observed exactly. Outcomes and covariates may be missing because of nonresponse or attrition, reported only in brackets in household surveys, top-coded in administrative data, or observed through intervals generated by experimental and survey designs. In each case, the observed data leave ambiguity about the latent variables entering the regression, so that the parameters of interest are typically only partially identified.

Empirical work commonly addresses incomplete data through sample restrictions, imputation, or assumptions about the data generating process. These choices reflect the trade-off emphasized in Manski (2003) in his *law of decreasing credibility*: stronger assumptions can deliver more informative conclusions at the cost of credibility. This paper therefore asks: what does the maintained regression model identify from the incomplete data alone, without additional assumptions about how missing or interval observations are generated?

As formalized in Section 2, we study conditional moment models of the form

$$\mathbf{E}\left[m(Y, X, \theta_0) \mid X\right] = 0,$$

where some components of the non-conditioning variables Y, the conditioning variables X, or both are not observed exactly. This class includes, for instance, linear, probit, or logit regressions as well as more general econometric models characterized by conditional moment restrictions. Let $W \equiv (Y, X)$ denote the latent variables entering the model and let $W^O \equiv (Y_l, Y_u, X_l, X_u)$ denote the observed intervals. The observed data imply the containment restriction $Y \in [Y_l, Y_u]$ and $X \in [X_l, X_u]$ almost surely. Our setting nests the missing data case if the observed intervals span the full support of the respective variables.

Our central identifying restriction is deliberately weak. Beyond this containment requirement, we leave the process generating incomplete observations unrestricted: whether a variable is missing, as well as the location and width of an observed interval, may depend on the

latent variables themselves. In particular, we do not impose missingness-at-random or other restrictions on the relationship between the observation process and the latent variables. This approach gives a benchmark for assessing what additional assumptions contribute. Whether such weak restrictions yield informative conclusions is an empirical question, which our application addresses.

Our object of interest is the sharp identified set Θ_I^*: the set of parameter values for which a single completion of the latent data reproduces the distribution of the observed data, respects the reported intervals, and satisfies the conditional moment restriction.[1] Characterizing this set is difficult for two reasons. The conditional moment restriction is equivalent to infinitely many unconditional moment restrictions, and a single completion must satisfy all of them jointly. Existing procedures for missing covariates (Aucejo et al., 2017) check these restrictions separately and therefore deliver outer sets. In addition, the restriction conditions on covariates that are themselves incompletely observed. Random-set methods characterize sharp sets for moment restrictions that are unconditional or condition on observed variables, including best linear prediction with interval covariates (Beresteanu et al., 2011), but not a conditional mean restriction given an incompletely observed covariate.[2] We make three contributions.

First, we characterize the sharp identified set for the full conditional moment restriction in terms of the observed-data distribution alone. The key step is a single random set that encodes the entire conditional restriction: we interact the moment function with a countable class of instruments, evaluated at the latent covariates, and collect the resulting moments for all latent values consistent with the reported intervals. Weighting the instruments makes the infinitely many moments summable, so that this random set is bounded and well defined. A parameter belongs to the identified set if and only if the Aumann expectation of this random

[1] Θ_I^* contains θ_0 whenever the containment restriction and the maintained model hold. An empty identified set therefore rejects the model, whereas a nonempty set establishes only compatibility with at least one admissible latent distribution, not that the model holds under the true one.

[2] In the linear model, the difference is between $\mathbf{E}[(1, X)^\top (Y - \alpha - \beta X)] = 0$ and $\mathbf{E}[Y - \alpha - \beta X \mid X] = 0$ when X is only interval-observed.

set contains zero, or equivalently, if a family of support-function inequalities holds under the observed-data distribution. This extends the characterization of Beresteanu et al. (2011) to restrictions that condition on incompletely observed covariates and therefore involve infinitely many moments. The extension requires handling discontinuous instruments, such as indicators of cells partitioning the covariate space, which are standard in practice and enable exact linear-programming computation. With such instruments, the Aumann expectation equals only the closure of the set of moment vectors attainable by admissible completions; we show that this closure is harmless at zero.

Second, estimation and inference require additional arguments as the instrument collection expands. For set estimation, we establish a triangular-array law of large numbers for random sets of growing dimension and use it to verify the consistency conditions of Chernozhukov et al. (2007). For uniformly valid confidence regions for θ_0, we verify the manageability conditions required by Andrews and Shi (2017) for the support-function process indexed by an *uncountable* class of directions and the instrument class.

Third, we make the procedure operational. The theory involves infinitely many moments and an uncountable class of support-function directions; in practice we use finitely many of each, characterize the resulting outer sets, and show that inference remains valid even when the directions are selected from the data.[3] For linear, probit, and logit regressions, the nested minimax problem underlying the estimation criterion admits an exact linear-programming representation, so it can be solved with standard solvers. We propose two ways to choose directions for inference. An adaptive procedure selects informative directions on one half of the sample, tests on the other half, swaps the roles of the halves, and intersects the two confidence regions. A fixed procedure instead uses a prespecified set of directions and the full sample. The first concentrates on the most informative inequalities; the second uses all observations for testing. In Monte Carlo simulations, both maintain approximately nominal coverage and gain power as the sample size grows. Under random censoring, the power gap

[3]Finite approximations are standard in the many-moment inequality literature; see, for example, Andrews and Shi (2017).

relative to complete-case OLS narrows with sample size; under outcome-dependent censoring, our procedures remain reliable while conventional methods lose coverage.

Our empirical application examines how conclusions change as we progressively account for different forms of incomplete observation. We revisit Castillo et al. (2018), who use a multiple price list (MPL) to elicit discount rates and study their relationship with on-time high-school graduation. Because the offered rates lie on a discrete grid, even consistent responses reveal only an interval containing the latent discount rate. A conventional probit based on midpoint imputation yields a marginal effect of -0.086 with a 95% confidence interval of $[-0.152, -0.020]$. Allowing discount rates to vary within their elicited MPL intervals yields a 95% confidence set of $[-0.226, 0.020]$, approximately 27% wider than the corresponding midpoint-imputation benchmark under the same inference procedure, and renders the previously significant association inconclusive. Allowing for inconsistent switching patterns and missing graduation outcomes widens the confidence sets further. Nevertheless, our sharp approach remains substantially more informative than non-sharp outer approximations: for the basic MPL intervals, the procedure of Aucejo et al. (2017) yields $[-0.396, 0.403]$. Finally, sensitivity exercises show how the gains from more precise measurement depend on whether uncertainty is reduced for all respondents or eliminated for a subset. This illustrates how our framework can assess both the robustness of empirical conclusions and the potential value of additional identifying assumptions or improved data collection.

Our work is most closely related to the partial-identification literature on regressions with interval or missing covariates. Manski and Tamer (2002) study regression models with interval-observed variables and obtain sharp identification results under an auxiliary mean-independence restriction linking the latent variable to its observed interval. Asher et al. (2018) build on their work and impose further assumptions. In contrast, we retain only the containment restriction and otherwise leave the observation process unrestricted. Aucejo et al. (2017) study conditional moment models with missing covariates and develop estimation and inference procedures for outer sets, i.e., supersets of the sharp identified set. In

comparison, our work imposes very similar assumptions but delivers results for the sharp identified set and can handle interval-valued data. Schennach (2014) develops a more general entropy-based approach to latent-variable models and extends it to conditional moment restrictions through a growing collection of unconditional moments.[4] Our more specialized support-function approach instead exploits the structure of interval and missing observations directly, yielding sharp identification together with estimation, inference, and computational procedures tailored to regression models as delineated in Section 4.

A related literature obtains sharp identification when the relevant restrictions are finite-dimensional. Horowitz and Manski (1998) derive sharp worst-case bounds for conditional expectations of the form $\mathbf{E}[g(Y) \mid X \in A]$ when outcomes, covariates, or both may be missing, while Horowitz and Manski (2000) specialize this approach to randomized experiments and bound conditional response probabilities and treatment contrasts. In treatment-effect settings, Lee (2009) obtains sharp trimming bounds under random assignment and monotonicity of treatment-induced selection, with subsequent work extending this approach under related restrictions on selection. A distinct methodological strand uses random sets and support functions to characterize sharp identification from finitely many moment restrictions. Our approach builds directly on Beresteanu and Molinari (2008); Beresteanu et al. (2011), who characterize feasible moment predictions through Aumann expectations and support-function inequalities; Beresteanu et al. (2011) apply this framework to best linear prediction with interval data, improving on the computationally demanding characterization in Horowitz and Manski (2006). Chesher and Rosen (2017, 2020) use random sets to characterize identified sets in generalized instrumental variable models, including models with an interval-censored endogenous regressor, under restrictions conditional on observed instruments. More recently, Franguridi and Moon (2026) extend the same logic to GMM models with unrestricted, potentially endogenous missingness. In contrast, our restriction

[4] Applying this framework to our setting would still require developing estimation, inference, and computational procedures for the resulting growing-moment problem. Some of the tools developed in this paper for handling an expanding collection of moments could potentially also be used for such an implementation.

conditions on covariates that are themselves incompletely observed, which requires infinitely many moments and, in practice, discontinuous instruments. More broadly, the paper follows the partial-identification approach pioneered by Manski (2005, 2008); see Molinari (2020), Canay and Shaikh (2017), and Kline and Tamer (2023) for reviews.

A separate literature restores point identification by placing stronger restrictions on how incomplete observations are generated or by modeling the missing values themselves. Missing-data methods commonly invoke ignorability or missing-at-random assumptions and use likelihood, multiple imputation, or inverse-probability weighting (Rubin, 1976, 1987; Robins et al., 1995; Wooldridge, 2007); in longitudinal settings, related weighting approaches are routinely used to address attrition (Fitzgerald et al., 1998). Selection models instead specify how observation depends on latent outcomes or unobservables, ranging from the classical parametric correction of Heckman (1979) to semiparametric approaches such as Newey (2009). For more general forms of partial observation, Heitjan and Rubin (1991) formulate conditions under which the coarsening mechanism can be ignored. Other approaches exploit auxiliary information, validation samples, restrictions on reporting behavior, or explicit models for the latent values within observed intervals (Hsiao, 1983; Rigobon and Stoker, 2009; Pacini, 2018; Duan and Pelger, 2025; Freyberger et al., 2025; Bretschneider and Kaddoura, 2026). These methods can be considerably more informative when their assumptions are credible. Our identified set instead provides an assumption-minimal benchmark: each additional restriction can be viewed as eliminating latent completions from the larger set Γ of all completions feasible under the containment restriction; see Section 2.

The remainder of the paper is organized as follows. Section 2 derives the sharp identification result. Section 3 develops estimation and inference, and Section 4 presents the computational implementation. Section 5 studies finite-sample performance, Section 6 presents the empirical application, and Section 7 concludes.

Notation. Throughout the paper, ℓ_1 and ℓ_∞ denote the usual sequence spaces equipped with norms $|\cdot|_1$ and $|\cdot|_\infty$, respectively, and $\langle u, a\rangle$ denotes the dual pairing between $u \in \ell_\infty = (\ell_1)^*$ and $a \in \ell_1$. $h(A, u) \equiv \sup_{a\in A}\langle u, a\rangle$ denotes the support function of a set $A \subseteq \ell_1$. For a random set Q, $\mathbb{E}[Q] = \mathrm{cl}\,\{\mathbf{E}[q] : \ q \text{ is an integrable selection of } Q\}$ denotes its Aumann expectation, whereas $\mathbf{E}_F[Z]$ denotes the ordinary expectation of a random element Z under the probability law F, with the subscript suppressed when clear from context. We write $\rho_H(A, B)$ for the Hausdorff distance and $\vec{\rho}_H(A, B)$ for the directed Hausdorff distance. The symbols $A \oplus B$ and $\mathrm{co}(A)$ denote the Minkowski sum and convex hull, respectively. Finally, for a metric space $\mathcal{S}$, $\Delta(\mathcal{S})$ denotes the set of Borel probability measures on $\mathcal{S}$, and $\mathcal{L}_F(Z)$ denotes the law of Z under F.

2 Identification

Formally, we study conditional moment models of the form

$$\mathbf{E}_{F_0}\left[m(Y, X, \theta_0) \mid X\right] = 0 \qquad F_X\text{-a.s.} \tag{1}$$

Let $(\Omega, \mathcal{A}, F_0)$ denote the underlying probability space. The random variables Y and X take values in $\mathcal{Y} \subseteq \mathbb{R}^{d_y}$ and $\mathcal{X} \subseteq \mathbb{R}^{d_x}$, respectively. The known moment function $m : \mathbb{R}^{d_y} \times \mathbb{R}^{d_x} \times \Theta \to \mathbb{R}^{d_m}$ satisfies (1) at the true parameter $\theta_0 \in \Theta$, should it exist. We assume $d_y, d_x, d_m < \infty$; we allow Θ to be arbitrary for the purpose of identification; however, in Section 3 we restrict attention to $\Theta \subset \mathbb{R}^{d_\theta}$. The source of partial identification is not an incomplete model, but incomplete observation of the variables entering it.

Assumption 1. *Let $W \equiv (Y, X)$ take values in $\mathcal{W} \equiv \mathcal{Y} \times \mathcal{X}$, where $\mathcal{Y} \subset \mathbb{R}^{d_y}$ and $\mathcal{X} \subset \mathbb{R}^{d_x}$ are compact. Let $W^O = (Y_l, Y_u, X_l, X_u)$ be observed on the same probability space as W and take values in $\mathcal{W}^O \equiv \mathcal{Y}^2 \times \mathcal{X}^2$, with nonempty componentwise intervals almost surely.*

Assume

$$P_{F_0}\left[Y \in [Y_l, Y_u], X \in [X_l, X_u]\right] = 1.$$

We find it convenient at times to write this last condition as $P_{F_0}(W \in D(W^O)) = 1$*, for* $D(W^O) \equiv [Y_l, Y_u] \times [X_l, X_u]$*.*

Assumption 1 contains the central identifying stance of the paper. Apart from requiring the latent realization to lie inside its observed interval, we place no further restrictions on how incompleteness is generated. A fully observed value is represented by a degenerate interval, while a missing value is represented by the variable's full support. The purpose of the analysis is precisely to determine what follows before adding any stronger assumptions. Let us first define the identified set for our setting. Assumption 1 implicitly defines the set of admissible completions

$$\Gamma \equiv \left\{\eta \in \Delta(\mathcal{W} \times \mathcal{W}^O) : \eta_{\mathcal{W}^O} = P_{W^O}, \quad \eta\big(\{(w, w^o) : w \in D(w^o)\}\big) = 1\right\}$$

An element of Γ is a candidate joint law for the latent and observed data that has the observed marginal P_{W^O} and assigns probability one to latent values compatible with the reported intervals. Such a law cannot be ruled out from the observed data alone. Assumption 1 also guarantees nonemptiness, because $\eta_0 \equiv \mathcal{L}_{F_0}(W, W^O) \in \Gamma$.

The identified set being discussed is defined directly from the maintained model (1) and Assumption 1 as

$$\Theta_I^* \equiv \{\theta \in \Theta : \exists \eta \in \Gamma \text{ such that } \mathbf{E}_\eta[m(Y, X, \theta) \mid X] = 0 \quad \eta_X\text{-a.s.}\}. \tag{2}$$

The set Θ_I^* may be empty. This natural definition of the identified set may sometimes be redefined in convenient ways. In particular, Schennach (2014) considers a distance-based formulation for model (1) expressed with the condition $\|\mathbb{E}_\eta[m(Y, X, \theta)|X]\| = 0$, F_X-almost

surely. The precise relationship between the original identified set and that based on a distance-based formulation has not been previously examined to our knowledge. In Appendix Section A.3 we establish that in our setting the two identified sets coincide, show that the relationship between them can be expressed well through random sets, and provide a sufficient condition for equivalence. We do not discuss this distance-based formulation any further in the main text of this paper.

The characterization of Θ_I^* in terms of conditional expectations is inconvenient as the conditioning event is not observed. We begin by replacing the conditional equality by a (countably) infinite collection of unconditional equalities using instrument functions, as is standard in the conditional-moment literature; see, among others, Andrews and Shi (2013), Andrews and Shi (2017) and Aucejo et al. (2017). In Lemma 1 we establish that this can be performed without changing the identified set. The following assumption gives general conditions on the instrument class used, whereas in the Appendix Section A.2 we verify them for the class of weighted cubes used in our applications.

Assumption 2.

(i) *Let $\mathcal{H} = \{h_k : k \geq 1\}$ be a countable class of bounded Borel functions on $\mathcal{X}$ that span the class of continuous functions. Precisely, for every $f \in C(\mathcal{X})$ and every $\varepsilon > 0$, there exists $v_\varepsilon \in \text{span}(\mathcal{H})$ satisfying*

$$\|f - v_\varepsilon\|_\infty < \varepsilon.$$

(ii) *Let $\mathcal{G} = \{g_k : k \geq 1\}$ be the weighted instrument class defined by $g_k = \omega_k h_k$, where the deterministic weights $\omega_k > 0$ are chosen so that*

$$\lim_{K \to \infty} \sup_{x \in \mathcal{X}} \sum_{k > K} |g_k(x)| = 0.$$

In particular, weights decaying at the rate $O(k^{-\beta})$ for some $\beta > 1$ achieve this.

Part (i) ensures that the instruments reveal any nonzero conditional moment; note that

span($\mathcal{H}$) = span($\mathcal{G}$). This mimics Assumption MFA in Andrews and Shi (2017) and Assumption CI in Andrews and Shi (2013), which both require the instrument class to be rich enough that the set of x violating the conditional moments has positive measure if and only if some instrument reveals that fact.[5] Part (ii) ensures that the stacked moment vector lives in ℓ_1 and, combined with a uniform integrability condition (supplied by Assumption 3), is also compact. This setting handles the technical difficulties of the space and ensures finite truncations are uniformly accurate.

We reformulate the identified set using the unconditional-moment set

$$\Theta_I \equiv \left\{ \theta \in \Theta : \begin{array}{c} \exists \eta \in \Gamma \text{ such that} \\ \int_{\mathcal{W}\times\mathcal{W}^O} m_j(y,x,\theta) g_k(x) \eta(dy,dx,dw^o) = 0, \\ j = 1,\ldots,d_m, \quad k \geq 1, \end{array} \right\}, \tag{3}$$

With this definition, we state the first auxiliary lemma.

Lemma 1. *Let $\mathcal{X}$ be compact and let Assumption 2(i) hold. Suppose that*

$$\mathbf{E}_\eta[\|m(Y,X,\theta)\|_\infty] < \infty \qquad \text{for all } (\theta,\eta) \in \Theta \times \Gamma.$$

Then, for every $(\theta,\eta) \in \Theta \times \Gamma$,

$$\mathbf{E}_\eta[m(Y,X,\theta) \mid X] = 0 \quad \eta_X\text{-a.s.} \qquad \Longleftrightarrow \qquad \int m_j(y,x,\theta) g_k(x)\, d\eta = 0 \quad \text{for every } (j,k).$$

Consequently, $\Theta_I = \Theta_I^*$.

Furthermore, given our weights $\omega_k > 0$, one could equivalently define Θ_I and this lemma using the unweighted class $\mathcal{H}$ with elements $h_k(x)$ instead of $g_k(x)$.

It is noted that the compactness and integrability conditions in Lemma 1 are ensured by Assumptions 1 and 3, respectively. Lemma 1 is the well-known property that a conditional

[5] These formulations are based on what Stinchombe and White (1998) refer to as *comprehensively revealing*.

moment equality is equivalent to an infinite family of unconditional moment equalities, and it isolates the central feasibility requirement: one admissible completion must satisfy every unconditional restriction simultaneously. Aucejo et al. (2017) derive a similar result in their Lemma 1 and note, as we do, that verifying the existence of such an η is extremely difficult and hard to make progress on in this direct formulation. Andrews and Shi (2017) and Chernozhukov et al. (2017) likewise work with infinite families of moment (in-)equalities, but typically assume that the variables entering those moments are observable, whereas here η is not identified. The proof is in Appendix A. An immediate implication is instrument-class invariance: every class satisfying Assumption 2 generates the same population set Θ_I^*.

Lemma 1 removes searching over the unobserved conditioning event in the original model and thus makes a random-set formulation possible. For each candidate θ, making use of our weighted instruments $\mathcal{G}$, we collect all moments in an ℓ_1-valued random set and ask whether its expectation contains zero. This logic follows the seminal paper by Beresteanu et al. (2011). We extend their framework to accommodate an infinite-dimensional moment vector and a discontinuous instrument map. In general, random-set methods were introduced to econometrics by Beresteanu and Molinari (2008); see Molchanov (2017); Molchanov and Molinari (2018) for textbook treatments.

For $\theta \in \Theta$, define

$$M_\theta(y, x) \equiv (m_j(y, x, \theta) g_k(x))_{j=1,\ldots,d_m,\ k\geq 1} \in \ell_1.$$

Let $\eta_0 = \mathcal{L}_{F_0}(W, W^O)$ denote the true joint law of the latent and observed variables. By Lemma 1, the model in (1) can equivalently be written as

$$\int_{\mathcal{W}\times\mathcal{W}^O} M_{\theta_0}(y, x)\, \eta_0(dy, dx, dw^o) = 0.$$

Assumption 1 implies $\eta_0 \in \Gamma$, but the observed data generally do not determine which element of Γ is the true joint law. For each candidate θ, we therefore consider the set of

attainable mean-moment vectors

$$I_\theta \equiv \left\{ \int_{\mathcal{W}\times\mathcal{W}^O} M_\theta(y,x)\,\eta(dy,dx,dw^o) : \eta \in \Gamma \right\}. \tag{4}$$

Thus,

$$\theta \in \Theta_I \quad \Longleftrightarrow \quad 0 \in I_\theta.$$

Checking this condition directly requires finding an admissible joint law that satisfies all moment restrictions simultaneously. To obtain a more tractable characterization, we first collect the moment vectors compatible with each observed interval:

$$Q_\theta(W^O) \equiv \mathrm{cl}_{\ell_1}\left\{ M_\theta(y,x) : (y,x) \in D(W^O) \right\}.$$

The set $Q_\theta(W^O)$ contains the moment vectors generated by feasible latent values, together with their limits in ℓ_1. We next show how its Aumann expectation characterizes whether I_θ contains zero, leading to a support-function criterion evaluated under the observed-data distribution. We note that $Q_\theta(W^O)$ depends only on the observed data.

Assumption 3.

(i) For every $\theta \in \Theta$, the map $(y,x) \mapsto m(y,x,\theta)$ is continuous on $\mathcal{W}$.

(ii) There exists a constant $C_1 < \infty$ such that

$$\sup_{(y,x,\theta)\in\mathcal{W}\times\Theta} \|m(y,x,\theta)\|_\infty \le C_1.$$

Assumption 3 is satisfied by common linear-mean, logit, and probit specifications on bounded domains. Because Assumption 1 makes $\mathcal{W}$ compact, continuity already gives boundedness for each fixed θ; Assumption 3 supplies a common bound over Θ and therefore

keeps the random sets integrably bounded under the conditions of Lemma 1. The Aumann expectations below are understood as the collection of expectations of ordinary random variables (Aumann, 1965).

Theorem 1. *Let Assumptions 1–3 hold. Then $Q_\theta(W^O)$ is almost surely a nonempty compact random set in ℓ_1, and $\mathbb{E}[Q_\theta(W^O)]$ is a nonempty closed convex subset of ℓ_1. Moreover,*

$$\theta \in \Theta_I^* \quad \Longleftrightarrow \quad 0 \in \mathbb{E}[Q_\theta(W^O)]. \tag{5}$$

Therefore,

$$\begin{aligned}
\Theta_I^* &= \left\{\theta \in \Theta : 0 \in \mathbb{E}[Q_\theta(W^O)]\right\} \\
&= \left\{\theta \in \Theta : \mathbf{E}_{P_{W^O}}[h(Q_\theta(W^O), u)] \geq 0 \quad \forall u \in B^\infty\right\} &(6)\\
&= \left\{\theta \in \Theta : \inf_{u \in B^\infty} \mathbf{E}_{P_{W^O}}[h(Q_\theta(W^O), u)] \geq 0\right\} &(7)\\
&= \left\{\theta \in \Theta : \inf_{u \in B^\infty} \mathbf{E}_{P_{W^O}}[\sup_{(x,y) \in D(W^O)} \langle M_\theta(y,x), u\rangle] \geq 0\right\}, &(8)
\end{aligned}$$

where $B^\infty := \{u \in \ell_\infty : \|u\|_\infty \leq 1\}$.

The proof is in Appendix B. Here we only sketch the arguments. First, we establish that the random-set construction is well defined and we relate it to the original feasibility problem. The appendix verifies the required random-set conditions, namely that Q_θ is measurable and integrably bounded. The Aumann expectation $\mathbb{E}[Q_\theta(W^O)]$ is therefore well defined and nonempty; it is closed by definition and convex.[6]

We next connect the random set to the original identification problem. By Lemma 1 and

[6] Aumann expectations on all appropriate non-atomic probability spaces are convex. Non-atomicity is often a trivial requirement. Without loss of generality the Aumann expectation is taken on the uniform enrichment of the original probability space described in Appendix B. A similar construction is used in the proof of the Artstein Selection Theorem in Molchanov (2017).

the definition of I_θ:

$$\theta \in \Theta_I^* \quad \Longleftrightarrow \quad \theta \in \Theta_I \quad \Longleftrightarrow \quad 0 \in I_\theta.$$

The remaining step is to relate the attainable set I_θ to the Aumann expectation of $Q_\theta(W^O)$. This is also where the main infinite-dimensional subtlety enters. If M_θ were continuous, compactness of $D(W^O)$ would imply that $M_\theta(D(W^O))$ is already closed, and the means of selections of Q_θ could be matched directly with admissible completion laws. Here, however, the instruments g_k need only be (Borel) measurable. Thus M_θ may be discontinuous and $M_\theta(D(W^O))$ need not be closed. Consequently, one cannot in general assert that $\mathbb{E}[Q_\theta(W^O)] = I_\theta$. Instead, the Appendix proves $\mathbb{E}[Q_\theta(W^O)] = \mathrm{cl}_{\ell_1} I_\theta$.

In general, $\mathrm{cl}_{\ell_1} I_\theta$ may contain vectors that are not themselves generated by any admissible completion. That is, $\mathrm{cl}_{\ell_1} I_\theta \neq I_\theta$; identification, however, only asks whether the origin is attainable. The key additional lemma establishes that $0 \in \mathrm{cl}_{\ell_1} I_\theta \iff 0 \in I_\theta$, i.e. closure is harmless precisely at this point.

Combining the preceding steps gives

$$\theta \in \Theta_I^* \quad \Longleftrightarrow \quad 0 \in I_\theta \quad \Longleftrightarrow \quad 0 \in \mathrm{cl}_{\ell_1} I_\theta \quad \Longleftrightarrow \quad 0 \in \mathbb{E}[Q_\theta(W^O)].$$

Having completed the difficult work, as in Beresteanu and Molinari (2008) or Beresteanu et al. (2011), the Hahn–Banach separation theorem implies that the Aumann expectation contains zero if and only if its support function is nonnegative for every $u \in B^\infty$. The support-function identity for Aumann expectations gives

$$h\left(\mathbb{E}[Q_\theta(W^O)], u\right) = \mathbf{E}_{P_{W^O}}\left[h(Q_\theta(W^O), u)\right],$$

which yields the characterization in Theorem 1.

The language of support functions transforms the original problem into a geometric one that involves a nested minimax optimization problem:

$$\Theta_I^* = \{\theta \in \Theta : \inf_{u \in B^\infty} \mathbf{E}_{P_{W^O}}[\sup_{(x,y) \in D(W^O)} \langle M_\theta(y,x), u \rangle] \geq 0\}$$

In general this minimax problem is nonconvex and non-trivial to solve. It is frequently the case that, for a fixed $u \in B^\infty$, the inner problem, that is, $h(Q_\theta, u) = \sup_{(x,y) \in D(W^O)} \langle M_\theta(y,x), u \rangle$, is easy to solve and, depending on the form of Equation (1), convex. However, the potential computability of the inner problem is not sufficient to claim superiority in formulating the problem through support functions, and we do not make such a claim. We note two properties of the characterization that have computational appeal.

First, Theorem 1 defines the identified set as an infinite collection of inequalities depending only on observables. Thus, existing inference procedures, e.g. Andrews and Shi (2017) or Chernozhukov et al. (2019), can be used for consistency and inference arguments. We discuss the details in the following section.

Second, although the general problem is computationally demanding, several observations simplify the problem substantially. First, Theorem 1 implies that we can exclude parameter candidates by finding counterexample-directions. In practice, this means we can search a large parameter space effectively by first screening out candidate θ based on a collection of fixed directions. Second, as we will show in the respective implementation section, the nested optimization problem reduces to a linear program for the important subclass of monotone index models.

While the dual space of ℓ_1 is ℓ_∞, suggesting the natural use of the ℓ_∞-unit ball, the compactness of $\mathbb{E}[Q_\theta(W^O)]$ allows one to consider other unit balls B^p, which may offer computational advantages. The choice of unit ball is equivalent for identification, and we conjecture the same for consistency and inference, though the proofs would need to be extended accordingly. As the related literature discussed in Section 1 makes clear, our formulation also

has computational benefits and drawbacks. In particular, the scale of $\langle M_\theta(y,x), u\rangle$ has no intrinsic interpretation beyond determining whether zero belongs to $\mathbb{E}[Q_\theta(W^O)]$; we do not assign substantive or finite-sample meaning to a positive distance from zero. Nor do we claim that our choice of instrument class $\mathcal{G}$, norm $\|\cdot\|_\infty$, or support-function representation is optimal for estimation or the construction of critical values. We favor the formulation for its geometric clarity:

$$\theta \in \Theta_I^* \quad \Longleftrightarrow \quad 0 \in \mathbb{E}[Q_\theta(W^O)].$$

Section 3 develops its computational and inferential implementation.

3 Estimation and Computation

3.1 Estimator and Confidence Sets

Theorem 1 is pointwise in θ and requires neither compactness of Θ[7] nor continuity of $m(\cdot)$ in θ. For estimation and inference the following assumptions must be introduced. Compactness of Θ in Assumption 4 and the uniform Lipschitz bound in Assumption 5 control the sample criterion uniformly over θ, as required for the Hausdorff-consistency result in Theorem 2. The coverage proof for Theorem 3 uses a common moment envelope and the decay of the weighted instruments to obtain relative compactness of the covariance kernels over $(\theta, P) \in \mathcal{F}$ for which Assumptions 1–3 are sufficient. Its sequential argument takes limits of covariance kernels and requires no continuity of m in θ.

Assumption 4. *The parameter space Θ is a nonempty compact subset of $\mathbb{R}^{d_\theta}$.*

Assumption 5. *$m(y,x,\theta)$ is uniformly Lipschitz continuous. That is, there exists a constant*

[7] In fact, no restriction is put on Θ whatsoever for the purposes of identification.

$L < \infty$ *such that, for every* $\theta, \theta' \in \Theta$,

$$\sup_{(y,x)\in\mathcal{W}} \|m(y,x,\theta) - m(y,x,\theta')\|_\infty \leq L\|\theta - \theta'\|.$$

The norm over Θ *is inconsequential.*

Theorem 1 characterizes the sharp identified set through the population criterion

$$T(\theta) \equiv -\inf_{u\in B^\infty} \mathbf{E}\left[h(Q_\theta(W^O), u)\right].$$

Since $0 \in B^\infty$, $T(\theta) \geq 0$, and Theorem 1 implies $\Theta_I^* = \{\theta \in \Theta : T(\theta) = 0\}$.

The criterion $T(\theta)$ is not directly computable because it involves countably many moment coordinates and an infinite-dimensional set of directions. For estimation, we approximate the conditional moment restriction by retaining only a finite number of instruments. Let $\pi_K : \ell_1 \to \mathbb{R}^{d_m \cdot K}$ denote the truncation map that selects the coordinates indexed by (j,k), $j = 1, \ldots, d_m$, $k = 1, \ldots, K$. Let $\lambda_K : \mathbb{R}^{d_m K} \to \ell_1$ denote the zero-padding embedding that maps a finite vector into ℓ_1 by keeping its first $d_m K$ coordinates and setting all remaining coordinates equal to zero. Define the truncated moment mappings (random vectors) and truncated random sets as[8]

$$M_{\theta,K}(y,x) \equiv \pi_K M_\theta(y,x)$$

$$Q_{\theta,K}(w^o) \equiv \mathrm{cl}_{\mathbb{R}^{d_m K}}\, M_{\theta,K}(D(w^o)) = \pi_K Q_\theta(w^o)$$

[8]For comparisons with the infinite-dimensional set $Q_\theta(w^o) \subset \ell_1$, and to ensure that set distances, such as the Hausdorff distance, are well defined, we require the embedding

$$\widetilde{Q}_{\theta,K}(w^o) = \lambda_K Q_{\theta,K}(w^o) = \lambda_K \pi_K Q_\theta(w^o),$$

where π_K and λ_K applied to sets denote their image under the corresponding point maps.

Let $B_K^\infty \equiv \{u \in \mathbb{R}^{d_m K} : \|u\|_\infty \leq 1\}$. The finite-dimensional population criterion is defined as

$$T_K(\theta) \equiv \vec{\rho}_H\left(0, \mathbb{E}[Q_{\theta,K}(W^O)]\right) = -\inf_{u \in B_K^\infty} h\left(\mathbb{E}[Q_{\theta,K}(W^O)], u\right) = -\inf_{u \in B_K^\infty} \mathbf{E}\left[h(Q_{\theta,K}(W^O), u)\right].$$

The corresponding finite-dimensional outer approximation to the identified set is $\Theta_{I,K} \equiv \{\theta \in \Theta : T_K(\theta) = 0\}$.

Since every admissible completion satisfying the conditional moment restriction also satisfies any finite collection of the unconditional moment restrictions, $\Theta_I^* \subseteq \Theta_{I,K}$ for every K.

Given an i.i.d. sample $\{W_i^O\}_{i=1}^n$, define the sample Aumann average[9]

$$\widehat{\mathbb{E}}_n[Q_{\theta,K}] \equiv \text{co}\left(\frac{1}{n}\bigoplus_{i=1}^n Q_{\theta,K}(W_i^O)\right)$$

The sample criterion is

$$\widehat{T}_{n,K}(\theta) \equiv \vec{\rho}_H\left(0, \widehat{\mathbb{E}}_n[Q_{\theta,K}]\right) = -\inf_{u \in B_K^\infty} h\left(\widehat{\mathbb{E}}_n[Q_{\theta,K}], u\right) = -\inf_{u \in B_K^\infty} \frac{1}{n}\sum_{i=1}^n h(Q_{\theta,K}(W_i^O), u).$$

For a deterministic sequence $K(n) \to \infty$, we use $\widehat{T}_n(\theta) \equiv \widehat{T}_{n,K(n)}(\theta)$. Thus an estimator based on the criterion function of distance between 0 and $\mathbb{E}[Q_{\theta,K(n)}(W^O)]$ is the contour set

$$\widehat{\Theta}_n \equiv \left\{\theta \in \Theta : a_n \widehat{T}_{n,K(n)}(\theta) \leq \tau_n\right\}.$$

With this we can state the result concerning consistency of the proposed estimator.

Theorem 2. *Let Assumptions 1–5 hold. Suppose that $\Theta_I^* \neq \emptyset$. Let $K(n) \to \infty$, and let $a_n \equiv \sqrt{\frac{n}{d_m K(n) + d_\theta}}$. Let $\frac{d_m K(n)}{n} \to 0$, $\tau_n \to \infty$ and $\frac{\tau_n}{a_n} \to 0$. Then $\rho_H(\widehat{\Theta}_n, \Theta_I^*) \xrightarrow{p} 0$.*

For the proof of Theorem 2 we first establish a law of large numbers for random sets with growing dimension; see Appendix C. The estimator is built on the result of Chernozhukov

[9] In contrast to the population Aumann expectation, the finite-sample Minkowski average need not be convex in the most general case. We therefore convexify it in defining its empirical analogue. This does not alter any support-function inequality because $h(A, u) = h(\text{co}(A), u)$ for every u.

et al. (2007) for constructing set estimators using a criterion function approach.

The sequence $K(n)$ controls the finite-dimensional approximation of the infinite collection of unconditional moment restrictions. Since each of the $K(n)$ retained instruments is interacted with the d_m components of the moment function, the dimension of the finite random set is $d_m K(n)$. $K(n) \to \infty$ makes the deterministic truncation error vanish, while $d_m K(n)/n \to 0$ ensures that the sampling error from estimating this growing-dimensional random set vanishes. Thus $K(n)$ grows to include more moment restrictions, but slowly enough to control sampling error. We note that since d_m and d_θ are fixed, the rate conditions could be equivalently stated as $\frac{K(n)}{n} \to 0$ and $a_n \asymp \sqrt{\frac{n}{K(n)}}$. The estimator above is obtained by accepting parameter values for which the finite-sieve sample violation is small.

3.2 Inference

We construct confidence sets using the many-moment procedure of Andrews and Shi (2017). The procedure is well suited to the present setting because the support-function inequalities are indexed by the uncountable direction class B^∞; however, we note that alternative procedures exist, e.g., Chernozhukov et al. (2017), which might also be applied to our results. The Andrews and Shi (2017) implementation requires a method for evaluating the supremum over B^∞; the computational section discusses finite and truncated approximations to this problem. We do not consider here the additional error caused by terminating a global optimization routine at an inexact solution, but such ϵ-minimax solutions may be considered, as for example in Aradillas Fernández et al. (2025).

For the closed infinite-dimensional unit ball under the $\|\cdot\|_\infty$ norm, write $u = (u_{jk}) \in B^\infty$, with coordinates indexed in the same order as M_θ. Define

$$\widetilde{m}(w^o, \theta, u) \equiv h(Q_\theta(w^o), u) = \sup_{(y,x)\in D(w^o)} \sum_{j=1}^{d_m} \sum_{k\geq 1} u_{jk} g_k(x) m_j(y, x, \theta) \in \mathbb{R}^1$$

Let

$$\mathcal{U} \equiv B^{\infty} = \{u = (u_{jk})_{j=1,\ldots,d_m,\ k\geq 1} : \|u\|_{\infty} \leq 1\}.$$

Thus each $u \in \mathcal{U}$ is one direction for the complete stacked moment vector and indexes one scalar support-function inequality. The same feasible (y, x) enters every term of the sum inside the supremum, as the latent completion must solve all moments simultaneously. We can slightly rewrite Theorem 1 to bring it into the notation of Andrews and Shi (2017):

$$\Theta_I = \Theta_I^* = \left\{\theta \in \Theta : \mathbf{E}_{P_{W^O}}[\widetilde{m}(W^O, \theta, u)] \geq 0 \quad \forall u \in \mathcal{U}\right\}.$$

The null hypothesis that $\theta \in \Theta_I^*$ can be stated as

$$H_0(\theta): \qquad \mathbf{E}_{P_{W^O}}[\widetilde{m}(W^O, \theta, u)] \geq 0 \quad \text{for every} \quad u \in \mathcal{U}.$$

Finite direction classes and their inferential interpretation are discussed below. The auxiliary conditioning-instrument class in Andrews and Shi (2017) is not needed here, as the functions $\tilde{m}$ already encode the original conditioning restriction of model (1) through Lemma 1, and the support direction u indexes the resulting unconditional inequalities.[10]

Following Andrews and Shi (2017), we use the scalar normalization $\sigma_F(\theta) = \widehat{\sigma}_n(\theta) \equiv 1$. Define the sample mean

$$\overline{m}_n(\theta, u) \equiv \frac{1}{n}\sum_{i=1}^{n} \widetilde{m}(W_i^O, \theta, u),$$

the corresponding sample variance

$$\widehat{\Sigma}_n(\theta, u) \equiv \frac{1}{n}\sum_{i=1}^{n} \left(\widetilde{m}(W_i^O, \theta, u) - \overline{m}_n(\theta, u)\right)^2,$$

[10] See the first paragraph of page 276 in Andrews and Shi (2017).

and its regularized version

$$\overline{\Sigma}_n(\theta, u) \equiv \widehat{\Sigma}_n(\theta, u) + \epsilon, \qquad \epsilon > 0.$$

The fixed ridge is the scalar specialization of the Andrews and Shi (2017) regularization. Since $\sigma_F(\theta) = \widehat{\sigma}_n(\theta) \equiv 1$, we use their benchmark choice $\epsilon = 1/20$.[11]

For the single lower-bound moment inequality indexed by $u \in \mathcal{U}$, the S_1 function of Andrews and Shi (2017) reduces to

$$S_1(s, \Sigma) \equiv \left[\frac{s}{\sqrt{\Sigma}}\right]_-^2, \qquad [x]_- \equiv \max\{-x, 0\}.$$

The negative part appears because our null hypothesis H_0 is that each moment should be weakly greater than zero. The corresponding test statistic is

$$T_n^{AS}(\theta) \equiv \sup_{u \in \mathcal{U}} S_1\left(\sqrt{n}\,\overline{m}_n(\theta, u), \overline{\Sigma}_n(\theta, u)\right).$$

We use the generalized moment-selection critical value of Andrews and Soares (2010), as implemented by Andrews and Shi (2017).

Step 1. Set

$$\kappa_n \equiv (0.3 \log n)^{1/2}, \qquad B_n \equiv \left(\frac{0.4 \log n}{\log\log n}\right)^{1/2},$$

and define the scalar standardized slackness measure

$$\xi_n(\theta, u) \equiv \kappa_n^{-1} \frac{\sqrt{n}\overline{m}_n(\theta, u)}{\sqrt{\overline{\Sigma}_n(\theta, u)}}$$

[11] A direction-adaptive regularization would be an alternative, e.g., $\epsilon(u) = \epsilon_0 \sum_{k \geq 1} |u_k|$. This addresses the concern that, for directions with very small support-function variance, a fixed ridge may dominate the empirical variance. The alternative is an open question, however, because the ridge can approach zero over the direction class. Its validity requires an additional argument.

The corresponding GMS adjustment is

$$\overline{\varphi}_n(\theta,u) \equiv \sqrt{\overline{\Sigma}_n(\theta,u)}B_n\mathbf{1}\{\xi_n(\theta,u) > 1\}.$$

Step 2. Generate B_{boot} nonparametric bootstrap samples $\{W_{i,b}^{O,*} : i = 1,\ldots,n\}$, $b = 1,\ldots,B_{\text{boot}}$, by sampling with replacement from the empirical distribution. For every b, calculate

$$\begin{aligned}
\overline{m}_{n,b}^*(\theta,u) &\equiv \frac{1}{n}\sum_{i=1}^n \widetilde{m}(W_{i,b}^{O,*},\theta,u),\\
\widehat{\Sigma}_{n,b}^*(\theta,u) &\equiv \frac{1}{n}\sum_{i=1}^n \left\{\widetilde{m}(W_{i,b}^{O,*},\theta,u) - \overline{m}_{n,b}^*(\theta,u)\right\}^2,\\
\overline{\Sigma}_{n,b}^*(\theta,u) &\equiv \widehat{\Sigma}_{n,b}^*(\theta,u) + \epsilon.
\end{aligned}$$

Step 3. Form the recentered bootstrap statistic

$$T_{n,b}^{AS,*}(\theta) \equiv \sup_{u\in\mathcal{U}}\left[\frac{\sqrt{n}\left\{\overline{m}_{n,b}^*(\theta,u) - \overline{m}_n(\theta,u)\right\} + \overline{\varphi}_n(\theta,u)}{\sqrt{\overline{\Sigma}_{n,b}^*(\theta,u)}}\right]_-^2.$$

Step 4. Let $q_{n,1-\alpha+\eta}^{AS,*}(\theta)$ be the empirical $(1-\alpha+\eta)$-quantile of $\{T_{n,b}^{AS,*}(\theta) : b = 1,\ldots,B_{\text{boot}}\}$ and define

$$c_{n,1-\alpha}^{AS}(\theta) \equiv q_{n,1-\alpha+\eta}^{AS,*}(\theta) + \eta, \qquad 0 < \eta < \alpha.$$

The parameter η is part of the infinitesimal uniformity adjustment in Andrews and Shi (2017). The theoretical results retain a fixed $0 < \eta < \alpha$. Following their implementation convention, η can be set equal to zero in computation.

The confidence set obtained by test inversion is

$$CS_n^{AS}(1-\alpha) \equiv \left\{\theta \in \Theta : T_n^{AS}(\theta) \leq c_{n,1-\alpha}^{AS}(\theta)\right\}.$$

The inference result requires a strengthening of the tail condition in Assumption 2 (ii). Assumption 2 (ii) is sufficient to ensure that the stacked moment map is ℓ_1-valued and that finite truncations approximate the full moment vector uniformly. To control the complexity of the infinite-dimensional class of support directions, we additionally require a polynomial decay rate for the individual instrument coordinates.

Assumption 6. *Define*

$$a_k \equiv \sup_{x \in \mathcal{X}} |g_k(x)|, \qquad k \geq 1.$$

For some $\beta > 3/2$, $C_a \equiv \sup_{k \geq 1} k^\beta a_k < \infty$.

For many common applications of instrument classes, like those of indicator functions, the unweighted class has the property that $\sup_{x \in \mathcal{X}} |h_k(x)| \leq 1$. If, additionally, $d_m = 1$, then, for example, the researcher may define $g_k(x) \equiv k^{-\beta} h_k(x)$ which will satisfy Assumption 6 when $\beta > 3/2$. The polynomial decay condition above is stronger than Assumption 2 (ii) in general and provides the entropy bound required for a technical requirement used to deal with an uncountable number of moment inequalities called *manageability* (Pollard, 1990). For the uniform coverage statement, we introduce additional notation. Let $\mathcal{P}$ denote the class of observed-data laws on $\mathcal{W}^O$ over which inference is required to be valid. For each $P \in \mathcal{P}$, let $\Theta_I^*(P)$ denote the identified set under P. By Theorem 1,

$$\Theta_I^*(P) = \left\{\theta \in \Theta : \mathbf{E}_P\left[\widetilde{m}(W^O, \theta, u)\right] \geq 0 \quad \forall u \in \mathcal{U}\right\}.$$

Define $\mathcal{F}^+ \equiv \Theta \times \mathcal{P}$ and $\mathcal{F} \equiv \{(\theta, P) \in \mathcal{F}^+ : \theta \in \Theta_I^*(P)\}$.

Theorem 3. *Let Assumptions 1–3 and 6 hold. Then, for every* $\alpha \in (0,1)$,

$$\liminf_{n\to\infty} \inf_{(\theta,P)\in\mathcal{F}} P\left\{\theta \in CS_n^{AS}(1-\alpha)\right\} \geq 1-\alpha.$$

Moreover, for every fixed $P \in \mathcal{P}$ *and every fixed* $\theta \notin \Theta_I^*(P)$, $P\left\{T_n^{AS}(\theta) > c_{n,1-\alpha}^{AS}(\theta)\right\} \to 1$.

The restriction $\beta > 3/2$ ensures that the entropy integral for the infinite-dimensional direction class is finite. The proof is given in Appendix D. Theorem 3 gives uniformly valid marginal coverage over all null pairs $(\theta, P) \in \mathcal{F}$. It does not imply simultaneous coverage of the entire identified set $\Theta_I^*(P)$ or Hausdorff convergence of a fixed-level confidence set.

4 Implementation

This section discusses the computational implementation of the identification, estimation, and inference procedures developed above. We first show that, for an important class of regression models, the finite-dimensional estimation criterion admits an exact linear programming representation. We then discuss outer approximations obtained by restricting the direction class and present two practical strategies for constructing confidence sets using finite collections of directions.

4.1 Exact LP Representation for Monotone-Index Models

For scalar moment functions, the finite-dimensional estimation criterion $\widehat{T}_{n,K}(\theta)$ from Section 3.1 can be computed exactly by linear programming for an important class of models: monotone-index models, including linear, probit, and logit regressions. Formally, let $d_m = d_y = 1$ and $m(y,x,\theta) = y - f(x,\theta)$ with $E[Y|X] = f(x,\theta_0)$ at the true parameter. For each fixed θ, assume $f(\cdot,\theta)$ is continuous and coordinatewise monotone on the feasible rectangles considered below.

Without loss of generality, the covariate domain is $[0,1]^{d_x}$. For our instrument class we use

the weighted cube instruments of Appendix A.2. Consider retaining K weighted indicators $g_k(x) = \omega_k \mathbf{1}\{x \in C_k\}$. Throughout this subsection, we work within the truncated problem $K < \infty$. As $\{C_k\}_{k=1}^K$ overlap, let $\{C_\ell^*\}_{\ell=1}^L$ be the common refinement of their partitions, such that $\forall x, x \in C_\ell^*$ for exactly one ℓ. Define $c_\ell \in \mathbb{R}^K$ as the mapping between $\{C_k\}$ and $\{C_\ell^*\}$ by

$$c_{\ell k} \equiv \omega_k \mathbf{1}\{C_\ell^* \subseteq C_k\}, \qquad k = 1, \ldots, K.$$

Algebraically we have the observation $u^\top g(x) = c_\ell^\top u$ for each $x \in C_\ell^*$. For each observation, denote the domain of an observation i within the set C_ℓ^* as

$$R_{i\ell} \equiv \mathrm{cl}\big(D_i \cap (\mathbb{R} \times C_\ell^*)\big), \quad \ell \in S_i.$$

Per our assumption of monotone-index models, $m(\cdot, \cdot, \theta)$ is continuous and coordinatewise monotone on each $R_{i\ell}$. Its extrema on $R_{i\ell}$ are then the corner values

$$m_{i\ell}^-(\theta) \equiv \min_{z \in \mathrm{vert}(R_{i\ell})} m(z, \theta), \qquad\qquad m_{i\ell}^+(\theta) \equiv \max_{z \in \mathrm{vert}(R_{i\ell})} m(z, \theta).$$

Because the instruments are constant on each refined cell, the observation-level support function satisfies

$$h(Q_{\theta,K}(W_i^O), u) = \max_{\ell \in S_i} \max_{s \in \{-,+\}} \big\{ m_{i\ell}^s(\theta) c_\ell^\top u \big\}, \qquad u \in B_K^\infty = [-1, 1]^K. \tag{9}$$

Each term inside the maximum is linear in u. Consequently, $-\widehat{T}_{n,K}(\theta)$ is exactly the optimal value of the epigraph linear program

$$\begin{aligned} \min_{u\in\mathbb{R}^K,\, t\in\mathbb{R}^n} \quad & \frac{1}{n}\sum_{i=1}^{n} t_i \\ \text{subject to} \quad & -1 \le u_k \le 1, \qquad k = 1,\dots,K, \\ & t_i \ge m^s_{i\ell}(\theta)\, c_\ell^\top u, \qquad i = 1,\dots,n, \quad \ell \text{ s.t } R_{i,\ell} \neq \emptyset, \quad s \in \{-,+\}. \end{aligned} \tag{10}$$

The program has $K + n$ variables and $2\sum_i |S_i|$ epigraph inequalities, in addition to the box constraints. In particular, the constraints are imposed on the original direction u, not separately on the induced coefficients $c_\ell^\top u$. The representation is exact for each finite K; its derivation is given in Appendix E. The negative of the program's optimal value is the estimation criterion $\widehat{T}_{n,K}(\theta)$. The studentized statistic in Section 3.2 is a different optimization problem because its variance denominator depends on the direction u. The GMS adjustment (Andrews and Shi, 2017) enters the bootstrap statistic used to construct its critical value. The program above does not directly compute either statistic.

4.2 Construction of Outer Sets

Theorem 1 characterizes Θ_I^* through the inequalities $\mathbf{E}_{P_{W^O}}[\widetilde{m}(W^O,\theta,u)] \ge 0$ for all directions $u \in \mathcal{U} = B^\infty$, where $\widetilde{m}(w^o,\theta,u) = h(Q_\theta(w^o),u)$. Any subcollection of these inequalities defines an *outer set*, that is, a set containing Θ_I^*. These population outer sets can be tightened by adding directions. Under the conditions of Corollary 2, the corresponding tests control asymptotic size. Adding directions also changes the calibrated critical value, so the resulting confidence sets need not be nested in a given sample. For $\mathcal{V} \subseteq \mathcal{U}$, define

$$\Theta_O(\mathcal{V}) \equiv \left\{\theta \in \Theta : \ \mathbf{E}_{P_{W^O}}[\widetilde{m}(W^O,\theta,u)] \ge 0 \ \ \forall u \in \mathcal{V}\right\}.$$

Truncation of the instrument class is a special case where dropped instruments are not tested by the support function: $u_k = 0$. Since support functions $h(\cdot, \cdot)$ must come from the same space, for $u \in \mathcal{U}$, let $\Pi_K u \equiv \lambda_K \pi_K u$ retain the first K instrument coordinates and set the remaining coordinates to zero. Let $B_K \equiv \Pi_K \mathcal{U}$. Since $h(Q_\theta(w^o), \Pi_K u) = h(Q_{\theta,K}(w^o), \pi_K u)$, inequalities indexed by $\mathcal{V} \subseteq B_K$ involve only the finite-dimensional sets $Q_{\theta,K}$. In particular, $\Theta_O(B_K)$ coincides with the outer set $\Theta_{I,K}$ of Section 3.1. We write $\Theta_O^K(\mathcal{V}) \equiv \Theta_O(\Pi_K \mathcal{V})$.

Corollary 1. *For all $\mathcal{V} \subseteq \mathcal{V}' \subseteq \mathcal{U}$ and every K,*

$$\Theta_I^* \subseteq \Theta_O(\mathcal{V}') \subseteq \Theta_O(\mathcal{V}) \qquad \text{and} \qquad \Theta_I^* \subseteq \Theta_O^K(\mathcal{V}).$$

If, in addition, $\Pi_K \mathcal{V} \subseteq \mathcal{V}$, then $\Theta_O(\mathcal{V}) \subseteq \Theta_O^K(\mathcal{V})$.

Proof. By Theorem 1, the first claim is trivial. The second claim is a special case, since $\Pi_K \mathcal{V} \subseteq \mathcal{U}$. We note one caution in algebra. Even though $\Pi_K \mathcal{U} = \Pi_K B^\infty = B_K$, for $\mathcal{V} \subset \mathcal{U}$, it is not necessarily true that $\Pi_K \mathcal{V} \subset \mathcal{V}$, though again noting $\Pi_K \mathcal{V} \subset \mathcal{U}$. □

Examples of sets closed under truncation are $\mathcal{U}$, B_K, and the coordinate directions $\{\pm e_j\}$ used in Appendix F. Note that when truncating coordinate directions (instruments) $\Pi_K e_j \in \{e_j, 0\}$, where $h(A, 0) = 0$ adds no information.

Under the decay condition of Theorem 3, $(\mathcal{U}, \rho_a)$ is totally bounded. Thus it admits a finite ϵ-net for each $\epsilon > 0$. See Appendix D and Lemma 13 to construct a finite-dimensional grid $\mathcal{U}_\xi \subseteq \mathcal{U}$ such that $\min_{v \in \mathcal{U}_\xi} \rho_a(u, v) \leq \xi/2$ for every $u \in \mathcal{U}$. Combined with the Lipschitz bound $|\widetilde{m}(w^o, \theta, u) - \widetilde{m}(w^o, \theta, v)| \leq M_0 \rho_a(u, v)$, which holds uniformly in θ and P_{W^o}, we note:

$$\inf_{u \in \mathcal{U}} \mathbf{E}[\widetilde{m}(W^O, \theta, u)] \;\geq\; \min_{v \in \mathcal{U}_\xi} \mathbf{E}[\widetilde{m}(W^O, \theta, v)] - \frac{M_0 \xi}{2}.$$

For inference with a direction set $\mathcal{V} \subseteq \mathcal{U}$, let $CS_n^{AS}(1 - \alpha; \mathcal{V})$ be defined as $CS_n^{AS}(1 - \alpha)$ in Section 3.2, with every supremum over $\mathcal{U}$ replaced by a supremum over $\mathcal{V}$. Let $\Theta_O(\mathcal{V})(P)$

denote $\Theta_O(\mathcal{V})$ under the observed-data law P, and let $\mathcal{F}_{\mathcal{V}} \equiv \{(\theta, P) \in \mathcal{F}^+ : \theta \in \Theta_O(\mathcal{V})(P)\}$. By Corollary 1, $\mathcal{F} \subseteq \mathcal{F}_{\mathcal{V}}$ and $\Theta_I \subseteq \Theta_O(\mathcal{V})(P)$.

Corollary 2. *Let the conditions of Theorem 3 hold.*

(a) Coverage holds uniformly over direction sets and null pairs:

$$\liminf_{n\to\infty} \inf_{\mathcal{V}\subseteq\mathcal{U}} \inf_{(\theta,P)\in\mathcal{F}_{\mathcal{V}}} P\big\{\theta \in CS_n^{AS}(1-\alpha;\mathcal{V})\big\} \geq 1-\alpha.$$

In particular, since $\mathcal{F} \subseteq \mathcal{F}_{\mathcal{V}}$, the bound holds uniformly over $(\theta, P) \in \mathcal{F}$ and $\mathcal{V} \subseteq \mathcal{U}$.

(b) For fixed $\mathcal{V} \subseteq \mathcal{U}$, $P \in \mathcal{P}$ and $\theta \notin \Theta_O(\mathcal{V})(P)$, $P\{\theta \notin CS_n^{AS}(1-\alpha;\mathcal{V})\} \to 1$.

(c) If $K(n) \to \infty$, then $CS_n^{AS}(1-\alpha; B_{K(n)})$ has coverage at least $1-\alpha$ uniformly over $\mathcal{F}$, by (a). It also excludes each fixed $\theta \notin \Theta_I^*(P)$ with probability approaching one. No rate restriction on $K(n)$ is required.

The proof can be found in Appendix D. For a single fixed finite direction set, *manageability*, a technical condition of Andrews and Shi (2017), follows from its finite cardinality without Assumption 6. For uniformity over arbitrarily varying direction sets, we retain Assumption 6 and the entropy bound of Appendix D. In (c), the confidence level does not improve as $K(n)$ grows; it remains at least $1-\alpha$. What tightens is the collection of tested inequalities, so that every fixed $\theta \notin \Theta_I^*$ is eventually excluded.

Finally, a particularly interesting connection to the related literature is given in Appendix F. It turns out that, up to minor implementation choices and the fact that they consider missing data instead of interval data, the two outer sets proposed in Aucejo et al. (2017) are special cases of our estimator with particularly chosen direction sets $\mathcal{V}$. In particular, in the Appendix we show that taking $\mathcal{V} = \{\pm e_k\}_{k\leq K}$ consisting of basis vectors retrieves their first outer set construction. The intuition is simple: for their first outer set construction, Aucejo et al. (2017) solve the identification problem by only checking $\forall k, \exists \eta \in \Gamma$ s.t $E_\eta[m(\cdot,\theta)g_k(\cdot)] = 0$. We note that $\Theta_I = \{\theta : \exists \eta \text{ s.t } \forall k, E_\eta[m(\cdot,\theta)g_k(\cdot)] = 0\}$ and for

any proposition $\mathbf{P}$ one can write the logical proposition $(\exists \eta \text{ s.t } \forall k, \mathbf{P_k}) \Rightarrow (\forall k \exists \eta \text{ s.t } \mathbf{P_k})$. The second outer set instead integrates out the missing covariate. In Appendix F we show that, when all covariates are integrated out, this corresponds to $\mathcal{V} = \{\pm \mathbf{1}_K\}$, which attaches equal weight to all cubes. Aucejo et al. (2017) note that $E[m(\cdot, \theta)|X = x] = 0 \Rightarrow E[m(\cdot, \theta)] = 0$, so instrument functions can receive equal weight everywhere. If some instruments are observed pointwise (non-interval) then these could be *integrated out*. In general, it is interesting that the direction set $\mathcal{V}$ in the *dual* space may offer some geometric intuition for support function constructions.

In the simulations and the application, whenever we refer to Aucejo et al. (2017), we mean their estimator extended to the interval data case via our implementation and inference procedure.

4.3 Finite Sample Implementation

The linear program in Section 4.1 exactly minimizes the sample support function, whereas inference in Section 3.2 uses the studentized criterion $T_n^{AS}(\theta)$. Nevertheless, LP solutions to the unstudentized problem may provide a good direction bank $\mathcal{V}$ for assessing computationally convenient candidates for inference. More generally, approximating the infinite-dimensional direction class by finite subcollections is standard in the moment-inequality literature; see, for example, Andrews and Shi (2017). The validity of inference with data-adaptively selected directions follows from Corollary 2 when the direction bank is selected on an independent subsample and the studentized statistic and bootstrap critical value are computed using that bank. We therefore consider two computational strategies: an adaptive procedure based on sample splitting and a fixed procedure using a prespecified direction bank. Their finite-sample performance is examined in Section 5.

For the first implementation strategy, we use Corollary 2(a) to select informative directions in a data-adaptive way while preserving validity. The procedure is as follows:

1. Randomly split the sample into two subsamples, denoted A and B, with sizes n_A, n_B of

approximately equal size.

2. Using the n_A observations of A, find extremum directions u_θ at a small number of values of θ. Specifically, determine $\widehat{\mathcal{V}}_A = \{u_\theta : u_\theta = \arg\min_{u \in B_K} \frac{1}{n_A} \sum_{i=1}^{n_A} h(Q_{\theta,K}(W_i^O), u), \theta \in \{\theta_1, \ldots\} \subset \Theta\}$. For a particular θ and with $K < \infty$ this problem can be solved using the provided program (see Section 4.1). Theoretically, there is no guarantee that this is a good approximation to the true criterion $\hat{T}_{n_B}^{AS}(\theta)$; however, continuity of the criterion intuitively suggests it might be. We note that any direction bank that is a subset of B_K will define an outer set and thus deliver correct size asymptotically.

3. Compute the confidence set on B with direction set $\widehat{\mathcal{V}}_A$. The test statistic $\hat{T}_{n_B}^{AS}(\theta)$ and the bootstrap values $\{\hat{T}_{n_B}^{AS,b}(\theta)\}_{b=1}^{B_{boot}}$ are used to construct a confidence set $CS_{n_B}^{AS}(1-\alpha/2; \widehat{\mathcal{V}}_A)$, such that if $\theta \in \Theta_I$ or $\theta \in \Theta_O(\widehat{\mathcal{V}}_A)$ then $\lim_{n\to\infty} \Pr(\theta \in CS_{n_B}^{AS}(1-\alpha/2; \widehat{\mathcal{V}}_A)) \geq 1-\alpha/2$. We elaborate on this result below.

4. Reverse the roles of A and B, and intersect the two confidence sets, so that, if $\theta \in \Theta_I$, by a Bonferroni argument $\lim_{n\to\infty} \Pr(\theta \in \bigcap_{j\in\{A,B\}} CS_{n_j}^{AS}(1-\alpha/2; \widehat{\mathcal{V}}_{-j})) \geq 1-\alpha$.

Because $\widehat{\mathcal{V}}_A$ is a function of A alone, it is independent of B. Writing n_B for the size of B, conditioning on A gives, for every $(\theta, P) \in \mathcal{F}$,

$$\begin{aligned} P\big\{\theta \in CS_{n_B}^{AS}(1-\alpha/2; \widehat{\mathcal{V}}_A)\big\} &= \mathbf{E}\Big[P\big\{\theta \in CS_{n_B}^{AS}(1-\alpha/2; \widehat{\mathcal{V}}_A) \,\big|\, A\big\}\Big] \\ &\geq \inf_{\mathcal{V}\subseteq\mathcal{U}} P^{n_B}\big\{\theta \in CS_{n_B}^{AS}(1-\alpha/2; \mathcal{V})\big\}. \end{aligned}$$

By Corollary 2(a), the limit inferior of the right-hand side is at least $1-\alpha/2$, uniformly over $\mathcal{F}$. This is where uniformity over $\mathcal{V}$ matters: a guarantee for each fixed direction set separately would not cover a bank that changes with the sample. We believe this procedure avoids treating directions selected from the same observations as fixed, while retaining most of the computational advantage of concentrating the bootstrap on a small set of informative directions. A potential shortcoming of this approach is that splitting the sample might affect

inference in finite samples.

The second complementary strategy for choosing a direction bank consists of considering an extended collection of ex-ante prespecified directions.

Specifically, we consider an extended fixed direction bank consisting of the global directions $\pm\mathbf{1}_J$, the coordinate directions $\pm e_{rj}$, contiguous block directions $\pm\sum_{j=a}^{b} e_{rj}$, and single-sign-change directions $\pm(\sum_{j=1}^{k} e_{rj} - \sum_{j=k+1}^{K_r} e_{rj})$. The latter two classes are constructed separately within each resolution r, where K_r denotes its number of cells. Exact duplicates are removed. Since the directions are prespecified, inference uses the full sample without sample splitting.

For the remainder of this paper we will refer to the split-sample procedure as the *adaptive* implementation, as it chooses the direction set specifically for the data and model studied, whereas the prespecified direction implementation is referred to as the *fixed* implementation.

5 Monte Carlo Simulation

We study the finite-sample performance of the proposed confidence sets in the linear regression model

$$Y_i = \theta_0 + \theta_1 X_i + \varepsilon_i, \qquad X_i \sim U[0,1], \qquad (\theta_0, \theta_1) = (0,1),$$

where ε_i is a centered Gaussian draw clipped to a large compact support. Outcomes are always observed, while covariates may be observed only through intervals

$$[X_{i,l}, X_{i,u}] = [\max\{0, X_i - ws_i\}, X_i], \qquad s_i \sim U[0.5, 1.5].$$

We first examine coverage and confidence-set geometry under three designs with varying degrees of interval observation, before turning to a point-identified design to study power against alternatives.

The first design features random interval observation, independent of (X_i, Y_i), with probability $p = 0.10$ and interval-width parameter $w = 0.1$. In the second design, all covariates are interval-observed ($p = 1$) and $w = 0.1$. The third design allows outcome-dependent interval observation with $p = 0.30$ and $w = 0.25$: the probability of interval observation is $1.6p$ for $Y_i \leq 0.5$ and $0.4p$ otherwise. By symmetry, the unconditional probability remains p.

For each design, we conduct 250 Monte Carlo replications with $n \in \{500, 1000, 5000\}$ observations and construct 90% confidence sets using 399 bootstrap draws. We compare our (Adaptive) and (Fixed) implementations, defined in Section 4.3, with an extension of the outer-set procedure of Aucejo et al. (2017) (ABH) to the interval case as well as conventional OLS confidence ellipses based on midpoint imputation (OLS MID) or complete cases (OLS CC). We report empirical coverage and summarize confidence-set geometry using Minkowski averages across replications.

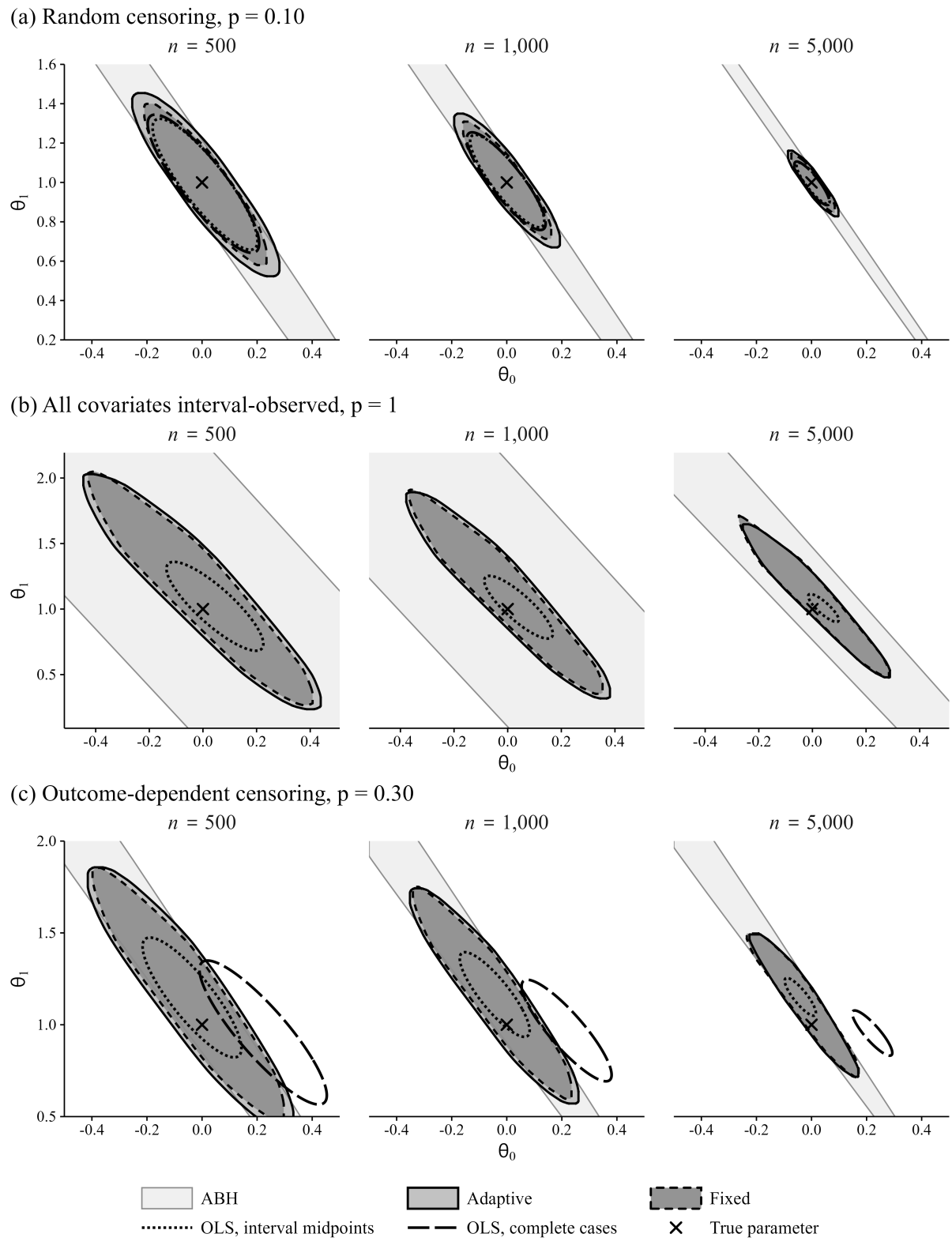


Figure 1: Minkowski-Averaged Confidence Sets across Simulations

Note: Minkowski averages of the joint 90% confidence sets for (θ_0, θ_1) across the Monte Carlo replications of Table 1. The ABH sets extend beyond the plotted range.

Figure 1 displays the Minkowski-averaged 90% confidence sets across the three simulation designs, while Table 1 reports empirical coverage. Under random censoring with 10% of covariates interval-observed (a), our adaptive and fixed procedures perform similarly to conventional OLS, with confidence sets contracting rapidly as the sample size increases. The ABH outer-set procedure also becomes more informative, but its confidence regions retain an elongated, band-like shape, reflecting substantially weaker joint restrictions on the intercept and slope. By exploiting these restrictions jointly, our procedure produces considerably tighter confidence sets that better distinguish the two coefficients.

The difference becomes more pronounced when all covariates are interval-observed (b). The ABH confidence regions remain relatively wide, whereas our procedure continues to provide informative joint restrictions on the regression parameters. Nevertheless, our confidence sets remain substantially wider than the midpoint-imputation OLS ellipses, even for $n = 5{,}000$. This reflects the fact that increasing the sample size reduces sampling uncertainty but does not eliminate the ambiguity inherent in interval measurement. Midpoint OLS, by contrast, becomes increasingly concentrated around the parameter implied by its particular imputation rule. Since this imputation is incorrect in the simulated design, its empirical coverage deteriorates from 75.0% at $n = 500$ to only 9.9% at $n = 5{,}000$, as reported in Table 1.

Finally, outcome-dependent censoring (c) illustrates the potential failure of complete-case estimation. Because the probability of observing a covariate exactly depends on the outcome, complete-case OLS is subject to selection bias and its confidence regions become increasingly concentrated away from the true parameter. Its empirical coverage is only 3.6% at $n = 500$ and falls to zero for larger samples. Midpoint OLS also performs poorly, with coverage declining to 2.7% at $n = 5{,}000$. In contrast, our adaptive and fixed procedures retain 95.6% coverage at this sample size, despite the more severe and endogenous interval observation. Taken together, the simulations illustrate that conventional estimators can become increasingly precise without becoming more reliable, whereas our procedure accounts for the uncertainty generated by incomplete data.

Table 1: Empirical Coverage by DGP and Method

n	Adaptive	Fixed	ABH	OLS MID	OLS CC
(a) Random censoring, $p = 0.10$					
500	0.920	0.964	0.936	0.920	0.896
1,000	0.912	0.960	0.908	0.920	0.916
5,000	0.960	0.980	0.904	0.872	0.900
(b) All covariates interval-observed, $p = 1$					
500	0.980	0.980	0.940	0.750	—
1,000	0.990	1.000	0.960	0.580	—
5,000	0.956	0.945	0.945	0.099	—
(c) Outcome-dependent censoring, $p = 0.30$					
500	0.976	0.984	0.948	0.760	0.036
1,000	0.978	0.991	0.964	0.476	0.000
5,000	0.956	0.956	0.920	0.027	0.000

Note: Share of Monte Carlo replications in which the joint 90% confidence set contains the true parameter $(0, 1)$. Adaptive and Fixed are our procedures (Section 4.3), with 399 bootstrap draws; ABH is the outer set of Aucejo et al. (2017) extended to interval data; OLS MID and OLS CC are joint 90% F-based confidence ellipses from OLS with midpoint imputation and complete cases. OLS CC is not available in design (b), where every covariate is interval-observed.

The preceding simulations illustrate the failure of conventional approaches and the robustness of our method. However, the true shape of the identified set is not known analytically, and the true parameter may lie in its interior. To provide a more stringent assessment of coverage and examine power against fixed local alternatives, we therefore consider a closely related design in which the regression parameter is point identified despite interval observation of some covariates.

We retain the same regression model and interval construction but restrict censoring to the lower half of the support of X_i. Let $C_i = 1$ indicate that X_i is interval-observed. Censoring can occur only when $X_i \leq 1/2$, while X_i is always observed exactly for $X_i > 1/2$. Per Equa-

tion (1), $E[m(\cdot,\theta)|X=x] = E[m(\cdot,\theta)|X>1/2] = 0$. Since X_i is observed exactly on the nondegenerate interval $(1/2,1]$, the model implies point identification or a rejection of the model. Since the model is simulated with $\theta_0 = (0,1)$ we therefore know that $\Theta_I^* = \{(0,1)\}$. The true parameter is therefore a known boundary point of the identified set. This allows us to evaluate finite-sample coverage at the boundary and power against nearby fixed alternatives.

We consider both random and outcome-dependent censoring as well as two levels of the overall frequency of missing data $p = P(C_i = 1)$. For censored observations, we use $w = 0.1$. Under random censoring, $\Pr(C_i = 1 \mid X_i \leq 1/2) = 2p$ ($\Pr(C_i = 1) = p$). For outcome-dependent censoring we set $\Pr(C_i = 1 \mid X_i \leq 1/2, Y_i \leq 0.25) = 3.2p$, and $\Pr(C_i = 1 \mid X_i \leq 1/2, Y_i > 0.25) = 0.8p$, implying $P(C_i = 1) = p$. We examine both $p \in \{0.05, 0.25\}$.

Figure 2 reports empirical inclusion frequencies over 200 Monte Carlo replications along the slope path $(0,\theta_1)$, with $\theta_1 \in [0.8,1]$, for $n \in \{1{,}000, 5{,}000, 20{,}000\}$. To avoid cluttering the graph, we compare our adaptive implementation with complete-case OLS; the fixed implementation yields similar curves. In Appendix F we compare to Aucejo et al. (2017) as well. The columns compare the baseline and more-censoring designs, while the rows distinguish random from outcome-dependent censoring.

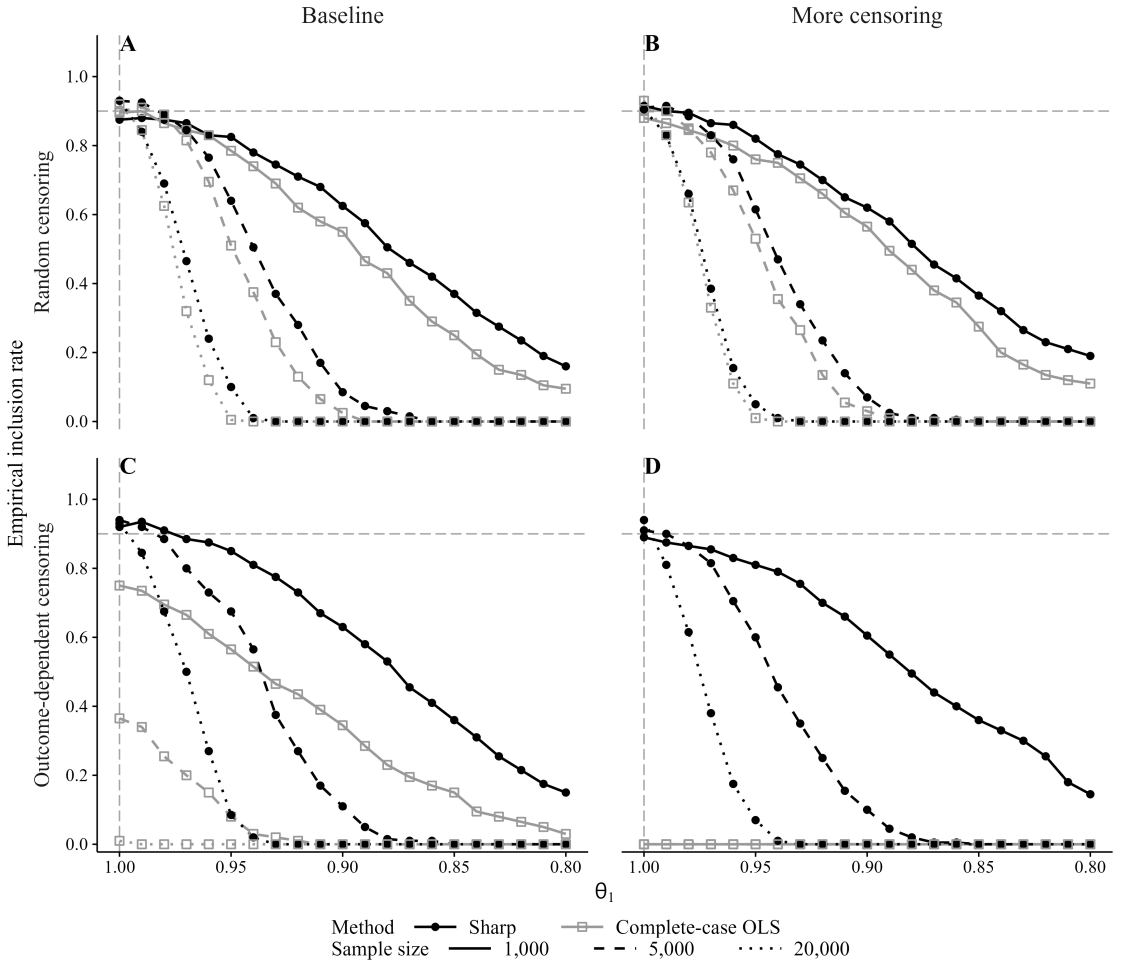


Figure 2: Empirical Inclusion Frequencies along the Slope Path

Note: Share of 200 Monte Carlo replications in which $(0, \theta_1)$ lies in the 90% confidence set, for the adaptive procedure (*Sharp*) and the joint Wald confidence set from complete-case OLS. The true parameter is $(0, 1)$, so the inclusion rate at $\theta_1 = 1$ is the coverage and the rates at $\theta_1 < 1$ are one minus the power. The horizontal dashed line marks the nominal level of 0.90. Baseline: $p = 0.05$; more censoring: $p = 0.25$.

Figure 2 illustrates the trade-off between robustness and power. Under random censoring, complete-case OLS is valid and generally more powerful than our procedure, since discarding interval-observed observations does not induce selection bias. The lower power of the adaptive procedure in this benign design is therefore the price paid for remaining agnostic about the censoring mechanism. The power gap tends to narrow as the sample size increases and, in this design, under more severe censoring. The latter reflects the relative informational advantage of our approach, which retains interval-observed observations that complete-case OLS discards.

Importantly, our procedure continues to deliver approximately nominal coverage at the true parameter under both random and outcome-dependent censoring, while its power against fixed alternatives increases substantially with sample size. Under outcome-dependent censoring, however, more severe censoring makes complete-case OLS increasingly unreliable: its inclusion frequency at the true parameter deteriorates sharply, whereas our procedure retains coverage and comparable power to OLS.

6 Application

As an empirical illustration, we revisit Castillo et al. (2018), who elicit children's time preferences using a multiple price list (20 binary choices between \$49 in one month and a larger amount in seven months) and study how the resulting discount-rate measure predicts on-time high school graduation. In a multiple price list, respondents face a sequence of binary intertemporal choices in which one option is held fixed while the alternative (here, the later payment) varies monotonically. Under standard rationality assumptions, choices should exhibit a single cutoff: the respondent selects the sooner option for lower later amounts and then switches once to the later option when the offered premium becomes sufficiently large. The setting is attractive for our purposes for three reasons. First, multiple price lists are widely used in experimental and applied microeconomics, making the example relevant to a broad class of studies. The design inherently generates interval-valued covariates: even with a single switch, because only a finite number of questions are asked and researchers want to use rounded dollar values, the discount rate is only known to lie within a bracket determined by two adjacent rows. Second, respondents sometimes answer inconsistently, switching multiple times between the earlier and later payments. Such respondents have been found to differ systematically in observable characteristics (see Yu et al. (2021) for a discussion and potential design-based solutions). Treating these responses as random error or using the interval between the first and last switch generates heterogeneous interval lengths that may be

correlated with unobservables, raising concerns for identification strategies relying on mean independence under interval measurement (Manski and Tamer, 2002). Third, graduation outcomes are missing for roughly 30% of participants due to attrition or departure from the district. This motivated inverse-probability weighting in the original analysis and provides a natural setting for methods that accommodate missing outcomes alongside interval measurement.

In their original analysis, to deal with observing multiple switches, Castillo et al. (2018) use an algorithm that maps inconsistent responses into consistent single-switch patterns, which we call their corrected switching pattern. Because the correction need not be unique, for robustness they consider several alternatives. The resulting discount-rate intervals are then represented by their midpoints in the regression.

Our approach instead accounts explicitly for the uncertainty within these intervals. We consider three progressively more permissive specifications assuming a probit model of graduation on an intercept and the potentially interval-observed discount rate. First, taking the corrected switching patterns as given, we allow discount rates to vary within the survey-implied intervals of width 0.0735 (*MPL intervals*). Second, for respondents with inconsistent choices, we allow discount rates to lie anywhere between their first and last switches (*+ Inconsistent responses*). Finally, we also include respondents with missing graduation outcomes for whom discount-rate information is available and who were dropped from the original analysis (*+ Missing outcomes*).

Table 2 reports results for two specifications. Panel A considers the baseline probit model with the discount rate X as the only covariate. Panel B examines robustness to including the additional covariates W used by Castillo et al. (2018) and applying inverse-propensity weighting as in their original analysis. For this specification, we retain instruments that depend only on the discount rate X, rather than the full covariate vector (X, W). Since this restricts the class of moment inequalities, it yields an outer set to the identified set of the fully

conditioned model.[12] Although theoretically a richer instrument class could incorporate W, a fully interacted partition of its 16 covariates would generate up to $2^{16} = 65{,}536$ cells even if all were binary, before accounting for the partition of X. With only $n = 547$ observations, most cells would contain few or no observations, leading to imprecisely estimated moment inequalities and potentially unstable inference.

Table 2 reports 95% confidence sets for the marginal effect of the discount rate on the probability of on-time graduation, evaluated at a common reference value. In Panel A, the conventional probit yields a marginal effect of -0.086 with a confidence interval of $[-0.152, -0.020]$, excluding zero. To distinguish the effect of interval measurement from that of the inference procedure, the *Midpoint X* specification treats discount rates as exactly observed at their imputed midpoints but applies our projection-based inference procedure using Andrews and Shi (2017). The resulting confidence sets are already wider than the conventional probit interval, reflecting the change in inference method rather than uncertainty about the latent discount rates. Relative to this benchmark, accounting for the MPL intervals increases confidence-set width by approximately 27%, yielding $[-0.226, 0.020]$ for the adaptive and $[-0.216, 0.018]$ for the fixed implementation. Both now include zero. Allowing for inconsistent responses widens the sets considerably further, while incorporating respondents with missing graduation outcomes increases the sample from 547 to 790 but introduces additional ambiguity, resulting in substantially wider confidence sets.

The comparison with Aucejo et al. (2017) highlights the gains from exploiting the joint moment restrictions underlying sharp identification. For the basic MPL intervals, the ABH outer-set procedure yields $[-0.396, 0.403]$, substantially wider than either of our confidence sets, with even larger differences when inconsistent responses and missing outcomes are incorporated. The adaptive and fixed implementations, by contrast, deliver broadly similar

[12] For computational tractability, we do not search over a grid of nuisance parameters γ associated with W. Instead, we fix $\hat{\gamma}$ at the coefficients estimated from a conventional probit using midpoint-imputed discount rates, so that $W^\top \hat{\gamma}$ enters as an offset. The estimated propensity weights are likewise treated as fixed. Consequently, the reported confidence sets do not account for first-stage estimation uncertainty, and the formal coverage guarantees of Corollary 2 do not extend to Panel B.

results. Panel B shows that the qualitative pattern persists when additional covariates and inverse-propensity weighting are included: conventional probit inference excludes zero, while accounting for interval measurement produces wider sets containing zero.

Table 2: Marginal Effects at a Common Reference Discount Rate

Specification	n	Adaptive	Fixed	ABH
Panel A: Without controls				
Conventional probit	547		-0.086 $[-0.152, -0.020]$	
Midpoint X	547	$[-0.194, 0.000]$	$[-0.185, -0.003]$	$[-0.223, 0.039]$
MPL intervals	547	$[-0.226, 0.020]$	$[-0.216, 0.018]$	$[-0.396, 0.403]^*$
+ Inconsistent responses	547	$[-0.446, 0.160]$	$[-0.465, 0.211]$	$[-0.638, 0.587]^*$
+ Missing outcomes	790	$[-0.471, 0.439]$	$[-0.518, 0.465]$	$[-8.872, 3.446]^*$
Panel B: With controls				
Conventional probit	547		-0.091 $[-0.174, -0.008]$	
Midpoint X	547	$[-0.269, 0.045]$	$[-0.221, 0.008]$	$[-0.237, 0.044]$
MPL intervals	547	$[-0.283, 0.097]$	$[-0.258, 0.051]$	$[-0.733, 1.096]^*$
+ Inconsistent responses	547	$[-0.528, 0.355]$	$[-0.446, 0.328]$	$[-7.059, 4.991]^*$
+ Missing outcomes	790	$[-0.407, 0.557]$	$[-0.458, 0.581]$	$[-8.872, 3.446]^*$

Note: Entries are projections of joint 95% confidence sets for (θ_0, θ_1) onto the marginal effect $\phi(\theta_0 + \theta_1\bar{x} + \bar{c})\,\theta_1$. Following Castillo et al. (2018), the marginal effect is evaluated at the sample mean of the discount rate: $\bar{x} = 0.799$ in Panel A and the IPW-weighted mean $\bar{x} = 0.823$ in Panel B for the 547 observations with observed graduation status, and the unweighted mean $\bar{x} = 0.797$ over all 790 observations in the *+ Missing outcomes* rows. Controls enter as an offset centered at their (weighted) sample means, so $\bar{c} = 0$. Because the parameter is set-identified, no point estimate is reported. The *Conventional probit* row is the exception: it reports the estimated marginal effect from a standard probit with midpoint discount rates (Panel A: unweighted, no controls; Panel B: IPW with the controls of Castillo et al. (2018)) and its marginal 95% delta-method confidence interval based on heteroskedasticity-robust standard errors. Projection intervals are conservative for the scalar marginal effect relative to such marginal intervals. In Panel B, the control coefficients and propensity weights are estimated in a first step and held fixed, so these confidence sets do not account for first-step uncertainty and the coverage guarantee of Corollary 2 does not apply (see footnote 12). *The ABH confidence set reaches the boundary of the parameter grid, so the reported interval is truncated.

The application illustrates how even modest uncertainty about incompletely observed covariates, here a binning of 7.35 percentage points, can affect empirical conclusions. Relative to the midpoint benchmark, accounting for the intervals inherent to the MPL design increases the width of the adaptive 95% confidence set by approximately 27%. An additional question for applied researchers is what improvements in data quality and/or additional assumptions would be needed to obtain more informative confidence sets. Figure 3 explores this by progressively reducing the ambiguity in the observed discount rates. The horizontal axes of both panels represent the same aggregate reduction in interval width. Panel A narrows every MPL interval symmetrically around its midpoint, whereas Panel B replaces the intervals of a randomly selected share of respondents by their midpoints, treating these discount rates as exactly observed. These exercises represent either a shrinking of the intervals or having fewer interval observations—both a weakening of the midpoint-imputation assumption.

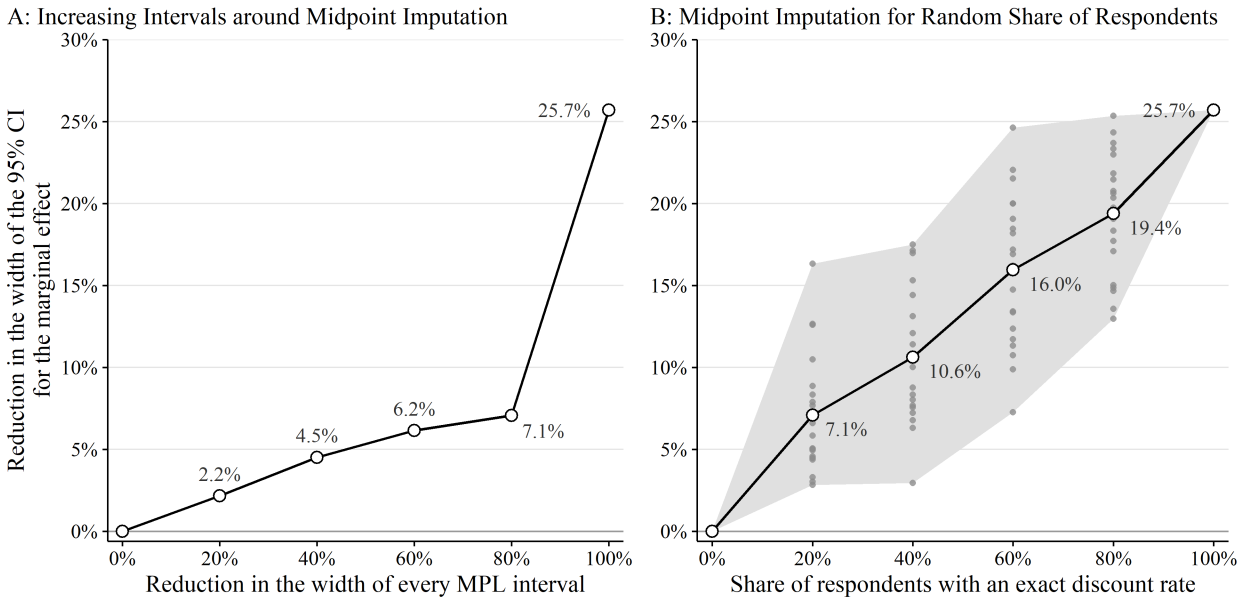


Figure 3: The Value of More Precise Covariate Measurement

Note: Reduction in the width of the projected 95% confidence interval for the marginal effect, relative to the *MPL intervals* specification in Panel B of Table 2 (fixed direction bank). Panel B: the line is the mean over 20 random draws of the respondents who become exactly observed, the dots are the individual draws, and the shaded area is the minimum–maximum range across draws.

The results reveal substantial differences between these two ways of reducing measure-

ment uncertainty. Narrowing every interval by 80% reduces the width of the confidence interval by only 7.1%, whereas replacing 80% of the intervals by their midpoints yields a reduction of 19.4%. Both approaches converge to the same 25.7% reduction when all interval uncertainty is eliminated. The relationship is notably nonlinear in Panel A, where moving from point observations to 20% length (equivalently 1.47 p.p.) produces a pronounced decrease in precision. The variation across random subsamples in Panel B also illustrates that the gains from more precise measurement depend on which observations become more informative. Such sensitivity exercises can inform the design of future surveys by comparing the potential benefits of more precise measurement for all respondents with those of obtaining more detailed information for a subset.

7 Conclusion

Conditional moment equality models are central to empirical economics, but the variables on which they condition are often not fully observed. This paper studies this problem without replacing the latent variables by imputations, conditioning on the observed intervals, or imposing missing-at-random or mean-independence assumptions. We characterize the sharp identified set: the set of all parameter values consistent with the maintained conditional moment model and the incomplete data alone. The resulting object provides a worst-case benchmark against which stronger identifying assumptions and imputation-based procedures can be assessed.

The main identification result rewrites the conditional moment restriction as a countable collection of unconditional moments and embeds these restrictions in an infinite-dimensional random set. A support-function characterization then converts the existence of an admissible completion of the latent data into a collection of moment inequalities. This extends the random-set approach to a setting in which incomplete continuous covariates generate infinitely many restrictions. We also show how the characterization can be operationalized

through finite-dimensional sieve approximations, yielding a Hausdorff-consistent estimator and uniformly valid confidence regions. For monotone-index models, including linear, probit, and logit regressions, the finite approximations are computationally tractable.

The simulations illustrate both the benefits and costs of robustness to incomplete observation. Under outcome-dependent censoring, conventional methods can suffer severe coverage failures, whereas our procedure retains approximately nominal coverage. In a point-identified design, our procedure also gains power against fixed alternatives as the sample size increases, although robustness entails a power cost when complete-case analysis is valid.

The empirical application demonstrates that even modest data limitations can affect substantive conclusions. In the multiple-price-list setting, accounting for the intervals inherent in the survey increases confidence-set width by approximately 27% relative to the midpoint benchmark, rendering a previously significant association inconclusive. Inconsistent responses and missing outcomes introduce further uncertainty, while our sharp approach yields substantially tighter confidence sets than non-sharp outer approximations. Finally, the sensitivity exercises illustrate how the gains from more precise measurement depend on whether uncertainty is reduced for all respondents or eliminated for a subset. Such exercises can help guide future survey design and data collection by quantifying the potential value of more informative observations.

Several extensions remain open. The exact linear-programming representation relies on scalar moments with a monotone index; vector-valued moment functions require a different computational approach. Further, the linear program evaluates the unstudentized criterion and is used here to select directions at finitely many parameter values; it could potentially be replaced by an optimization routine that maximizes the studentized statistic over directions directly, at the expense of computational efficiency. Confidence sets for scalar functionals such as marginal effects are obtained here by projecting joint confidence sets, which is conservative; methods for subvector inference, such as Bugni et al. (2017) and Kaido et al. (2019), could deliver shorter intervals. Finally, the choice of instruments and directions

affects finite-sample power, and data-driven rules for this choice are left for future work.

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A Proofs of Section 2

A.1 Proof of Lemma 1

For compact notation, let $\mathsf{C}(\theta, \eta)$ denote the statement that

$$\mathbf{E}_\eta[m(Y, X, \theta) \mid X] = 0 \qquad \eta_X\text{-a.s.},$$

and let $\mathsf{U}(\theta, \eta)$ denote the statement that

$$\int m_j(y, x, \theta) g_k(x)\, \eta(dy, dx, dw^o) = 0 \qquad \forall j,\ k.$$

Proof. Fix $(\theta, \eta) \in \Theta \times \Gamma$.

Suppose $\mathsf{C}(\theta, \eta)$ holds and set

$$\mu_j(x) \equiv \mathbf{E}_\eta[m_j(Y, X, \theta) \mid X = x].$$

Iterated expectations give

$$\int m_j(y, x, \theta) g_k(x)\, d\eta = \int \mu_j(x) g_k(x)\, \eta_X(dx) = 0$$

for every (j, k). Thus $\mathsf{U}(\theta, \eta)$ holds.

Suppose $\mathsf{U}(\theta, \eta)$ holds and define $\mu_j(X) = \mathbf{E}_\eta[m_j(Y, X, \theta) \mid X]$. By linearity, the equality holds for every $v \in \text{span}(\mathcal{G})$. Fix $f \in C(\mathcal{X})$. Assumption 2(i) supplies $v_r \in \text{span}(\mathcal{G})$ with $\|f - v_r\|_\infty \to 0$. For all sufficiently large r,

$$|\mu_j v_r| \le (\|f\|_\infty + 1)|\mu_j|.$$

Dominated convergence therefore gives

$$\int \mu_j(x) f(x)\, \eta_X(dx) = \lim_{r\to\infty} \int \mu_j(x) v_r(x)\, \eta_X(dx) = 0 \qquad \forall f \in C(\mathcal{X}).$$

Let ν_j be the finite signed measure with density μ_j relative to η_X. Since $\mathcal{X}$ is compact, uniqueness in the Riesz representation theorem implies from the preceding display that $\nu_j = 0$. Thus $\mu_j = 0$ η_X-almost surely for every j, and $\mathsf{C}(\theta, \eta)$ holds.

We have proved the fixed-pair equivalence

$$\mathsf{C}(\theta, \eta) \quad \Longleftrightarrow \quad \mathsf{U}(\theta, \eta). \tag{FP}$$

Both identified sets quantify over the same $\eta \in \Gamma$, with the appropriate integrability included in C and U. Therefore $\Theta_I = \Theta_I^*$. □

A.2 Class of Weighted Cubes

The following proposition introduces the instrument class used and establishes that the class satisfies Assumption 2. The choice of weighted cubes allows the exact linear program epigraph formulation described in the main text. For this reason, it is the selected class in our implementations; however, we note that many other classes would also fulfill the conditions needed for our theoretical results.

Proposition 1. *For the cube construction, let* $\mathcal{X} = [0,1]^{d_x}$*, following Andrews and Shi (2013). Let* $r_0 \geq 1$ *be an integer. For each* $r \geq r_0$ *let* $N_r \equiv \sum_{s=r_0}^{r} (2s)^{d_x}$*, with* $N_{r_0-1} = 0$*,*

and for $a = (a_1, \ldots, a_{d_x}) \in \{1, \ldots, 2r\}^{d_x}$*, define*

$$I_{a_j,r} \equiv \begin{cases} [0, (2r)^{-1}], & a_j = 1, \\ ((a_j - 1)/(2r), a_j/(2r)], & a_j = 2, \ldots, 2r, \end{cases} \qquad C_{a,r} \equiv \prod_{j=1}^{d_x} I_{a_j,r},$$

$$h_{a,r}(x) \equiv \mathbf{1}\{x \in C_{a,r}\}, \qquad \omega_{a,r} \equiv w(r) = (c_0 + N_r^{\beta})^{-1},$$

$$g_{a,r}(x) \equiv \omega_{a,r} h_{a,r}(x).$$

If $c_0 > 0$*,* $\beta > 1$*, and the pairs* (a, r) *are enumerated by increasing resolution, the resulting classes* $\mathcal{H} = \{h_{a,r}\}$ *and* $\mathcal{G} = \{g_{a,r}\}$ *satisfy Assumption 2.*

Proof. At every resolution r, the cubes $\{C_{a,r} : a \in \{1, \ldots, 2r\}^{d_x}\}$ form a partition of $[0,1]^{d_x}$ under the stated boundary convention. At each x, exactly one indicator $h_{a,r}$ is active per resolution, so

$$\sup_{x \in [0,1]^{d_x}} \sum_{r=r_0}^{\infty} \sum_{a} |g_{a,r}(x)| = \sum_{r=r_0}^{\infty} (c_0 + N_r^{\beta})^{-1} < \infty.$$

The series converges because $N_r \geq r - r_0 + 1$ and $\beta > 1$. Let $R(K)$ be the largest resolution completed among the first K instruments, with $R(K) = r_0 - 1$ if none is complete. Enumeration by increasing resolution implies $R(K) \to \infty$. At any x, at most one of the remaining indicators is active at each unresolved resolution; hence the tail is bounded uniformly by

$$\sum_{r > R(K)} (c_0 + N_r^{\beta})^{-1} \longrightarrow 0.$$

This proves Assumption 2(ii).

For Part (i), fix $f \in C([0,1]^{d_x})$ and $\varepsilon > 0$. Uniform continuity supplies a $\delta > 0$ such that $|f(x) - f(x')| < \varepsilon$ whenever $\|x - x'\|_2 < \delta$. Choose r with $\sqrt{d_x}/(2r) < \delta$; this quantity

bounds the diameter of every $C_{a,r}$. Select $x_{a,r} \in C_{a,r}$ and define

$$v_r(x) \equiv \sum_a f(x_{a,r}) h_{a,r}(x).$$

The sum is finite, so $v_r \in \text{span}(\mathcal{H}) = \text{span}(\mathcal{G})$, and the diameter bound gives $\|f - v_r\|_\infty < \varepsilon$. □

The positive weights do not alter this finite-span approximation, because $g_{a,r} = \omega_{a,r} h_{a,r}$ with $\omega_{a,r} > 0$ implies $\text{span}(\mathcal{H}) = \text{span}(\mathcal{G})$. Their role is to place the infinite stack in ℓ_1 and control its tail.

For inference, the same weights also satisfy Assumption 6 when $\beta > 3/2$. To see this, let $(a(k), r(k))$ be the cell and resolution of instrument k in this enumeration, so that $(h_k, \omega_k, g_k) = (h_{a(k),r(k)}, \omega_{a(k),r(k)}, g_{a(k),r(k)})$ and $N_{r(k)-1} < k \leq N_{r(k)}$. Recall $a_k \equiv \sup_{x \in \mathcal{X}} |g_k(x)|$ from Assumption 6. Since each cube indicator h_k has supremum one,

$$a_k = \omega_k = (c_0 + N_{r(k)}^\beta)^{-1} \leq N_{r(k)}^{-\beta} \leq k^{-\beta},$$

so that $C_a \leq 1$.

For truncation through resolution R, let $K = N_R$ and let $\Pi_K v$ retain the first K instruments in each primitive-moment block and zero-pad the rest. Uniformly over w^o, θ, and $v \in \mathcal{U} = B^\infty$,

$$\begin{aligned} |\widetilde{m}(w^o, \theta, v) - \widetilde{m}(w^o, \theta, \Pi_K v)| &\leq d_m C_1 \sup_{x \in \mathcal{X}} \sum_{k > K} |g_k(x)| \\ &= d_m C_1 \sum_{r > R} (c_0 + N_r^\beta)^{-1}. \end{aligned}$$

Exactly one cube indicator is active at each resolution. Since $N_r \asymp r^{d_x+1}$, the last expression is $O(R^{1-\beta(d_x+1)})$.

A.3 Exact feasibility and closure-based identification

This subsection compares the exact set Θ_I with a distance-based formulation and shows that they coincide under the paper's assumptions. The point is not that I_θ must be closed; it need not be. Rather, any sequence of admissible moment vectors converging to zero yields an admissible law that attains zero.

In the notation of this paper, the distance-based formulation associated with Schennach (2014, Equation (3)) is

$$\Theta_I^{\text{Sch}} \equiv \left\{ \theta \in \Theta : \inf_{\eta \in \Gamma} \left\| \int M_\theta(y,x)\, \eta(dy, dx, dw^o) \right\|_1 = 0 \right\}. \tag{11}$$

By (4),

$$\inf_{\eta \in \Gamma} \left\| \int M_\theta \, d\eta \right\|_1 = \text{dist}_{\ell_1}(0, I_\theta).$$

Distance to a set equals distance to its closure, so

$$\Theta_I^{\text{Sch}} = \{\theta \in \Theta : 0 \in \text{cl}_{\ell_1} I_\theta\}. \tag{12}$$

Because Γ is nonempty,

$$\Theta_I \subseteq \Theta_I^{\text{Sch}},$$

so that

$$\Theta_I^{\text{Sch}} \setminus \Theta_I = \{\theta \in \Theta : 0 \in \text{cl}_{\ell_1} I_\theta \text{ and } 0 \notin I_\theta\}.$$

Proposition 2 (Equivalence in the present model)**.** *Under Assumptions 1–3,*

$$\Theta_I = \Theta_I^{\text{Sch}}.$$

By (12), $\Theta_I^{\text{Sch}} = \{\theta \in \Theta : 0 \in \text{cl}_{\ell_1} I_\theta\}$, and Appendix Lemma 6 gives $0 \in I_\theta$ if and only if $0 \in \text{cl}_{\ell_1} I_\theta$. The two sets therefore coincide. Lemma 6 is proved in Appendix B.

For comparison, the following general observation gives a familiar sufficient condition under which exact and distance-based definitions agree.

Proposition 3 (A general compactness condition)**.** *Let Γ' be a nonempty compact topological space and suppose that $b_\theta : \Gamma' \to \ell_1$, or $\mathbb{R}^L$ for finite L, is well defined for every $\theta \in \Theta$. Define*

$$\Theta^{\text{exact}} \equiv \{\theta : b_\theta(\gamma) = \mathbf{0} \text{ for some } \gamma \in \Gamma'\},$$
$$\Theta^{\text{dist}} \equiv \{\theta : \inf_{\gamma \in \Gamma'} \|b_\theta(\gamma)\|_1 = 0\}.$$

If $\gamma \mapsto \|b_\theta(\gamma)\|_1$ is lower semicontinuous on Γ' for every θ, then $\Theta^{\text{exact}} = \Theta^{\text{dist}}$.

Proof. The inclusion $\Theta^{\text{exact}} \subseteq \Theta^{\text{dist}}$ is immediate. Fix $\theta \in \Theta^{\text{dist}}$ and for each $n \geq 1$ define

$$L_n(\theta) \equiv \left\{\gamma \in \Gamma' : \|b_\theta(\gamma)\|_1 \leq n^{-1}\right\}.$$

Because $\inf_{\gamma \in \Gamma'} \|b_\theta(\gamma)\|_1 = 0$, every $L_n(\theta)$ is nonempty. Lower semicontinuity makes each $L_n(\theta)$ closed in Γ' (Beer, 1993), and $L_{n+1}(\theta) \subseteq L_n(\theta)$. Compactness and the finite-intersection property therefore give $\bigcap_{n \geq 1} L_n(\theta) \neq \varnothing$. For any γ^* in this intersection, $0 \leq \|b_\theta(\gamma^*)\|_1 \leq n^{-1}$ for every n, so $b_\theta(\gamma^*) = \mathbf{0}$ and $\theta \in \Theta^{\text{exact}}$. □

In the present model $b_\theta(\eta) = \int M_\theta \, d\eta$. If every instrument g_k is continuous, then M_θ is a uniform ℓ_1-limit of continuous truncations and is therefore continuous, so $\eta \mapsto \| \int M_\theta \, d\eta\|_1$ is continuous and satisfies the lower-semicontinuity condition of Proposition 3. This is a sufficient condition, not the route used for Theorem 1. Continuity of the instruments would

substantially simplify some parts of the following identification proof; see the remarks in the respective section. For the general theory developed in this paper we do not rely on continuity.

The multiresolution cube geometry follows Andrews and Shi (2013); the ℓ_1 coordinate weights are imposed here for Assumption 2(ii) and Theorem 3. The offset c_0 is a positive normalization or tuning constant, not an identifying restriction. Setting $c_0 = 100$ and $\beta = 2$ retains the tuning constants in Andrews and Shi (2013); here they enter $w(r) = (c_0 + N_r^\beta)^{-1}$ to satisfy the enumerated decay condition. No restriction is imposed on the probability that X lies on a cube boundary.

B Proof of Theorem 1

B.1 Roadmap and probability-space convention

The proof has five steps. Lemma 2 establishes that the moment image is a measurable integrably bounded random compact set. Lemma 3 identifies admissible laws with laws of feasible latent selections. Lemma 4 records compactness and convexity of the admissible-law set. Lemma 5 connects the Aumann expectation to the attainable mean-moment set I_θ. Lemma 6 shows that taking the closure does not alter membership at zero. The main proof then applies the convexity and support-function results of Molchanov (2017), similar to Beresteanu et al. (2011).

The probability space carrying the true latent variables $(\Omega, \mathcal{A}, F_0)$ is only a representation of their joint law. An element $\eta \in \Gamma$ should not be understood as a perturbation of the true law $\eta_0 = \mathcal{L}_{F_0}(W, W^O)$ on that particular space. Rather, it represents an alternative complete data-generating process having the observed marginal P_{W^O} and assigning probability one to latent values compatible with the observed set. Because the maintained restrictions and the definition of Γ depend only on joint laws, we may, without changing the model or its identified set, work throughout this appendix on the completed enrichment of the original probability

space obtained by adjoining an independent $V \sim \text{Unif}[0,1]$. All original random variables are identified with their lifts to the enriched space, which preserves their joint laws and therefore also preserves the conditional moment restriction in (1). Selections and Aumann expectations below are understood on this enriched space, while the original notation is retained for simplicity. The uniform variable allows every conditional probability kernel represented by an element of Γ to be realized jointly with W^O, as established in Lemma 3.[13] The enriched probability space is nonatomic, so the Aumann expectations used below are convex by Molchanov (2017, Theorem 2.1.30). Thus their population convexity does not rely on nonatomicity of the original probability space or on the i.i.d. sampling assumption; the latter remains in force for the sampling and asymptotic results.

Recall that π_K is the truncation map and λ_K is the corresponding zero-padding embedding defined in Section 3.

B.2 Auxiliary lemmas

Lemma 2. *The map $M_\theta : \mathcal{W} \to \ell_1$ is bounded and Borel measurable. Moreover,*

$$Q_\theta(W^O) = \text{cl}_{\ell_1} M_\theta(D(W^O))$$

is an integrably bounded, almost surely nonempty random compact set in ℓ_1. For every finite K, $Q_{\theta,K} \equiv \pi_K Q_\theta$ is a random compact set in $\mathbb{R}^{d_m K}$.

Proof. Index the coordinates by (j,k) and write

$$M_\theta(y,x)_{j,k} = m_j(y,x,\theta) g_k(x).$$

[13] This enrichment follows the construction in the proof of Molchanov (2017, Theorem 1.4.8, pp. 82–83); see also Molchanov (2017, Definition 1.4.7 and Example 1.4.9, pp. 82–83).

Assumptions 2 and 3 give

$$\begin{aligned}\|M_\theta(y,x)\|_1 &\le \sum_{j=1}^{d_m}\sum_{k\ge 1}|m_j(y,x,\theta)||g_k(x)| \\ &\le d_m C_1 \sup_{x'\in\mathcal{X}}\sum_{k\ge 1}|g_k(x')| < \infty, \end{aligned} \tag{13}$$

uniformly in (y,x). Thus M_θ is bounded and ℓ_1-valued.

Let $M^0_{\theta,K} = \lambda_K \pi_K M_\theta$. The finite-coordinate map $\pi_K M_\theta : \mathcal{W} \to \mathbb{R}^{d_m K}$ is Borel because its coordinates are products of continuous and Borel functions; composing it with the continuous embedding λ_K shows that $M^0_{\theta,K}$ is Borel. Moreover,

$$\sup_{(y,x)\in\mathcal{W}} \|M_\theta(y,x) - M^0_{\theta,K}(y,x)\|_1 \le d_m C_1 \sup_{x\in\mathcal{X}}\sum_{k>K}|g_k(x)| \longrightarrow 0. \tag{14}$$

A pointwise limit of Borel maps into a metric space is Borel, so M_θ is Borel.

The same tail bound makes the range totally bounded. Given $\varepsilon > 0$, choose K so that the bound in (14) is less than $\varepsilon/2$. The bounded set $M^0_{\theta,K}(\mathcal{W})$ lies in a finite-dimensional subspace and has a finite $\varepsilon/2$-net. That net is an ε-net for $M_\theta(\mathcal{W})$. Since ℓ_1 is complete, $\mathrm{cl}_{\ell_1} M_\theta(\mathcal{W})$ is compact. Each $Q_\theta(w^o)$ is a closed subset of this common compact set and is nonempty because $D(w^o)$ is almost surely nonempty under Assumption 1. Thus its values are almost surely compact and nonempty.

It remains to verify random-set measurability on Ω. Let $O \subseteq \ell_1$ be open. Then $B = M_\theta^{-1}(O)$ is Borel, and

$$\begin{aligned} H_B &\equiv \{w^o : D(w^o) \cap B \neq \varnothing\} \\ &= \mathrm{proj}_{\mathcal{W}^O}\left\{(w,w^o) \in \mathrm{Gr}(D) : w \in B\right\}. \end{aligned}$$

The set inside the projection is Borel in a product of Polish spaces, so H_B is analytic and therefore P_{W^O}-universally measurable. Closure does not change whether a set meets an open

set, and hence

$$\{\omega : Q_\theta(W^O(\omega)) \cap O \neq \varnothing\} = \{\omega : W^O(\omega) \in H_B\}.$$

The right-hand side belongs to $\mathcal{A}$ because the probability space is complete. This proves Effros measurability of $\omega \mapsto Q_\theta(W^O(\omega))$. The envelope (13) gives integrable boundedness. Finally, the continuous image of a compact random set under π_K is a compact random set, which establishes the finite-dimensional claim. □

Lemma 3. *Let $\mathcal{S} \equiv \mathcal{Y} \times \mathcal{X}$ and let $\mathcal{W}_o$ denote the state space of W_o. Suppose $\mathcal{S} \times \mathcal{W}_o$ is Polish and $D : \mathcal{W}_o \rightrightarrows \mathcal{S}$ is nonempty compact-valued with closed graph. Let Γ be as defined in Section 2. Equivalently, for $\mathrm{Gr}(D) \equiv \{(s, w) \in \mathcal{S} \times \mathcal{W}_o : s \in D(w)\}$, we have*

$$\Gamma = \{\eta \in \Delta(\mathcal{S} \times \mathcal{W}_o) : \eta_{W_o} = P_{W_o},\ \eta(\mathrm{Gr}(D)) = 1\}.$$

Under the enriched probability-space convention described above, any measurable selection $S' = (Y', X')$ of $D(W_o)$ induces an element $\eta \equiv \mathcal{L}(S', W_o) \in \Gamma$. Conversely, for every $\eta \in \Gamma$, there exists a measurable selection $S^\eta = (Y^\eta, X^\eta)$ of $D(W_o)$ such that $\mathcal{L}(S^\eta, W_o) = \eta$. In other words, $\Gamma = \{\mathcal{L}(S, W_o) : S(\omega) \in D(W_o(\omega))\ a.s.\}$.

Proof. First let $S' = (Y', X')$ be a measurable selection of $D(W_o)$ and define $\eta \equiv \mathcal{L}(S', W_o)$. Then $\eta_{W_o} = P_{W_o}$ by construction. Moreover, $F_0(S' \in D(W_o)) = 1$, so $\eta(\mathrm{Gr}(D)) = 1$. Thus $\eta \in \Gamma$.

Conversely, fix $\eta \in \Gamma$. Define

$$\mathbb{D}(\omega) := D(W_o(\omega)) \times \{W_o(\omega)\} \subseteq \mathcal{S} \times \mathcal{W}_o.$$

By Assumption 1, $\mathbb{D}$ is a nonempty random compact set. We verify Artstein's inequalities. Let $F \subseteq \mathcal{S} \times \mathcal{W}_o$ be closed. Since $\eta(\mathrm{Gr}(D)) = 1$, $\eta(F) = \eta(F \cap \mathrm{Gr}(D))$. If $(s, w_o) \in F \cap \mathrm{Gr}(D)$,

then $s \in D(w_o)$ and $(s, w_o) \in F$. Hence $F \cap (D(w_o) \times \{w_o\}) \neq \varnothing$. Therefore,

$$\mathbb{I}\{(s, w_o) \in F \cap \mathrm{Gr}(D)\} \leq \mathbb{I}\{F \cap (D(w_o) \times \{w_o\}) \neq \varnothing\}.$$

Integrating with respect to η gives

$$\eta(F) \leq \int \mathbb{I}\{F \cap (D(w) \times \{w\}) \neq \varnothing\} \, \eta_{W_o}(dw).$$

Because $\eta_{W_o} = P_{W_o}$,

$$\eta(F) \leq F_0 \left(F \cap (D(W_o) \times \{W_o\}) \neq \varnothing\right) = F_0 \left(\mathbb{D} \cap F \neq \varnothing\right).$$

Thus η satisfies Artstein's inequalities for $\mathbb{D}$. By Artstein's selectionability theorem (Molchanov, 2017, Theorem 1.4.8, pp. 82–83), η is selectionable with respect to the law of $\mathbb{D}$. More specifically, the construction in the proof of that theorem yields a measurable map f_η such that, using the independent $V \sim \mathrm{Unif}[0, 1]$ supplied by the enriched probability space, the random element $Z^\eta \equiv f_\eta(V, \mathbb{D})$ satisfies $Z^\eta \in \mathbb{D}$ almost surely and $\mathcal{L}(Z^\eta) = \eta$. Since $\mathbb{D} = D(W_o) \times \{W_o\}$, the last component of Z^η equals W_o almost surely. Writing $Z^\eta = (Y^\eta, X^\eta, W_o)$ therefore gives $(Y^\eta, X^\eta) \in D(W_o)$ F_0-a.s. and $\mathcal{L}(Y^\eta, X^\eta, W_o) = \eta$. □

This law-selection bridge is the formal random-set foundation for the sharpness statement in the main text. It plays a role analogous to that of Assumption 2.4 in Beresteanu et al. (2011), but here the existence of the relevant selection is derived from the definition of Γ and the enriched probability-space convention rather than imposed as an additional model assumption.

Lemma 4. *Under Assumption 1, the set Γ is nonempty, convex, compact, and metrizable under weak convergence.*

Proof. Nonemptiness follows from $\eta_0 = \mathcal{L}_{F_0}(W, W^O) \in \Gamma$. The ambient space $\mathcal{W} \times \mathcal{W}^O$ is a compact metric space under Assumption 1, so its space of probability measures is compact

and metrizable under weak convergence (Bogachev, 2007, Section 8.6). It remains to show that Γ is closed.

Let $\eta_n \in \Gamma$ and $\eta_n \Rightarrow \eta$. Continuity of the coordinate projection gives $(\eta_n)_{W^O} \Rightarrow \eta_{W^O}$, so $\eta_{W^O} = P_{W^O}$. Since $\mathrm{Gr}(D)$ is closed, the Portmanteau theorem gives

$$1 = \limsup_{n\to\infty} \eta_n(\mathrm{Gr}(D)) \leq \eta(\mathrm{Gr}(D)) \leq 1.$$

Thus $\eta \in \Gamma$, proving closedness and hence compactness. Finally, the fixed-marginal and probability-one support restrictions defining Γ are affine. Therefore every mixture of two elements of Γ also belongs to Γ, so Γ is convex. □

The preceding lemmata are sufficient to establish that Q_θ is well defined and to connect the admissible completion laws in Γ to feasible latent selections. It remains to relate the attainable mean-moment set I_θ generated by Γ to the Aumann expectation of $Q_\theta(W^O)$. Two closure operations must be handled carefully. First, because M_θ need only be Borel measurable, $M_\theta(D(w^o))$ need not be closed, so $Q_\theta(w^o)$ is defined as its closure. Second, the Aumann expectation is the closure of the set of selection means. Lemma 5 shows that the combined effect of these operations is $\mathbb{E}[Q_\theta(W^O)] = \mathrm{cl}_{\ell_1} I_\theta$, while Lemma 6 shows that this closure does not affect identification because it does not alter membership at zero.
Let

$$\mathcal{A}_\theta \equiv \left\{\mathbf{E}[q] : q \text{ is an integrable selection of } Q_\theta(W^O)\right\}$$

denote the unclosed set of selection means. Recall that I_θ was defined in (4) and that $\mathbb{E}[Q_\theta(W^O)] = \mathrm{cl}_{\ell_1} \mathcal{A}_\theta$.

Lemma 5. *For every $\theta \in \Theta$, $I_\theta \subseteq \mathcal{A}_\theta \subseteq \mathrm{cl}_{\ell_1} I_\theta$, and consequently $\mathbb{E}[Q_\theta(W^O)] = \mathrm{cl}_{\ell_1} \mathcal{A}_\theta = \mathrm{cl}_{\ell_1} I_\theta$.*

Proof. For the first inclusion, let $z \in I_\theta$. Then there exists $\eta \in \Gamma$ such that

$$z = \int M_\theta(w)\eta(dw, dw^o).$$

By Lemma 3, there exists a measurable $S^\eta \in D(W^O)$ whose joint law with W^O is η. Hence $q^\eta \equiv M_\theta(S^\eta)$ is an integrable selection of $Q_\theta(W^O)$ and

$$\mathbf{E}[q^\eta] = \int M_\theta(w)\eta(dw, dw^o) = z.$$

Thus $I_\theta \subseteq \mathcal{A}_\theta$.

For $\mathcal{A}_\theta \subseteq \mathrm{cl}_{\ell_1} I_\theta$, let q be an arbitrary integrable selection of $Q_\theta(W^O)$. By definition of Q_θ,

$$q(\omega) \in \mathrm{cl}_{\ell_1} M_\theta(D(W^O(\omega))) \qquad F_0\text{-a.s.}$$

The closure is relevant because M_θ is only Borel. Hence $q(\omega)$ need not admit an exact preimage in $D(W^O(\omega))$, although it admits admissible preimages whose moment vectors are arbitrarily close. Fix $r \geq 1$ and define the relation

$$\mathcal{R}_r \equiv \left\{((w^o, z), w) \in (\mathcal{W}^O \times \ell_1) \times \mathcal{W} : w \in D(w^o),\ ||M_\theta(w) - z||_1 < r^{-1}\right\}.$$

For $e = (w^o, z)$, let $\mathcal{R}_r^\circ(e) \equiv \{w : (e, w) \in \mathcal{R}_r\}$. Thus $\mathcal{R}_r^\circ(w^o, z)$ is the set of admissible latent values $w \in D(w^o)$ whose moment vectors approximate the target z within r^{-1}.

The relation $\mathcal{R}_r$ is Borel, because $\{(w^o, w) : w \in D(w^o)\} = \mathrm{Gr}(D)$ is closed, while $(w, z) \mapsto ||M_\theta(w) - z||_1$ is Borel because M_θ is Borel by Lemma 2 and the norm is continuous.

Next define

$$\mathsf{E} \equiv \mathcal{W}^O \times \ell_1, \quad C_r \equiv \mathrm{proj}_{\mathsf{E}} \mathcal{R}_r, \quad Z_q(\omega) \equiv (W^O(\omega), q(\omega)), \quad \mu \equiv \mathcal{L}_{F_0}(Z_q).$$

Since q is an integrable ℓ_1-valued selection, it is measurable, and hence Z_q is an E-valued

measurable random element. Moreover, E and $\mathcal{W}$ are standard Borel spaces.

Since $\mathcal{R}_r$ is Borel in $\mathsf{E} \times \mathcal{W}$, its projection C_r is analytic. Hence C_r is universally measurable and therefore belongs to the μ-completion $\overline{\mathcal{B}(\mathsf{E})}^{\mu}$.

By $q(\omega) \in \mathrm{cl}_{\ell_1} M_\theta(D(W^O(\omega)))$ F_0-a.s., for every r there exists an admissible $w \in D(W^O(\omega))$ such that $|M_\theta(w) - q(\omega)|_1 < r^{-1}$. Hence $\mathcal{R}_r^\circ(W^O(\omega), q(\omega)) \neq \varnothing$ F_0-a.s. Equivalently, $Z_q(\omega) \in C_r$ F_0-a.s. Thus the random pair $Z_q = (W^O, q)$ lies almost surely in the part of the relation on which an approximate inverse exists, and therefore $\mu(C_r) = 1$.

By the Jankov–von Neumann uniformization theorem (Kechris, 1995, Theorem 18.1), there exists a universally measurable map $\widetilde{\sigma}_r : C_r \to \mathcal{W}$ such that

$$\widetilde{\sigma}_r(e) \in \mathcal{R}_r^\circ(e) \qquad \text{for every } e \in C_r.$$

Fix $w_0 \in \mathcal{W}$ and extend $\widetilde{\sigma}_r$ to all of E by setting $\widetilde{\sigma}_r(e) = w_0$ for $e \notin C_r$. Since C_r and the inverse images of Borel sets under the selector are universally measurable, this extension remains universally measurable. In particular, it is measurable with respect to the μ-completion of $\mathcal{B}(\mathsf{E})$.

Define

$$S_{q,r}(\omega) \equiv \widetilde{\sigma}_r(Z_q(\omega)) = \widetilde{\sigma}_r(W^O(\omega), q(\omega)).$$

Then $S_{q,r}$ is $\mathcal{A}$-measurable. Indeed, for every $B \in \mathcal{B}(\mathcal{W})$, $\widetilde{\sigma}_r^{-1}(B)$ belongs to the μ-completion of $\mathcal{B}(\mathsf{E})$. Hence it differs from a Borel subset of E only by a μ-null set. Pulling back through Z_q therefore gives a set that differs from an $\mathcal{A}$-measurable set only by an F_0-null set. Completeness of $(\Omega, \mathcal{A}, F_0)$ yields measurability.

Since $Z_q \in C_r$ almost surely, the defining property of the selector gives

$$S_{q,r} \in D(W^O), \ ||M_\theta(S_{q,r}) - q||_1 < r^{-1} \qquad F_0\text{-a.s.}$$

Thus $S_{q,r}$ is an admissible latent completion whose moment vector approximates the selected vector q pointwise.

Let $\eta_r \equiv \mathcal{L}_{F_0}(S_{q,r}, W^O)$. By Lemma 3, $\eta_r \in \Gamma$. Moreover, the uniform ℓ_1 bound on M_θ implies that $M_\theta(S_{q,r})$ is Bochner integrable. Therefore

$$z_r \equiv \int M_\theta(w)\, \eta_r(dw, dw^o) = \mathbf{E}[M_\theta(S_{q,r})] \in I_\theta.$$

Using the above and the Bochner triangle inequality,

$$|z_r - \mathbf{E}[q]|_1 = |\mathbf{E}[M_\theta(S_{q,r}) - q]|_1 \ \leq \mathbf{E}\left[|M_\theta(S_{q,r}) - q|_1\right] \ < r^{-1}.$$

Hence $z_r \in I_\theta$ and $z_r \to \mathbf{E}[q]$ in ℓ_1. Therefore $\mathbf{E}[q] \in \mathrm{cl}_{\ell_1} I_\theta$. Since q was an arbitrary integrable selection of $Q_\theta(W^O)$, $\mathcal{A}_\theta \subseteq \mathrm{cl}_{\ell_1} I_\theta$. Together with the first inclusion, this proves the first part of the lemma.

Finally, taking norm closures yields $\mathrm{cl}_{\ell_1} \mathcal{A}_\theta = \mathrm{cl}_{\ell_1} I_\theta$. By definition of the Aumann expectation, $\mathbb{E}[Q_\theta(W^O)] = \mathrm{cl}_{\ell_1} \mathcal{A}_\theta$, which establishes the second claim of the lemma. □

Lemma 6. *Under Assumptions 1–3,*

$$0 \in \mathrm{cl}_{\ell_1} I_\theta \qquad \Longleftrightarrow \qquad 0 \in I_\theta. \tag{15}$$

Proof. Only the forward implication requires proof. Suppose $0 \in \mathrm{cl}_{\ell_1} I_\theta$. There are $\eta_n \in \Gamma$ such that

$$z_n \equiv \int M_\theta(w)\, \eta_n(dw, dw^o), \qquad \|z_n\|_1 \to 0.$$

By Lemma 4, pass to a subsequence with $\eta_n \Rightarrow \eta \in \Gamma$.

Fix $j \in \{1, \ldots, d_m\}$, $f \in C(\mathcal{X})$, and $\varepsilon > 0$. Assumption 2(i) gives

$$v_\varepsilon(x) = \sum_{k \in K_\varepsilon} a_k g_k(x), \qquad K_\varepsilon \subset \mathbb{N} \text{ finite}, \qquad \|f - v_\varepsilon\|_\infty < \varepsilon.$$

Let $B_\varepsilon = \sum_{k \in K_\varepsilon} |a_k| < \infty$. Because the (j,k) coordinate of z_n is $\int m_j g_k \, d\eta_n$,

$$\begin{aligned} \left| \int m_j(y,x,\theta) v_\varepsilon(x) \, d\eta_n \right| &= \left| \sum_{k \in K_\varepsilon} a_k \int m_j(y,x,\theta) g_k(x) \, d\eta_n \right| \\ &\leq B_\varepsilon \|z_n\|_1 \longrightarrow 0. \end{aligned} \tag{16}$$

Assumption 3 then gives

$$\left| \int m_j(y,x,\theta) f(x) \, d\eta_n \right| \leq B_\varepsilon \|z_n\|_1 + C_1 \varepsilon. \tag{17}$$

The function $(y,x,w^o) \mapsto m_j(y,x,\theta) f(x)$ is bounded and continuous. Weak convergence therefore permits passage to the limit in its integral. From (17),

$$\left| \int m_j(y,x,\theta) f(x) \, d\eta \right| \leq C_1 \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives

$$\int m_j(y,x,\theta) f(x) \, d\eta = 0 \qquad \forall f \in C(\mathcal{X}), \ \forall j. \tag{18}$$

Let $\mu_j(X) = \mathbf{E}_\eta[m_j(Y,X,\theta) \mid X]$. By iterated expectations, the finite signed measure with density μ_j relative to η_X integrates every $f \in C(\mathcal{X})$ to zero. The Riesz representation theorem therefore implies $\mu_j = 0$ η_X-almost surely for every j. The fixed-pair equivalence (FP), together with the coordinate identity for Bochner integrals, gives

$$\int M_\theta(w) \, \eta(dw, dw^o) = 0.$$

Because $\eta \in \Gamma$, this vector belongs to I_θ. Thus $0 \in I_\theta$.

Notice that the proof passes to the weak limit only through continuous test functions. It never requires the limiting law to assign zero probability to the discontinuity boundaries of the cube instruments. □

Remark 1. A different and simpler proof strategy could be employed under a continuous moment class, effectively making Lemma 5 and Lemma 6 superfluous. Suppose $M_\theta : \mathcal{W} \to \ell_1$ is continuous. Then $Q_\theta(W^O) = M_\theta(D(W^O))$ F_0-a.s. and $\mathcal{A}_\theta = I_\theta = \mathbb{E}[Q_\theta(W^O)]$. Moreover, I_θ is compact in ℓ_1.

Proof. Because $D(W^O)$ is compact and M_θ is continuous, $M_\theta(D(W^O))$ is compact and already closed. Thus the closure defining Q_θ is redundant.

Let q be an integrable selection of $Q_\theta(W^O)$ and define

$$H_q(\omega) \equiv \left\{ w \in D(W^O(\omega)) : M_\theta(w) = q(\omega) \right\}.$$

The correspondence is nonempty and random closed by the random-inverse theorem (Molchanov, 2017, Theorem 1.3.27). The Fundamental Selection Theorem (Molchanov, 2017, Theorem 1.4.1) supplies a measurable $S_q \in H_q$. Lemma 3 implies $\eta_q = \mathcal{L}_{F_0}(S_q, W^O) \in \Gamma$, and hence

$$\mathbf{E}[q] = \int M_\theta(w)\, \eta_q(dw, dw^o) \in I_\theta.$$

Therefore $\mathcal{A}_\theta \subseteq I_\theta$; the first inclusion in Lemma 5 gives the reverse inclusion.

To show closedness, define

$$T_\theta : \Gamma \to \ell_1, \qquad T_\theta(\eta) = \int M_\theta(w)\, \eta(dw, dw^o).$$

If $\eta_n \Rightarrow \eta$, the first $d_m K$ coordinates converge for every fixed K because they are bounded and continuous. The tail bound (14) holds uniformly over Γ. Finite-coordinate convergence

followed by the uniform tail bound gives $\|T_\theta(\eta_n) - T_\theta(\eta)\|_1 \to 0$. Thus T_θ is continuous. Lemma 4 makes Γ compact, so $I_\theta = T_\theta(\Gamma)$ is compact and closed. Since $\mathbb{E}[Q_\theta(W^O)] = \mathrm{cl}_{\ell_1} \mathcal{A}_\theta$, all claimed equalities follow. □

B.3 Main Proof of Theorem 1

Proof. Lemma 2 establishes the asserted random-set properties. The Fundamental Selection Theorem (Molchanov, 2017, Theorem 1.4.1) supplies a measurable selection of $Q_\theta(W^O)$; the common envelope in (13) makes that selection integrable. Hence $\mathbb{E}[Q_\theta(W^O)]$ is nonempty. By the coordinate identity for Bochner integrals and the fixed-pair equivalence (FP),

$$\theta \in \Theta_I^* \iff \theta \in \Theta_I \iff 0 \in I_\theta.$$

Lemmas 5 and 6 give

$$0 \in \mathbb{E}[Q_\theta(W^O)] \iff 0 \in \mathrm{cl}_{\ell_1} I_\theta \iff 0 \in I_\theta.$$

This proves (5) and the first line of (6).

The set $\mathbb{E}[Q_\theta(W^O)]$ is closed by Molchanov (2017, Definition 2.1.21). It is convex because the extended probability space is nonatomic and Molchanov (2017, Theorem 2.1.30) applies to the integrably bounded random closed set $Q_\theta(W^O)$ in the separable Banach space ℓ_1. Equivalently, after Lemma 5, convexity also follows from convexity of Γ and affinity of $\eta \mapsto \int M_\theta \, d\eta$.

By Molchanov (2017, Theorem 2.1.35),

$$h(\mathbb{E}[Q_\theta(W^O)], u) = \mathbb{E}_{P_{W^O}}[h(Q_\theta(W^O), u)] \qquad \forall u \in \ell_\infty.$$

For a nonempty closed convex subset C of ℓ_1, the Hahn–Banach separation theorem gives

$$0 \in C \quad \Longleftrightarrow \quad h(C,u) \geq 0 \qquad \forall u \in (\ell_1)^* = \ell_\infty.$$

Positive homogeneity of $u \mapsto h(C,u)$ permits restriction to B^∞. Combining the last two displays proves the final equality in (6). □

C Proof of Theorem 2

This section first proves a triangular law of large numbers for random sets. We then proceed by stating and proving several auxiliary lemmata, which are needed to verify the high-level conditions in Chernozhukov et al. (2007) for our case and thus establish consistency of the proposed estimator.

C.1 Triangular Law of Large Numbers for Random Sets

Before stating the consistency result for the proposed estimator, we introduce an auxiliary lemma establishing a law of large numbers for random sets growing at rate p_n, which might be of independent interest and is a crucial part of the proof.

Lemma 7. *Let p_n denote the ambient dimension of the n-th row of a triangular array, with $p_n \to \infty$. For each n, let $X_{n,1}, \ldots, X_{n,n}$ be i.i.d. nonempty compact random subsets of $\mathbb{R}^{p_n}$, and let X_n denote a generic copy from the n-th row and suppose they are uniformly bounded by a deterministic constant b. Then*

$$\rho_H\left(\operatorname{co}\left[\frac{1}{n}\bigoplus_{i=1}^{n} X_{n,i}\right], \mathbb{E}[\operatorname{co} X_n]\right) = O_p\left(\sqrt{\frac{p_n}{n}}\right).$$

If, in addition, the Aumann expectation satisfies $\mathbb{E}[X_n] = \mathbb{E}[\operatorname{co} X_n]$*, then*

$$\rho_H\left(\frac{1}{n}\bigoplus_{i=1}^{n} X_{n,i}, \mathbb{E}[X_n]\right) = O_p\left(\sqrt{\frac{p_n}{n}} + \frac{p_n}{n}\right).$$

Consequently, if $p_n/n \to 0$*, then*

$$\rho_H\left(\frac{1}{n}\bigoplus_{i=1}^{n} X_{n,i}, \mathbb{E}[X_n]\right) = O_p\left(\sqrt{\frac{p_n}{n}}\right).$$

Proof. Let $\hat{X}_n \equiv \frac{1}{n}\bigoplus_{i=1}^{n} X_{n,i}$. First, let us control the convexified average. Note that $\operatorname{co}(\hat{X}_n) = \frac{1}{n}\bigoplus_{i=1}^{n} \operatorname{co}(X_{n,i})$. Using $h(\operatorname{co}(\hat{X}_n), u) = h(\hat{X}_n, u) = \frac{1}{n}\sum_{i=1}^{n} h(X_{n,i}, u)$ and $h(\mathbb{E}[\operatorname{co}(X_n)], u) = \mathbf{E}[h(\operatorname{co}(X_n), u)] = \mathbf{E}[h(X_n, u)]$, we can rewrite the Hausdorff distance between the convexified average and the Aumann expectation as the empirical process supremum:

$$\rho_H\left(\operatorname{co}(\hat{X}_n), \mathbb{E}[\operatorname{co}(X_n)]\right) = \sup_{u \in \mathbb{B}_n^*} \left|\frac{1}{n}\sum_{i=1}^{n} h(X_{n,i}, u) - \mathbf{E}[h(X_n, u)]\right|$$

where $\mathbb{B}_n^*$ denotes the finite-dimensional dual unit ball corresponding to the ambient space $\mathbb{R}^{p_n}$.

Define the support function class as $\mathcal{H}_n \equiv \{f_u : \ f_u(X) = h(X, u), u \in \mathbb{B}_n^*\}$, thus the above is akin to $\sup_{h \in \mathcal{H}_n} |(P_n - P_{W^O})h|$, which is controlled by the entropy of the function class $\mathcal{H}_n$. From Lemma 4.10 in Pisier (1989) we have $N(\mathbb{B}_n^*, \delta, ||\cdot||_\infty) \leq (1 + \frac{2}{\delta})^{p_n}$, which we combine with Lipschitz continuity of the support function, i.e. $|f_u(X) - f_v(X)| = |h(X, u) - h(X, v)| \leq \sup_{x \in X} |\langle x, u - v\rangle| \leq \sup_{x \in X} \|x\|_1 \|u - v\|_\infty \leq b\|u - v\|_\infty$, where b is the uniform deterministic bound from the statement of the lemma. Hence $||f_u - f_v||_{L_2(P_{W^O})} \leq b||u - v||_\infty$ and we have $N(\epsilon, \mathcal{H}_n, L_2(P_{W^O})) \leq N(\frac{\epsilon}{b}, \mathbb{B}_n^*, ||\cdot||_\infty)$, thus

$$\log N(\epsilon, \mathcal{H}_n, L_2(P_{W^O})) \leq p_n \log\left(1 + \frac{2b}{\epsilon}\right) \leq p_n \log\left(\frac{Cb}{\epsilon}\right)$$

By Dudley's entropy theorem (Dudley, 1967), we have

$$\begin{aligned}\mathbb{E}\left[\sup_{u\in\mathbb{B}_n^*}\left|\frac{1}{n}\sum_{i=1}^{n}h(X_{n,i},u)-\mathbf{E}[h(X_n,u)]\right|\right] &\le \frac{C}{\sqrt{n}}\int_0^b\sqrt{p_n\log\left(\frac{Cb}{\epsilon}\right)}d\epsilon\\ &= C\sqrt{\frac{p_n}{n}}b\int_0^1\sqrt{\log(C/t)}dt\\ &\le Cb\sqrt{\frac{p_n}{n}}\end{aligned}$$

where the equality uses a change of variable and the last inequality uses $\int_0^1\sqrt{\log(C/t)}dt<\infty$. Using Markov's inequality it follows that

$$\sup_{u\in\mathbb{B}_n^*}\left|\frac{1}{n}\sum_{i=1}^{n}h(X_{n,i},u)-\mathbf{E}[h(X_n,u)]\right| = O_p\left(\sqrt{\frac{p_n}{n}}\right).$$

Hence,

$$\rho_H\left(\mathrm{co}(\hat{X}_n),\mathbb{E}[\mathrm{co}(X_n)]\right) = O_p\left(\sqrt{\frac{p_n}{n}}\right).$$

It remains to pass from the convexified average back to the original, potentially non-convex, Minkowski average. For each n, the sets $X_{n,i}$ are compact subsets of the finite-dimensional space $\mathbb{R}^{p_n}$. Therefore, applying the finite-dimensional Shapley–Folkman–Starr theorem row by row gives

$$\rho_H\left(\hat{X}_n,\mathrm{co}(\hat{X}_n)\right)\le\frac{p_n}{n}\max_{1\le i\le n}\mathrm{diam}_1(X_{n,i})\le 2b\frac{p_n}{n}.$$

If $\mathbb{E}[X_n]=\mathbb{E}[\mathrm{co}(X_n)]$, then by the triangle inequality,

$$\begin{aligned}\rho_H\left(\hat{X}_n,\mathbb{E}[X_n]\right) &\le \rho_H\left(\hat{X}_n,\mathrm{co}(\hat{X}_n)\right)+\rho_H\left(\mathrm{co}(\hat{X}_n),\mathbb{E}[\mathrm{co}(X_n)]\right)\\ &= O\left(\frac{p_n}{n}\right)+O_p\left(\sqrt{\frac{p_n}{n}}\right).\end{aligned}$$

Thus,

$$\rho_H\left(\frac{1}{n}\bigoplus_{i=1}^{n} X_{n,i}, \mathbb{E}[X_n]\right) = O_p\left(\sqrt{\frac{p_n}{n}} + \frac{p_n}{n}\right).$$

If $p_n/n \to 0$, then $p_n/n = o(\sqrt{p_n/n})$, and therefore

$$\rho_H\left(\frac{1}{n}\bigoplus_{i=1}^{n} X_{n,i}, \mathbb{E}[X_n]\right) = O_p\left(\sqrt{\frac{p_n}{n}}\right).$$

□

C.2 Auxiliary Lemmata

The first auxiliary lemma establishes a uniform version of the triangular law of large numbers for the random sets established above.

Lemma 8. *Let Assumptions 1–5 hold. Let $K_n \to \infty$, and let $Q_{\theta,K_n} \subset \mathbb{R}^{d_m K_n}$ denote the truncation of Q_θ to the first K_n instruments $g_1, \ldots, g_{K_n}$. For $i = 1, \ldots, n$, write $Q_{\theta,K_n,i} \equiv Q_{\theta,K_n}(W_i^O)$. Suppose that $\frac{d_m K_n}{n} \to 0$. Then*

$$\sup_{\theta\in\Theta} \rho_H\left(\frac{1}{n}\bigoplus_{i=1}^{n} Q_{\theta,K_n,i}, \mathbb{E}[Q_{\theta,K_n}]\right) = O_p\left(\sqrt{\frac{d_m K_n + d_\theta}{n}}\right).$$

Since d_m and d_θ are fixed, this is equivalently

$$\sup_{\theta\in\Theta} \rho_H\left(\frac{1}{n}\bigoplus_{i=1}^{n} Q_{\theta,K_n,i}, \mathbb{E}[Q_{\theta,K_n}]\right) = O_p\left(\sqrt{\frac{K_n}{n}}\right).$$

Proof. Let

$$\hat{Q}_{n,\theta,K_n} \equiv \frac{1}{n}\bigoplus_{i=1}^{n} Q_{\theta,K_n,i}.$$

First, control the convexified average. Since $Q_{\theta,K_n} \subset \mathbb{R}^{d_m K_n}$, the support-function represen-

tation of the Hausdorff distance for compact convex sets gives

$$\rho_H\left(\mathrm{co}\left(\hat{Q}_{n,\theta,K_n}\right), \mathbb{E}[\mathrm{co}(Q_{\theta,K_n})]\right) = \sup_{u\in B_{K_n}} \left| \frac{1}{n}\sum_{i=1}^{n} h(Q_{\theta,K_n,i}, u) - \mathbb{E}[h(Q_{\theta,K_n}, u)] \right|,$$

where

$$B_{K_n} \equiv \{u \in \mathbb{R}^{d_m K_n} : \|u\|_\infty \le 1\}.$$

Here we used $h(\mathrm{co}(A), u) = h(A, u)$, $h(n^{-1} \oplus_i A_i, u) = n^{-1}\sum_i h(A_i, u)$, and

$$h(\mathbb{E}[\mathrm{co}(Q_{\theta,K_n})], u) = \mathbb{E}[h(Q_{\theta,K_n}, u)].$$

Define the support-function class

$$\mathcal{H}_n^\Theta \equiv \left\{ f_{\theta,u}(W_i^O) = h(Q_{\theta,K_n}(W_i^O), u) : \theta \in \Theta,\ u \in B_{K_n} \right\}.$$

By Assumptions 2 and 3, there exists $b < \infty$, independent of n, such that

$$\sup_{\theta\in\Theta} \sup_{q\in Q_{\theta,K_n}} \|q\|_1 \le b.$$

Indeed, one may take

$$b = d_m C_1 \sup_{x\in\mathcal{X}} \sum_{k=1}^{\infty} |g_k(x)| < \infty.$$

Hence the class has envelope b. Moreover, for $u, v \in B_{K_n}$,

$$|h(Q_{\theta,K_n}, u) - h(Q_{\theta,K_n}, v)| \le b\|u - v\|_\infty.$$

Assumption 5 and the summability condition in Assumption 2 imply that, for some constant $\bar{L} < \infty$ independent of n,

$$|h(Q_{\theta,K_n}, u) - h(Q_{\theta',K_n}, u)| \le \bar{L}\|\theta - \theta'\|.$$

Therefore, covering both B_{K_n} and Θ gives

$$\log N(\epsilon, \mathcal{H}_n^{\Theta}, L_2(P_{W^O})) \leq d_m K_n \log\left(\frac{Cb}{\epsilon}\right) + d_\theta \log\left(\frac{C\bar{L}}{\epsilon}\right).$$

By Dudley's entropy theorem,

$$\begin{aligned}\mathbb{E}\left[\sup_{\theta\in\Theta}\sup_{u\in B_{K_n}}\left|\frac{1}{n}\sum_{i=1}^{n}\{h(Q_{\theta,K_n,i},u) - \mathbb{E}[h(Q_{\theta,K_n},u)]\}\right|\right] &\leq \frac{C}{\sqrt{n}}\int_0^b \sqrt{d_m K_n \log\left(\frac{Cb}{\epsilon}\right) + d_\theta \log\left(\frac{C\bar{L}}{\epsilon}\right)}d\epsilon \\ &\leq C\sqrt{\frac{d_m K_n + d_\theta}{n}}.\end{aligned}$$

Hence, by Markov's inequality,

$$\sup_{\theta\in\Theta}\rho_H\left(\mathrm{co}\left(\hat{Q}_{n,\theta,K_n}\right), \mathbb{E}[\mathrm{co}(Q_{\theta,K_n})]\right) = O_p\left(\sqrt{\frac{d_m K_n + d_\theta}{n}}\right).$$

It remains to remove convexification. For each n and each θ, $Q_{\theta,K_n,i}$ is a compact subset of the finite-dimensional space $\mathbb{R}^{d_m K_n}$. Thus the finite-dimensional Shapley–Folkman–Starr theorem gives, uniformly in θ,

$$\rho_H\left(\hat{Q}_{n,\theta,K_n}, \mathrm{co}\left(\hat{Q}_{n,\theta,K_n}\right)\right) \leq \frac{d_m K_n}{n}\max_{1\leq i\leq n}\mathrm{diam}_1(Q_{\theta,K_n,i}) \leq 2b\frac{d_m K_n}{n}.$$

By the convexification identity for the Aumann expectation,

$$\mathbb{E}[Q_{\theta,K_n}] = \mathbb{E}[\mathrm{co}(Q_{\theta,K_n})].$$

Therefore, by the triangle inequality,

$$\begin{aligned}\sup_{\theta\in\Theta}\rho_H\left(\hat{Q}_{n,\theta,K_n},\mathbb{E}[Q_{\theta,K_n}]\right) &\le \sup_{\theta\in\Theta}\rho_H\left(\hat{Q}_{n,\theta,K_n},\mathrm{co}\left(\hat{Q}_{n,\theta,K_n}\right)\right)\\ &\quad+\sup_{\theta\in\Theta}\rho_H\left(\mathrm{co}\left(\hat{Q}_{n,\theta,K_n}\right),\mathbb{E}[\mathrm{co}(Q_{\theta,K_n})]\right)\\ &= O\left(\frac{d_mK_n}{n}\right)+O_p\left(\sqrt{\frac{d_mK_n+d_\theta}{n}}\right).\end{aligned}$$

Hence

$$\sup_{\theta\in\Theta}\rho_H\left(\frac{1}{n}\bigoplus_{i=1}^{n}Q_{\theta,K_n,i},\mathbb{E}[Q_{\theta,K_n}]\right)=O_p\left(\sqrt{\frac{d_mK_n+d_\theta}{n}}+\frac{d_mK_n}{n}\right).$$

If $d_mK_n/n\to 0$, the second term is absorbed into the first. Since d_m and d_θ are fixed, this yields

$$\sup_{\theta\in\Theta}\rho_H\left(\frac{1}{n}\bigoplus_{i=1}^{n}Q_{\theta,K_n,i},\mathbb{E}[Q_{\theta,K_n}]\right)=O_p\left(\sqrt{\frac{K_n}{n}}\right).$$

□

The following lemma links the deviation of directed Hausdorff distances to the undirected Hausdorff distances between two sets.

Lemma 9. *Let A and B be nonempty closed bounded subsets of a common normed space. Then*

$$\left|\vec{\rho}_H(0,A)-\vec{\rho}_H(0,B)\right|\le\rho_H(A,B).$$

Proof. By the triangle inequality for hemi-metrics, for all $q\in B$, $\vec{\rho}_H(0,A)\le\vec{\rho}_H(0,q)+$

$\vec{\rho}_H(q, A)$ (see Goubault-Larrecq, 2013, Lemma 6.2.16). Therefore,

$$\begin{aligned}\vec{\rho}_H(0, A) &\leq \min_{q \in B} \Big(\vec{\rho}_H(0, q) + \vec{\rho}_H(q, A) \Big) \leq \min_{q \in B} \Big(\rho_H(0, q) + \rho_H(q, A) \Big) \\ &\leq \vec{\rho}_H(0, B) + \rho_H(A, B).\end{aligned}$$

Hence $\vec{\rho}_H(0, A) - \vec{\rho}_H(0, B) \leq \rho_H(A, B)$. Reversing the inequality, for all $q \in A$ we similarly obtain $\vec{\rho}_H(0, B) \leq \vec{\rho}_H(0, q) + \vec{\rho}_H(q, B)$ so that

$$\begin{aligned}\vec{\rho}_H(0, B) &\leq \min_{q \in A} \Big(\vec{\rho}_H(0, q) + \vec{\rho}_H(q, B) \Big) \leq \min_{q \in A} \Big(\rho_H(0, q) + \rho_H(q, B) \Big) \\ &\leq \vec{\rho}_H(0, A) + \rho_H(A, B).\end{aligned}$$

Hence $\vec{\rho}_H(0, B) - \vec{\rho}_H(0, A) \leq \rho_H(A, B)$. Combining both inequalities gives the desired result.

□

Next we bound the deterministic approximation error from retaining only the first K instruments $(g_1, \ldots, g_K)$ instead of the full infinite instrument class.

Lemma 10. *Let Assumptions 1–3 hold. Let $Q_{\theta,K} \subset \mathbb{R}^{d_m K}$ denote the truncation of Q_θ to the first K instruments $g_1, \ldots, g_K$, and let $\widetilde{Q}_{\theta,K} \subset \ell_1$ denote its zero-padded embedding, $\widetilde{Q}_{\theta,K}(w^o) = \lambda_K Q_{\theta,K}(w^o) = \lambda_K \pi_K Q_\theta(w^o)$. Then*

$$\sup_{\theta \in \Theta} \Big| \vec{\rho}_H\big(0, \mathbb{E}[Q_\theta]\big) - \vec{\rho}_H\big(0, \mathbb{E}[\widetilde{Q}_{\theta,K}]\big) \Big| \to 0 \qquad \text{as } K \to \infty.$$

Moreover, since $\vec{\rho}_H\big(0, \mathbb{E}[\widetilde{Q}_{\theta,K}]\big) = \vec{\rho}_H\big(0, \mathbb{E}[Q_{\theta,K}]\big)$, the finite-dimensional criterion satisfies

$$\sup_{\theta \in \Theta} |T(\theta) - T_K(\theta)| \leq d_m C_1 \sup_{x \in \mathcal{X}} \sum_{k > K} |g_k(x)| \to 0.$$

Proof. Fix $\theta \in \Theta$. By Lemma 9, $\Big| \vec{\rho}_H\big(0, \mathbb{E}[Q_\theta]\big) - \vec{\rho}_H\big(0, \mathbb{E}[\widetilde{Q}_{\theta,K}]\big) \Big| \leq \rho_H\big(\mathbb{E}[Q_\theta], \mathbb{E}[\widetilde{Q}_{\theta,K}]\big)$. The right-hand side can be bounded using the support-function representation of the Haus-

dorff distance for closed convex bounded sets and the support-function representation of the Aumann expectation:

$$\begin{aligned}\rho_H\big(\mathbb{E}[Q_\theta],\mathbb{E}[\widetilde{Q}_{\theta,K}]\big) &= \sup_{u\in B^\infty}\Big|h(\mathbb{E}[Q_\theta],u)-h(\mathbb{E}[\widetilde{Q}_{\theta,K}],u)\Big| \\ &= \sup_{u\in B^\infty}\Big|\mathbb{E}_{P_{W^O}}[h(Q_\theta,u)]-\mathbb{E}_{P_{W^O}}[h(\widetilde{Q}_{\theta,K},u)]\Big| \\ &\le \mathbf{E}\left[\sup_{u\in B^\infty}\Big|h(Q_\theta,u)-h(\widetilde{Q}_{\theta,K},u)\Big|\right].\end{aligned}$$

The last inequality follows from $\sup_u|\mathbf{E}[Z_u]| \le \mathbf{E}[\sup_u|Z_u|]$. By Lipschitz continuity of support functions and $\|u\|_\infty \le 1$,

$$\sup_{u\in B^\infty}\Big|h(Q_\theta,u)-h(\widetilde{Q}_{\theta,K},u)\Big| \le \rho_H\big(Q_\theta,\widetilde{Q}_{\theta,K}\big).$$

Since $Q_\theta(W^O)$ and $\widetilde{Q}_{\theta,K}(W^O)$ are generated from the same feasible set $D(W^O)$, we have pathwise

$$\begin{aligned}\rho_H\big(Q_\theta(W^O),\widetilde{Q}_{\theta,K}(W^O)\big) &\le \sup_{(y,x)\in D(W^O)}\|M_\theta(y,x)-\lambda_K\pi_K M_\theta(y,x)\|_1 \\ &= \sup_{(y,x)\in D(W^O)}\sum_{j=1}^{d_m}\sum_{k>K}|m_j(y,x,\theta)||g_k(x)| \\ &\le d_m C_1 \sup_{x\in\mathcal{X}}\sum_{k>K}|g_k(x)|,\end{aligned}$$

where the last inequality follows from Assumption 3.

Combining the preceding displays yields, uniformly in θ,

$$\Big|\vec{\rho}_H\big(0,\mathbb{E}[Q_\theta]\big)-\vec{\rho}_H\big(0,\mathbb{E}[\widetilde{Q}_{\theta,K}]\big)\Big| \le d_m C_1 \sup_{x\in\mathcal{X}}\sum_{k>K}|g_k(x)|.$$

Taking the supremum over $\theta\in\Theta$ and using the weighted-tail condition in Assumption 2

gives

$$\sup_{\theta\in\Theta}\left|\vec{\rho}_H\big(0,\mathbb{E}[Q_\theta]\big)-\vec{\rho}_H\big(0,\mathbb{E}[\widetilde{Q}_{\theta,K}]\big)\right|\to 0.$$

Finally, under the zero-padding identification, $\vec{\rho}_H\big(0,\mathbb{E}[\widetilde{Q}_{\theta,K}]\big)=\vec{\rho}_H\big(0,\mathbb{E}[Q_{\theta,K}]\big)$ so the displayed bound also gives

$$\sup_{\theta\in\Theta}|T(\theta)-T_K(\theta)|\le d_m C_1\sup_{x\in\mathcal{X}}\sum_{k>K}|g_k(x)|\to 0$$

□

C.3 Main Proof of Theorem 2

We want to show $\rho_H(\widehat{\Theta}_n,\Theta_I^*)\xrightarrow{p}0$ by applying Theorem 3.1 in Chernozhukov et al. (2007). Their result rests on condition C.1 of the same paper, which we later verify for our setting. Define

$$a_n=\sqrt{\frac{n}{d_m K(n)+d_\theta}},\qquad \delta_K=\sup_{\theta\in\Theta}|T(\theta)-T_K(\theta)|,$$

and, for the verification of Chernozhukov et al. (2007) condition C.1(d), define

$$b_n=\left(\delta_{K(n)}+\sqrt{\frac{d_m K(n)+d_\theta}{n}}\right)^{-1}.$$

By Lemma 10 and Assumption 2, $\delta_{K(n)} \to 0$. Since $d_m K(n)/n \to 0$, it follows that $b_n \to \infty$. Recall,

$$
\begin{aligned}
T(\theta) &\equiv \vec{\rho}_H\left(0, \mathbb{E}[Q_\theta]\right), \\
T_{K(n)}(\theta) &\equiv \vec{\rho}_H\left(0, \mathbb{E}[Q_{\theta,K(n)}]\right), \\
\widehat{T}_n(\theta) &\equiv -\inf_{u \in B_{K(n)}} \frac{1}{n}\sum_{i=1}^{n} h(Q_{\theta,K(n),i}, u).
\end{aligned}
$$

Equivalently,

$$
\widehat{T}_n(\theta) = \vec{\rho}_H\left(0, \operatorname{co}\left[\frac{1}{n}\bigoplus_{i=1}^{n} Q_{\theta,K(n),i}\right]\right).
$$

We restate the estimator defined in the main text:

$$
\begin{aligned}
\widehat{\Theta}_n &= \left\{\theta \in \Theta : -\inf_{u \in B_{K(n)}} \frac{1}{n}\sum_{i=1}^{n} h(Q_{\theta,K(n),i}, u) \le \frac{\tau_n}{a_n}\right\} \\
&= \{\theta \in \Theta : a_n \widehat{T}_n(\theta) \le \tau_n\}.
\end{aligned}
$$

with $K(n) \to \infty$, $d_m K(n)/n \to 0$, $\tau_n \to \infty$, and $\tau_n/a_n \to 0$.

With these we are equipped to check conditions (a)–(e) in condition C.1 of Chernozhukov et al. (2007):

a) Θ is a compact subset of $\mathbb{R}^{d_\theta}$.

   Holds by Assumption 4.

b) $T(\theta)$ is continuous and $\inf_{\theta \in \Theta} T(\theta) = 0$

   By Lemma 9, $|T(\theta) - T(\theta')| \le \rho_H(\mathbb{E}[Q_\theta], \mathbb{E}[Q_{\theta'}])$. For any feasible (y, x), let $q_\theta =$

$M_\theta(y,x)$ and $q_{\theta'} = M_{\theta'}(y,x)$. Then

$$\begin{aligned}\|q_\theta - q_{\theta'}\|_1 &\leq \sum_{j=1}^{d_m}\sum_{k\geq 1}|m_j(y,x,\theta) - m_j(y,x,\theta')||g_k(x)| \\ &\leq \left[\sup_{x\in\mathcal{X}}\sum_{k\geq 1}|g_k(x)|\right]\sup_{(y,x)\in\mathcal{W}}\sum_{j=1}^{d_m}|m_j(y,x,\theta) - m_j(y,x,\theta')|.\end{aligned}$$

By Assumption 2, the bracketed term is finite, and by Assumption 5 the second factor converges to zero as $\theta' \to \theta$, uniformly on the compact domain. Thus

$$\rho_H\big(\mathbb{E}[Q_\theta], \mathbb{E}[Q_{\theta'}]\big) \to 0 \qquad \text{as } \theta' \to \theta,$$

and hence T is continuous.

c) $\hat{T}_n(\theta)$ is jointly measurable.

Not strictly necessary according to Chernozhukov et al. (2007). Nevertheless, it follows from the measurability of $Q_{\theta,K(n)}$, using the same arguments as in the proof of Theorem 1.

d) The sample criterion function is uniformly no smaller than the population function in large samples:

$$\begin{aligned}\sup_{\theta\in\Theta}\big(T(\theta) - \widehat{T}_n(\theta)\big)_+ &\leq \sup_{\theta\in\Theta}\left|T(\theta) - \widehat{T}_n(\theta)\right| \\ &= \sup_{\theta\in\Theta}\left|T(\theta) - T_{K(n)}(\theta) + T_{K(n)}(\theta) - \widehat{T}_n(\theta)\right| \\ &\leq \sup_{\theta\in\Theta}\left|T(\theta) - T_{K(n)}(\theta)\right| + \sup_{\theta\in\Theta}\left|T_{K(n)}(\theta) - \widehat{T}_n(\theta)\right|.\end{aligned}$$

The first term is the deterministic truncation error. By Lemma 10,

$$\sup_{\theta\in\Theta}\left|T(\theta) - T_{K(n)}(\theta)\right| \leq d_m C_1 \sup_{x\in\mathcal{X}}\sum_{k>K(n)}|g_k(x)| \equiv \delta_{K(n)} \to 0.$$

For the second term, recall that

$$T_{K(n)}(\theta) = \vec{\rho}_H\big(0, \mathbb{E}[Q_{\theta,K(n)}]\big), \qquad \widehat{T}_n(\theta) = \vec{\rho}_H\left(0, \operatorname{co}\left[\frac{1}{n}\bigoplus_{i=1}^{n} Q_{\theta,K(n),i}\right]\right).$$

Hence, by Lemma 8 and Lemma 9,

$$\begin{aligned}\sup_{\theta\in\Theta}\left|T_{K(n)}(\theta) - \widehat{T}_n(\theta)\right| &\le \sup_{\theta\in\Theta}\rho_H\left(\mathbb{E}[Q_{\theta,K(n)}], \operatorname{co}\left[\frac{1}{n}\bigoplus_{i=1}^{n} Q_{\theta,K(n),i}\right]\right)\\ &= O_p\left(\sqrt{\frac{d_m K(n) + d_\theta}{n}}\right).\end{aligned}$$

Therefore,

$$\begin{aligned}\sup_{\theta\in\Theta}\big(T(\theta) - \widehat{T}_n(\theta)\big)_+ &= O_p\left(\delta_{K(n)} + \sqrt{\frac{d_m K(n) + d_\theta}{n}}\right)\\ &= O_p(1/b_n).\end{aligned}$$

By Assumption 2, Lemma 10, and $d_m K(n)/n \to 0$, we have $b_n \to \infty$. The term $\delta_{K(n)}$ captures the deterministic approximation error from using finitely many instruments instead of the full infinite instrument class, while the second term captures the sampling error of the growing-dimensional random set estimator.

e) The sample criterion converges to the limit criterion function over the identified set Θ_I^*

at the $1/a_n$ rate.

$$\begin{aligned}
\sup_{\theta\in\Theta_I^*} \widehat{T}_n(\theta) &= \sup_{\theta\in\Theta_I^*} \vec{\rho}_H\left(0, \operatorname{co}\left[\frac{1}{n}\bigoplus_{i=1}^{n} Q_{\theta,K(n),i}\right]\right) \\
&\le \sup_{\theta\in\Theta_I^*} \vec{\rho}_H\big(0, \mathbb{E}[Q_{\theta,K(n)}]\big) \\
&\quad + \sup_{\theta\in\Theta_I^*} \rho_H\left(\mathbb{E}[Q_{\theta,K(n)}], \operatorname{co}\left[\frac{1}{n}\bigoplus_{i=1}^{n} Q_{\theta,K(n),i}\right]\right) \\
&= \sup_{\theta\in\Theta_I^*} \rho_H\left(\mathbb{E}[Q_{\theta,K(n)}], \operatorname{co}\left[\frac{1}{n}\bigoplus_{i=1}^{n} Q_{\theta,K(n),i}\right]\right) \\
&= O_p(1/a_n).
\end{aligned}$$

The first line is definitional. The second line follows from the inequality $\vec{\rho}_H(0,B) \le \vec{\rho}_H(0,A) + \rho_H(A,B)$, applied with $A = \mathbb{E}[Q_{\theta,K(n)}]$ and $B = \operatorname{co}\left[\frac{1}{n}\oplus_{i=1}^{n} Q_{\theta,K(n),i}\right]$. The third line uses the fact that if $\theta \in \Theta_I^*$, then $0 \in \mathbb{E}[Q_\theta]$. Lemmas 5 and 6 yield an $\eta \in \Gamma$ with $\int M_\theta \, d\eta = 0$, and Lemma 3 then produces an integrable selection $q_\theta(W^O) \in Q_\theta(W^O)$ satisfying $\mathbb{E}[q_\theta(W^O)] = 0$. Applying the coordinate projection $\pi_{K(n)}$ gives an integrable selection $\pi_{K(n)} q_\theta(W^O) \in Q_{\theta,K(n)}(W^O)$ satisfying $\mathbb{E}[\pi_{K(n)} q_\theta(W^O)] = \pi_{K(n)}\mathbb{E}[q_\theta(W^O)] = 0$. Therefore $0 \in \mathbb{E}[Q_{\theta,K(n)}]$ and $\vec{\rho}_H\big(0, \mathbb{E}[Q_{\theta,K(n)}]\big) = 0$.

Finally, the last rate follows from the convexified part of the uniform triangular-array law of large numbers and the definition

$$a_n = \sqrt{\frac{n}{d_m K(n) + d_\theta}}.$$

To complete the proof, after verifying condition C.1 in Chernozhukov et al. (2007), the result follows from Theorem 3.1(i) in Chernozhukov et al. (2007). Indeed, by part (e), $\mathcal{C}_n \equiv \sup_{\theta\in\Theta_I^*} a_n \widehat{T}_n(\theta) = O_p(1)$. Since $\tau_n \to \infty$, we have $\tau_n \ge \mathcal{C}_n$ with probability approaching one. Moreover, since $\tau_n/a_n \to 0$, the unscaled contour level vanishes. Therefore Theorem 3.1(i) of Chernozhukov et al. (2007) implies $\rho_H(\widehat{\Theta}_n, \Theta_I^*) \xrightarrow{p} 0$. □

D Proof of Theorem 3

This appendix verifies the high-level conditions of Andrews and Shi (2017), applies their Theorem 6.1, and adapts the proof of their Theorem 5.1 to the full class of null pairs. The principal technical step is to verify manageability of the process indexed by $\mathcal{U} = B^\infty$.

D.1 Manageability of the support-function process

Recall the uniform coordinate envelopes

$$a_k \equiv \sup_{x \in \mathcal{X}} |g_k(x)|, \qquad A \equiv \sum_{k=1}^{\infty} a_k.$$

The polynomial decay condition, Assumption 6, used in Theorem 3 gives $A < \infty$ and an explicit rate for its tail. Equip B^∞ with the normalized weighted semimetric

$$\rho_a(v, v') \equiv \frac{1}{d_m A} \sum_{\ell=1}^{d_m} \sum_{k=1}^{\infty} a_k |v_{\ell k} - v'_{\ell k}|.$$

Assumption 3 and the elementary inequality $|\sup_s f(s) - \sup_s g(s)| \le \sup_s |f(s) - g(s)|$ imply, uniformly over $w^o \in \mathcal{W}^O$, $\theta \in \Theta$, and $u, u' \in \mathcal{U}$,

$$\begin{aligned} |\widetilde{m}(w^o, \theta, u) - \widetilde{m}(w^o, \theta, u')| &\le C_1 \sum_{\ell=1}^{d_m} \sum_{k=1}^{\infty} a_k |u_{\ell k} - u'_{\ell k}| \\ &= M_0 \rho_a(u, u') \end{aligned}$$

with $M_0 \equiv d_m C_1 A$. Moreover, for every $u \in \mathcal{U}$,

$$\begin{aligned} |\widetilde{m}(w^o, \theta, u)| &\le C_1 \sum_{\ell=1}^{d_m} \sum_{k=1}^{\infty} a_k |u_{\ell k}| \\ &\le d_m C_1 A = M_0. \end{aligned}$$

Thus M_0 is a uniform scalar envelope for the support-function moment class.

Fix an arbitrary sequence $\{(\theta_n, P_n) \in \mathcal{F}^+ : n \geq 1\}$, and let $\{W^O_{n,i} : i \leq n\}$ be a row-wise i.i.d. triangular array with $W^O_{n,i} \sim P_n$. For a realization ω, define

$$\mathcal{M}_{n,\omega}(\theta_n) \equiv \left\{ \left(\widetilde{m}(W^O_{n,1}(\omega), \theta_n, u), \ldots, \widetilde{m}(W^O_{n,n}(\omega), \theta_n, u)\right)' : \ u \in \mathcal{U} \right\} \subset \mathbb{R}^n.$$

For a subset V of a metric space, let $N(\xi, V)$ and $D(\xi, V)$ denote its covering and packing numbers. Appendix B of Andrews and Shi (2017) records the standard inequalities

$$N(\xi, V) \leq D(\xi, V) \leq N(\xi/2, V).$$

The same reference restates Definition 7.9 of Pollard (1990): the triangular array is manageable with respect to an envelope array if there is a deterministic $\lambda : (0, 1] \to \mathbb{R}$ such that

$$\int_0^1 \sqrt{\log \lambda(\xi)}\, d\xi < \infty$$

and, for every $\xi \in (0, 1]$, every realization ω, every n, and every nonnegative n-vector α,

$$D(\xi \| \alpha \odot M_n \|_2, \alpha \odot \mathcal{M}_{n,\omega}(\theta_n)) \leq \lambda(\xi), \qquad M_n \equiv (M_0, \ldots, M_0)'.$$

The first lemma determines the number of instrument coordinates needed to control the coordinate-envelope tail.

Lemma 11. *Suppose $C_a \equiv \sup_{k \geq 1} k^\beta a_k < \infty$ for some $\beta > 1$. Equivalently, $a_k \leq C_a k^{-\beta}$ for every k. For $\xi \in (0, 1]$, let*

$$K_\xi \equiv \left\lceil \left(\frac{4C_a}{(\beta - 1)A\xi} \right)^{1/(\beta-1)} \right\rceil.$$

Then

$$\sum_{k>K_\xi} a_k \le \frac{A\xi}{4}$$

and $K_\xi = O\big(\xi^{-1/(\beta-1)}\big)$ *as* $\xi \downarrow 0$.

Proof. By the definition of C_a,

$$a_k \le C_a k^{-\beta} \qquad \text{for every } k \ge 1.$$

Hence, for every integer $K \ge 1$, the integral comparison for the decreasing function $x \mapsto x^{-\beta}$ gives

$$\begin{aligned}\sum_{k>K} a_k &\le C_a \sum_{k>K} k^{-\beta} \\ &\le C_a \int_K^\infty x^{-\beta}\, dx \\ &= \frac{C_a}{\beta-1} K^{1-\beta},\end{aligned}$$

where the last expression is finite because $\beta > 1$.

By definition,

$$K_\xi \ge \left(\frac{4C_a}{(\beta-1)A\xi}\right)^{1/(\beta-1)}.$$

Since $1-\beta < 0$, it follows that

$$\begin{aligned}K_\xi^{1-\beta} &\le \left(\frac{4C_a}{(\beta-1)A\xi}\right)^{-1} \\ &= \frac{(\beta-1)A\xi}{4C_a}.\end{aligned}$$

Therefore,

$$\sum_{k>K_\xi} a_k \le \frac{C_a}{\beta-1} K_\xi^{1-\beta} \le \frac{A\xi}{4}.$$

Finally, using $\lceil x \rceil \le x+1$,

$$K_\xi \le 1 + \left(\frac{4C_a}{(\beta-1)A}\right)^{1/(\beta-1)} \xi^{-1/(\beta-1)}.$$

Hence

$$K_\xi = O\left(\xi^{-1/(\beta-1)}\right) \qquad \text{as } \xi \downarrow 0.$$

□

The next lemma verifies the entropy-integrability requirement.

Lemma 12. *Let K_ξ be defined as in Lemma 11. For a sufficiently large constant $C_0 > 1$, define*

$$\lambda(\xi) \equiv \left(\frac{C_0}{\xi}\right)^{d_m K_\xi}, \qquad \xi \in (0,1].$$

If $\beta > 3/2$, then

$$\int_0^1 \sqrt{\log \lambda(\xi)}\, d\xi < \infty.$$

Proof. Let $q \equiv 1/(\beta-1)$. Lemma 11 implies that, for some finite constant C,

$$\log \lambda(\xi) = d_m K_\xi \log(C_0/\xi) \le C\xi^{-q} \log(C_0/\xi)$$

for all $\xi \in (0,1]$. Consequently,

$$\int_0^1 \sqrt{\log \lambda(\xi)}\, d\xi \le \sqrt{C} \int_0^1 \xi^{-q/2} \sqrt{\log(C_0/\xi)}\, d\xi.$$

Set $t = \log(1/\xi)$, so that $\xi = e^{-t}$ and $d\xi = -e^{-t}dt$. The right-hand side becomes

$$\sqrt{C} \int_0^\infty e^{-ct} \sqrt{t + \log C_0}\, dt, \qquad c \equiv 1 - \frac{q}{2} = \frac{2\beta - 3}{2(\beta - 1)}.$$

Because $C_0 > 1$,

$$\begin{aligned} \sqrt{t + \log C_0} &\le \sqrt{t} + \sqrt{\log C_0} \\ &\le 1 + t + \sqrt{\log C_0}. \end{aligned}$$

Hence, if $c > 0$,

$$\begin{aligned} \int_0^\infty e^{-ct} \sqrt{t + \log C_0}\, dt &\le \int_0^\infty e^{-ct} \left(1 + t + \sqrt{\log C_0}\right) dt \\ &= \frac{1 + \sqrt{\log C_0}}{c} + \frac{1}{c^2} < \infty. \end{aligned}$$

It follows that

$$\int_0^1 \sqrt{\log \lambda(\xi)}\, d\xi < \infty.$$

Finally, $c > 0$ is equivalent to $\beta > 3/2$. □

Lemma 13. *Assumption PS2 of Andrews and Shi (2017) holds in the present setting with*

$$\lambda(\xi) = \left(\frac{C_0}{\xi}\right)^{d_m K_\xi},$$

where K_ξ is the integer chosen in Lemma 11 so that the coordinate-envelope tail is at most $A\xi/4$.

Proof. Fix $\xi \in (0,1]$. Let $\Pi_{K_\xi} u$ retain the first K_ξ instrument coordinates in each primitive-moment block of u and set the remaining coordinates equal to zero. Cover $[-1,1]^{d_m K_\xi}$ in the sup norm by a grid $\mathcal{U}_\xi$ of radius $\xi/4$. For a constant $C_0 > 1$ independent of ξ, the grid may be chosen so that

$$|\mathcal{U}_\xi| \le \left(\frac{C_0}{\xi}\right)^{d_m K_\xi}.$$

For every $u \in \mathcal{U}$, choose $v \in \mathcal{U}_\xi$ within $\xi/4$ of $\Pi_{K_\xi} u$ in the sup norm and extend v by zeros. Then

$$\begin{aligned}\rho_a(u,v) &\le \frac{1}{d_m A} \sum_{\ell=1}^{d_m} \left\{ \sum_{k \le K_\xi} a_k |u_{\ell k} - v_{\ell k}| + \sum_{k > K_\xi} a_k |u_{\ell k}| \right\} \\ &\le \frac{\xi}{4} + \frac{\xi}{4} = \frac{\xi}{2}.\end{aligned}$$

Thus $\mathcal{U}_\xi$ is a $\xi/2$ cover of B^∞ under ρ_a.

Let $\alpha = (\alpha_1, \ldots, \alpha_n)'$ be nonnegative. The Lipschitz bound above implies

$$\begin{aligned}&\left\| \alpha \odot \left(\left(\widetilde{m}(W^O_{n,i}, \theta_n, u)\right)_{i=1}^n - \left(\widetilde{m}(W^O_{n,i}, \theta_n, v)\right)_{i=1}^n \right) \right\|_2 \\ &\qquad \le \rho_a(u,v) \|\alpha \odot M_n\|_2.\end{aligned}$$

The same direction grid therefore covers the sample paths. The packing–covering inequality from Andrews and Shi (2017) then gives

$$D(\xi \|\alpha \odot M_n\|_2, \alpha \odot \mathcal{M}_{n,\omega}(\theta_n)) \le \left(\frac{C_0}{\xi}\right)^{d_m K_\xi} = \lambda(\xi).$$

Lemma 12 verifies the required entropy integral. Hence the support-function process is manageable in the sense of Definition 7.9 of Pollard (1990). □

Since the preceding bounds are deterministic and do not depend on the sequence $\{P_n\}$,

the same manageability function λ applies to every sequence $\{(\theta_n, P_n) \in \mathcal{F}^+\}$.

D.2 Verification of the Andrews–Shi assumptions

We apply Andrews and Shi (2017) to the scalar process $\{\widetilde{m}(\cdot, \theta, u) : u \in \mathcal{U}\}$. In their notation there is one moment inequality, their index τ is $u \in \mathcal{U}$, the conditioning variable is degenerate, the instrument class is the singleton containing the constant function, and the weight function is a point mass on it. Their test statistic, GMS function and bootstrap critical value then reduce to $T_n^{AS}(\theta)$, $\overline{\varphi}_n(\theta, u)$ and $c_{n,1-\alpha}^{AS}(\theta)$. Recall the envelope and Lipschitz bounds established above, which hold uniformly in (w^o, θ):

$$|\widetilde{m}(w^o, \theta, u)| \le M_0, \qquad |\widetilde{m}(w^o, \theta, u) - \widetilde{m}(w^o, \theta, u')| \le M_0 \rho_a(u, u'). \tag{19}$$

For $(\theta, P) \in \mathcal{F}^+$ and $u, u' \in \mathcal{U}$ define

$$\begin{aligned}
\nu_{n,P}(\theta, u) &\equiv \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \big\{ \widetilde{m}(W_i^O, \theta, u) - \mathbf{E}_P[\widetilde{m}(W_i^O, \theta, u)] \big\}, \\
h_{1,n,P}(\theta, u) &\equiv \sqrt{n}\, \mathbf{E}_P[\widetilde{m}(W_i^O, \theta, u)], \\
h_{2,P}(\theta; u, u') &\equiv \mathrm{Cov}_P\big(\widetilde{m}(W_i^O, \theta, u), \widetilde{m}(W_i^O, \theta, u')\big).
\end{aligned}$$

Let $\widehat{h}_{2,n}(\theta; u, u')$ denote the sample analogue of $h_{2,P}(\theta; u, u')$. Then $\sqrt{n}\,\overline{m}_n(\theta, u) = \nu_{n,P}(\theta, u) + h_{1,n,P}(\theta, u)$ and $\overline{\Sigma}_n(\theta, u) = \widehat{h}_{2,n}(\theta; u, u) + \epsilon$. Moreover, $h_{1,n,P}(\theta, \cdot) \ge 0$ whenever $(\theta, P) \in \mathcal{F}$. Kernels on $\mathcal{U} \times \mathcal{U}$ are compared in the uniform metric $d(h, h') \equiv \sup_{u,u' \in \mathcal{U}} |h(u, u') - h'(u, u')|$.

(i) **PS1.** Under every $(\theta, P) \in \mathcal{F}^+$ the observations are i.i.d. and $\sigma_P(\theta) = 1$. By (19), M_0 is a deterministic envelope, so $\sup_{P \in \mathcal{P}} \mathbf{E}_P[M_0^{2+\delta}] < \infty$ for every $\delta > 0$.

(ii) **PS2.** This is Lemma 13, whose bound holds for every sequence in $\mathcal{F}^+$ with row-wise i.i.d. arrays. The map $w^o \mapsto \widetilde{m}(w^o, \theta, u) = h(Q_\theta(w^o), u)$ is measurable, being the support function of the random compact set of Lemma 2. By (19), suprema over $\mathcal{U}$

coincide with suprema over a countable ρ_a-dense subset.

(iii) **PS3.** With a degenerate conditioning variable, the moment inequality reads $\mathbf{E}_P[\widetilde{m}(W_i^O, \theta, u)] \geq 0$ for all $u \in \mathcal{U}$. By Theorem 1, this holds if and only if $\theta \in \Theta_I^*(P)$.

(iv) **SIG1 and SIG2.** Both hold identically, because $\widehat{\sigma}_n(\theta) = \sigma_P(\theta) = 1$.

(v) **MFA.** If $\theta \notin \Theta_I^*(P)$, Theorem 1 gives some $u_{\theta,P} \in \mathcal{U}$ with $\mathbf{E}_P[\widetilde{m}(W_i^O, \theta, u_{\theta,P})] < 0$. With a degenerate conditioning variable, this is a violation on a set of probability one.

(vi) **PS4 and PS5.** The constant instrument satisfies Assumption M of Andrews and Shi (2017), and S_1 satisfies their Assumptions S1 and S2 by Lemma 1 of Andrews and Shi (2013). Their Lemma D.2 derives PS4 and PS5 from these assumptions together with PS1 and PS2. As formulated there, PS4 and PS5 apply to every sequence (n_r, θ_r, P_r) with (n_r) strictly increasing, $(\theta_r, P_r) \in \mathcal{F}^+$ and $d(h_{2,P_r}(\theta_r; \cdot, \cdot), h_{2,0}) \to 0$, for *some* covariance kernel $h_{2,0}$, whether or not it is attained.

- PS4 gives $\nu_{n_r,P_r}(\theta_r, \cdot) \Rightarrow \nu_{h_{2,0}}$ and $d(\widehat{h}_{2,n_r}(\theta_r), h_{2,0}) \xrightarrow{p} 0$.
- PS5 gives the same two conclusions for the bootstrap process $\sqrt{n_r}(\overline{m}^*_{n_r} - \overline{m}_{n_r})(\theta_r, \cdot)$ and the bootstrap kernel $\widehat{h}^*_{2,n_r}(\theta_r)$. These hold conditionally on almost every sample path along which $d(\widehat{h}_{2,n_r}(\theta_r), h_{2,0}) \to 0$.

Next let us show relative compactness of the covariance kernels. Let $\mathcal{H}_{2,\mathcal{P}} \equiv \{h_{2,P}(\theta; \cdot, \cdot) : (\theta, P) \in \mathcal{F}\}$. Identify directions at zero ρ_a-distance. By (19), such directions generate identical moment functions, so all kernels are defined on the quotient. The quotient is compact. Given $\{u^{(r)}\} \subseteq \mathcal{U}$, a diagonal argument yields a subsequence converging coordinatewise to some $u \in \mathcal{U}$. For every K, this subsequence satisfies $\limsup_r \rho_a(u^{(r)}, u) \leq (2/A)\sum_{k>K} a_k$, which vanishes as $K \to \infty$ because $A < \infty$. Write $f_u \equiv \widetilde{m}(W_i^O, \theta, u)$. Expanding $h_{2,P}(\theta; u, u') - h_{2,P}(\theta; v, v') = \mathrm{Cov}_P(f_u - f_v, f_{u'}) + \mathrm{Cov}_P(f_v, f_{u'} - f_{v'})$ and applying

Cauchy–Schwarz together with (19) gives, uniformly over $(\theta, P) \in \mathcal{F}$,

$$|h_{2,P}(\theta; u, u')| \le M_0^2, \qquad |h_{2,P}(\theta; u, u') - h_{2,P}(\theta; v, v')| \le M_0^2\{\rho_a(u, v) + \rho_a(u', v')\}.$$

Hence $\mathcal{H}_{2,\mathcal{P}}$ is uniformly bounded and uniformly equicontinuous on the compact quotient of $\mathcal{U} \times \mathcal{U}$. By the Arzelà–Ascoli theorem, it is relatively compact under d, and its closure $\overline{\mathcal{H}}_{2,\mathcal{P}}$ is compact. Uniform limits preserve symmetry and positive semidefiniteness, so every element of $\overline{\mathcal{H}}_{2,\mathcal{P}}$ is a covariance kernel. Such a limit need not be attained by any $(\theta, P) \in \mathcal{F}$. For a kernel h on $\mathcal{U} \times \mathcal{U}$, define the semimetric $\rho_h(u, v) \equiv \{h(u, u) - 2h(u, v) + h(v, v)\}^{1/2}$.

- *Domination.* For $h \in \mathcal{H}_{2,\mathcal{P}}$, $\rho_h(u, v)$ is the $L_2(P)$ distance between f_u and f_v, so $\rho_h \le M_0\rho_a$ by (19). Since ρ_h is d-continuous in h, the bound extends to $\overline{\mathcal{H}}_{2,\mathcal{P}}$.

- *Entropy.* The proof of Lemma 13 shows $N(\varepsilon, \mathcal{U}, \rho_a) \le \lambda(2\varepsilon)$ for $\varepsilon \in (0, 1/2]$, and $\int_0^1 \sqrt{\log \lambda(\xi)}\, d\xi < \infty$ by Lemma 12.

Dudley's entropy bound (Dudley, 1967) therefore yields:

(G1) for every $h \in \overline{\mathcal{H}}_{2,\mathcal{P}}$ there is a mean-zero Gaussian process ν_h on $\mathcal{U}$ with covariance h and bounded, uniformly ρ_a-continuous sample paths, and $\lim_{\delta \downarrow 0} \sup_{h \in \overline{\mathcal{H}}_{2,\mathcal{P}}} \mathbf{E} \sup_{\rho_a(u,v) < \delta} |\nu_h(u) - \nu_h(v)| = 0$;

(G2) if $h_r, h \in \overline{\mathcal{H}}_{2,\mathcal{P}}$ and $d(h_r, h) \to 0$, then $\nu_{h_r} \Rightarrow \nu_h$ in $\ell^\infty(\mathcal{U})$, because the finite-dimensional laws converge and (G1) gives asymptotic uniform equicontinuity.

For $h \in \overline{\mathcal{H}}_{2,\mathcal{P}}$ and $h_1 : \mathcal{U} \to [0, \infty)$, define

$$T(h_1, h) \equiv \sup_{u \in \mathcal{U}} \left[\frac{\nu_h(u) + h_1(u)}{\sqrt{h(u, u) + \epsilon}} \right]_-^2.$$

This is a lower semicontinuous functional of the continuous path ν_h, hence a random variable. Let $c_0(h_1, h, 1 - \alpha)$ denote its $(1 - \alpha)$-quantile. When h_1 is data dependent, such as $\overline{\varphi}_n(\theta, \cdot)$, c_0 is evaluated with h_1 held fixed and ν_h independent of the data.

Finally, let ν, ν' be functions on $\mathcal{U}$ with sup norms at most B, let $s, s' \geq \sqrt{\epsilon}$, and let $\mu : \mathcal{U} \to [0, \infty)$ be arbitrary. Because $[x + \mu]_- \leq [x]_-$ and $|[a]_- - [b]_-| \leq |a - b|$,

$$\left| \sup_u \left[\tfrac{\nu(u)+\mu(u)}{s(u)} \right]_-^2 - \sup_u \left[\tfrac{\nu'(u)+\mu(u)}{s'(u)} \right]_-^2 \right| \leq \frac{2B}{\epsilon} \Big(\|\nu - \nu'\|_\infty + \frac{B}{\sqrt{\epsilon}} \|s - s'\|_\infty \Big). \tag{20}$$

The bound does not depend on μ. This is what permits nonnegative slackness functions that need not converge.

Theorem 5.1 of Andrews and Shi (2017) rests on their Lemma D.1, Theorem D.3(a) and Lemma D.4. All of these are *stated* for compact subsets of the attainable kernel class, and $\overline{\mathcal{H}}_{2,\mathcal{P}}$ need not be such a set. In their proofs, and in the proofs of Theorem 1 and Lemmas A2–A4 of Andrews and Shi (2013) to which they refer, compactness is used once: to pass from a near-extremal sequence of null pairs to a subsequence whose kernels converge uniformly to some covariance kernel. The subsequent steps use only PS4, PS5, monotonicity and continuity of S, GMS1, and $\nu_{h_{2,r}} \Rightarrow \nu_{h_{2,0}}$. The following lemma repeats those steps along a sequence that is kernel-convergent by hypothesis. The only properties of the limit it uses are (G1)–(G2), which hold on all of $\overline{\mathcal{H}}_{2,\mathcal{P}}$.

Lemma 14. *Let the conditions of Theorem 3 hold, let (n_r) be strictly increasing, and let $(\theta_r, P_r) \in \mathcal{F}$ satisfy $d\big(h_{2,P_r}(\theta_r; \cdot, \cdot), h_{2,0}\big) \to 0$ for some covariance kernel $h_{2,0}$. Then*

$$\liminf_{r \to \infty} P_r^{n_r} \big\{ \theta_r \in CS_{n_r}^{AS}(1 - \alpha) \big\} \geq 1 - \alpha.$$

Proof. We use the following abbreviations: $h_{1,r} \equiv h_{1,n_r,P_r}(\theta_r, \cdot) \geq 0$, $h_{2,r} \equiv h_{2,P_r}(\theta_r; \cdot, \cdot)$, $\nu_r \equiv \nu_{n_r,P_r}(\theta_r, \cdot)$, $\overline{\varphi}_r \equiv \overline{\varphi}_{n_r}(\theta_r, \cdot)$, $s_r(u) \equiv \{\overline{\Sigma}_{n_r}(\theta_r, u)\}^{1/2} \geq \sqrt{\epsilon}$, $s_{2,r}(u) \equiv \{h_{2,r}(u, u) + \epsilon\}^{1/2}$, $s_0(u) \equiv \{h_{2,0}(u, u) + \epsilon\}^{1/2}$, and $c_r \equiv c_0(h_{1,r}, h_{2,r}, 1 - \alpha)$. Since $h_{2,r} \in \mathcal{H}_{2,\mathcal{P}}$, the limit satisfies $h_{2,0} \in \overline{\mathcal{H}}_{2,\mathcal{P}}$. Hence (G1), (G2) and item (vi) apply, while $h_{1,r}$ need not converge. Fix $\delta \in (0, \eta)$.

(a) Oracle upper-tail bound: $\limsup_r P_r^{n_r} \{T_{n_r}^{AS}(\theta_r) > c_r + \delta\} \leq \alpha$. By PS4, $(\nu_r, s_r) \Rightarrow (\nu_{h_{2,0}}, s_0)$. By (G2), $\nu_{h_{2,r}} \Rightarrow \nu_{h_{2,0}}$, and $s_{2,r} \to s_0$ uniformly. By the almost sure representation

theorem (Pollard, 1990, Theorem 9.4), there exist two pairs of random elements:

- $(\tilde{\nu}_r, \tilde{s}_r)$ with the law of (ν_r, s_r), and $\tilde{\nu}_0$ with the law of $\nu_{h_{2,0}}$, such that $\|\tilde{\nu}_r - \tilde{\nu}_0\|_\infty + \|\tilde{s}_r - s_0\|_\infty \to 0$ almost surely;
- $\tilde{\nu}_r^G$ with the law of $\nu_{h_{2,r}}$, and $\tilde{\nu}_0'$ with the law of $\nu_{h_{2,0}}$, such that $\|\tilde{\nu}_r^G - \tilde{\nu}_0'\|_\infty \to 0$ almost surely.

Let $T_{r,0} \equiv \sup_u [(\nu_{h_{2,0}}(u) + h_{1,r}(u))/s_0(u)]_-^2$. The limits have bounded paths, so we may apply (20) with $\mu = h_{1,r}$ and B equal to the sup norm of the limit path plus one. This shows that $T_{n_r}^{AS}(\theta_r)$ and $T(h_{1,r}, h_{2,r})$ can each be coupled with a copy of $T_{r,0}$ such that the difference tends to zero almost surely. Therefore

$$P_r^{n_r}\{T_{n_r}^{AS}(\theta_r) > c_r + \delta\} \leq P\{T_{r,0} > c_r + \delta/2\} + o(1) \leq P\{T(h_{1,r}, h_{2,r}) > c_r\} + o(1) \leq \alpha + o(1).$$

This is the argument of Theorem D.3(a) of Andrews and Shi (2017) combined with Lemma A2 of Andrews and Shi (2013).

(b) GMS adjustment versus true slackness. Let $R_r \equiv \sup_u |\nu_r(u)|$, which is $O_p(1)$ by PS4 and the continuous mapping theorem. Consider a direction u.

- If $\xi_{n_r}(\theta_r, u) > 1$, then $\nu_r(u) + h_{1,r}(u) > \kappa_{n_r} s_r(u)$, so $h_{1,r}(u) > \kappa_{n_r} s_r(u) - R_r$, while $\overline{\varphi}_r(u) = B_{n_r} s_r(u)$.
- If $\xi_{n_r}(\theta_r, u) \leq 1$, then $\overline{\varphi}_r(u) = 0 \leq h_{1,r}(u)$.

Once $\kappa_{n_r} > B_{n_r}$, since $s_r \geq \sqrt{\epsilon}$, the inequality $\overline{\varphi}_r \leq h_{1,r}$ therefore holds simultaneously over $\mathcal{U}$ on the event $E_r \equiv \{R_r \leq (\kappa_{n_r} - B_{n_r})\sqrt{\epsilon}\}$. By GMS1, $(\kappa_{n_r} - B_{n_r})\sqrt{\epsilon} = \kappa_{n_r}(1 - B_{n_r}/\kappa_{n_r})\sqrt{\epsilon} \to \infty$, so $P_r^{n_r}(E_r) \to 1$. On E_r, every path of $\nu_{h_{2,r}}$ satisfies $[\nu_{h_{2,r}} + \overline{\varphi}_r]_- \geq [\nu_{h_{2,r}} + h_{1,r}]_-$ pointwise. Hence $T(\overline{\varphi}_r, h_{2,r}) \geq T(h_{1,r}, h_{2,r})$ pathwise for the same Gaussian process, and

$$c_0(\overline{\varphi}_r, h_{2,r}, 1 - \alpha) \geq c_r \qquad \text{on } E_r.$$

Replacing the true slackness by the smaller GMS adjustment weakens the inequalities in the Gaussian experiment. This enlarges the statistic and its quantile, which is the sense in which GMS is conservative.

(c) Bootstrap critical value: $P_r^{n_r}\{c^{AS}_{n_r,1-\alpha}(\theta_r) < c_0(\overline{\varphi}_r, h_{2,r}, 1-\alpha) + \delta\} \to 0$. By PS4, $d(\widehat{h}_{2,n_r}(\theta_r), h_{2,0}) \xrightarrow{p} 0$. It therefore suffices to argue along a further subsequence of an arbitrary subsequence on which this convergence holds almost surely. Fix such a sample path. Conditionally on it, $\overline{\varphi}_r$ is a fixed nonnegative function, and by PS5, $(\nu_r^*, s_r^*) \Rightarrow (\nu_{h_{2,0}}, s_0)$, where $\nu_r^* \equiv \sqrt{n_r}(\overline{m}^*_{n_r} - \overline{m}_{n_r})(\theta_r, \cdot)$ and $s_r^*(u) \equiv \{\overline{\Sigma}^*_{n_r}(\theta_r, u)\}^{1/2}$. Repeat the argument of (a) with $(\nu_r^*, s_r^*, \overline{\varphi}_r)$ in place of $(\nu_r, s_r, h_{1,r})$, and let P^* denote bootstrap probability and T_r^* the bootstrap statistic. For every $\xi > 0$ and all constants x_r,

$$\limsup_r \big[P^*\{T_r^* \le x_r\} - P\{T(\overline{\varphi}_r, h_{2,r}) \le x_r + \xi\}\big] \le 0.$$

Now take $x_r \equiv c_0(\overline{\varphi}_r, h_{2,r}, 1-\alpha) + \delta - \eta$ and $\xi \in (0, \eta - \delta)$.

- Since $x_r + \xi$ lies below the $(1-\alpha)$-quantile, $P\{T(\overline{\varphi}_r, h_{2,r}) \le x_r + \xi\} < 1 - \alpha$.
- Hence $P^*\{T_r^* \le x_r\} < 1 - \alpha + \eta$ for all large r, so x_r lies below the conditional $(1-\alpha+\eta)$-quantile.
- Therefore $c^{AS}_{n_r,1-\alpha}(\theta_r) > x_r + \eta = c_0(\overline{\varphi}_r, h_{2,r}, 1-\alpha) + \delta$ for all large r.

Bounded convergence yields the claim. This is Lemma D.4 of Andrews and Shi (2017); it uses the fixed $\eta > \delta$.

Combining (a), (b) and (c),

$$P_r^{n_r}\{\theta_r \notin CS^{AS}_{n_r}(1-\alpha)\} \le P_r^{n_r}\{T^{AS}_{n_r}(\theta_r) > c_r + \delta\} + P_r^{n_r}(E_r^c) + o(1) \le \alpha + o(1).$$

□

Finally, let us establish uniformity by contradiction. Suppose the coverage claim of Theorem 3 fails. Then there exist $\delta_0 > 0$, a strictly increasing sequence (n_r), and pairs

$(\theta_r, P_r) \in \mathcal{F}$ with $P_r^{n_r}\{\theta_r \in CS_{n_r}^{AS}(1-\alpha)\} \leq 1-\alpha-\delta_0$ for all r. By relative compactness of $\mathcal{H}_{2,\mathcal{P}}$, a subsequence satisfies $d(h_{2,P_r}(\theta_r;\cdot,\cdot), h_{2,0}) \to 0$ for some $h_{2,0} \in \overline{\mathcal{H}}_{2,\mathcal{P}}$, which Lemma 14 contradicts. Hence

$$\liminf_{n\to\infty} \inf_{(\theta,P)\in\mathcal{F}} P^n\{\theta \in CS_n^{AS}(1-\alpha)\} \geq 1-\alpha.$$

The kernel convergence used here is extracted from relative compactness, which rests only on (19). Neither θ_r nor $h_{2,0}$ needs to correspond to a limiting pair, so no continuity of $\theta \mapsto h_{2,P}(\theta;\cdot,\cdot)$ is used.

The second claim of Theorem 3 concerns power against fixed alternatives. Fix $P \in \mathcal{P}$ and $\theta \in \Theta \setminus \Theta_I^*(P)$. Theorem 6.1 of Andrews and Shi (2017) is a fixed-alternative result, not restricted to compact kernel sets. It is a corollary of their Theorem E.1, whose conditions all hold here:

- PS4 and PS5, by item (vi) applied to the constant sequence;
- MFA by item (v), and SIG2 by item (iv);
- CI and MQ, because the single constant instrument detects the violation in (v) and the weight function is a point mass;
- S1, S3 and S4, because S_1 is nonincreasing, positive if and only if its argument is negative, and homogeneous of degree two.

Therefore

$$P^n\left\{T_n^{AS}(\theta) > c_{n,1-\alpha}^{AS}(\theta)\right\} \longrightarrow 1.$$

This proves Theorem 3. □

D.3 Finite and truncated direction classes

Proof of Corollary 2. Throughout, direction sets are nonempty, and the restricted statistics and conditional bootstrap quantiles are assumed measurable.

(a) Consider a strictly increasing sequence (n_r) and a sequence $(\theta_r, P_r, \mathcal{V}_r)$ with $\mathcal{V}_r \subseteq \mathcal{U}$ and $(\theta_r, P_r) \in \mathcal{F}_{\mathcal{V}_r} \subseteq \mathcal{F}^+$. The envelope bound, the covariance Lipschitz bound and the domination $\rho_h \leq M_0 \rho_a$ used in the proof of Theorem 3 hold uniformly on $\mathcal{F}^+$. Consequently, the covariance-kernel family $\{h_{2,P}(\theta; \cdot, \cdot) : (\theta, P) \in \mathcal{F}^+\}$ is relatively compact by the same Arzelà–Ascoli argument, and (G1)–(G2) hold on its closure. Suppose that the kernels $h_{2,P_r}(\theta_r; \cdot, \cdot)$ converge uniformly.

Apply PS4 and PS5, which are stated on $\mathcal{F}^+$, to the empirical and bootstrap processes on the full index set $\mathcal{U}$. The comparison bound (20) remains valid when its supremum is restricted to $\mathcal{V}_r$. Its right-hand side is bounded by the same full-space process and covariance-estimation errors, independently of $\mathcal{V}_r$. Thus these errors vanish along the sequence even when $\mathcal{V}_r$ changes with r. Moreover, $(\theta_r, P_r) \in \mathcal{F}_{\mathcal{V}_r}$ gives $h_{1,n_r,P_r}(\theta_r, u) \geq 0$ for every $u \in \mathcal{V}_r$. This is the nonnegativity used in the oracle-tail and GMS comparisons in Steps (a) and (b) of Lemma 14. The bootstrap-quantile comparison in Step (c) uses only $\overline{\varphi}_{n_r} \geq 0$. These comparisons therefore apply with every supremum restricted to $\mathcal{V}_r$, including in the Gaussian statistic $T(\cdot, \cdot)$ and its quantile c_0. The bound $R_r = \sup_{u \in \mathcal{U}} |\nu_r(u)| = O_p(1)$ in Step (b) is still taken over the full index set. The fixed critical-value adjustment η supplies the same quantile margin as in that lemma.

It follows that the sequential coverage bound holds along every such triple sequence with uniformly convergent kernels. If the uniform bound in (a) failed, there would exist $\delta_0 > 0$, a strictly increasing sequence (n_r), and triples as above whose coverage probabilities are at most $1 - \alpha - \delta_0$ for all r. Relative compactness supplies a kernel-convergent subsequence, contradicting the sequential bound. This proves (a).

(b) Fix P, $\theta \notin \Theta_O(\mathcal{V})(P)$, and a direction $u_0 \in \mathcal{V}$ with $\mathbf{E}_P[\widetilde{m}(W^O, \theta, u_0)] < 0$. The law of large numbers makes the test statistic over $\mathcal{V}$ diverge at order n. Using the same per-

direction GMS adjustments and bootstrap draws, each bootstrap statistic over $\mathcal{V}$ is bounded by its counterpart over $\mathcal{U}$. The latter has an $O_p(1)$ conditional critical value under the fixed law P, by the bootstrap-process bound used in the proof of Theorem E.1 of Andrews and Shi (2017). Hence the critical value over $\mathcal{V}$ is $O_p(1)$, and the rejection probability tends to one.

(c) Coverage follows from (a), since $B_{K(n)} \subseteq \mathcal{U}$ and $\mathcal{F} \subseteq \mathcal{F}_{B_{K(n)}}$. For power, fix P and $\theta \notin \Theta_I^*(P)$, and choose $u_0 \in \mathcal{U}$ such that $\mathbf{E}_P[\widetilde{m}(W^O, \theta, u_0)] = -2c < 0$. Put $u_n = \Pi_{K(n)} u_0 \in B_{K(n)}$. By (19),

$$\left|\mathbf{E}_P[\widetilde{m}(W^O, \theta, u_0)] - \mathbf{E}_P[\widetilde{m}(W^O, \theta, u_n)]\right| \leq d_m C_1 \sum_{k > K(n)} a_k \longrightarrow 0.$$

Thus $\mathbf{E}_P[\widetilde{m}(W^O, \theta, u_n)] \leq -c$ for all sufficiently large n. PS4, applied to the constant sequence (θ, P) on $\mathcal{U}$, gives

$$\sup_{u \in \mathcal{U}} \left|\overline{m}_n(\theta, u) - \mathbf{E}_P[\widetilde{m}(W^O, \theta, u)]\right| = O_p(n^{-1/2}),$$

so $\overline{m}_n(\theta, u_n) \leq -c/2$ with probability approaching one. Since $\overline{\Sigma}_n \leq M_0^2 + \epsilon$, the statistic over $B_{K(n)}$ is then at least $nc^2/\{4(M_0^2 + \epsilon)\}$.

With the same per-direction GMS adjustments and bootstrap draws, each bootstrap statistic over $B_{K(n)}$ is bounded by the corresponding statistic over $\mathcal{U}$. Their conditional critical values have the same ordering, and the full-space critical value is $O_p(1)$ by the bound used in the proof of Theorem E.1 of Andrews and Shi (2017). Therefore θ is excluded with probability approaching one, without a rate restriction on $K(n)$. □

E Derivation of the Exact LP Representation

We derive equations (9) and (10). Throughout, $d_m = d_y = 1$, the covariate domain is $[0,1]^{d_x}$, and the feasible set for observation i is the nonempty rectangle

$$D_i = [y_{iL}, y_{iU}] \times \prod_{q=1}^{d_x} [x_{iqL}, x_{iqU}].$$

The argument applies to rectangular feasible domains on which the moment function has the stated corner-extremum property. In particular, additional nonrectangular feasibility restrictions cannot be ignored.

The primitive cube indicators and their positive weights are given in Appendix A.2. In the notation of Assumption 2, the weighted instrument is $g_k = \omega_k h_k$, where $h_k(x) = \mathbf{1}\{x \in C_k\}$. For a complete truncation through resolution R, the retained instruments are indexed by (a, r) with $r_0 \le r \le R$, and $K = N_R = \sum_{r=r_0}^{R} (2r)^{d_x}$; the argument also applies if truncation stops partway through a resolution.

All cube partitions use the same boundary convention: each coordinate interval is open on the left and closed on the right, except that the first interval contains zero. For each coordinate q, collect the endpoints of all retained cube sides in that coordinate, together with 0 and 1, and sort the distinct values as

$$0 = \xi_{q0} < \xi_{q1} < \cdots < \xi_{qL_q} = 1.$$

Define $I^*_{q1} = [0, \xi_{q1}]$ and $I^*_{qj} = (\xi_{q,j-1}, \xi_{qj}]$ for $j \ge 2$. The common refined cells are the rectangles

$$C^*_\ell \equiv \prod_{q=1}^{d_x} I^*_{q,\ell_q}, \qquad \ell = (\ell_1, \ldots, \ell_{d_x}), \quad 1 \le \ell_q \le L_q.$$

Their total number is $L = \prod_{q=1}^{d_x} L_q$. Every original cube C_k is a union of refined cells, and $g_k(x)$ is constant on each C_ℓ^*.

To make the construction explicit when $d_x = 1$, the resolution-r partition consists of $2r$ bins with endpoints $j/(2r)$, $j = 0, \ldots, 2r$. The common refinement is obtained simply by sorting the union of these endpoints over the retained resolutions. Thus the original multi-resolution instruments become step functions on a single refined grid.

Write $g(x) = (g_1(x), \ldots, g_K(x))^\top$, and define

$$c_{\ell k} = \omega_k \mathbf{1}\{C_\ell^* \subseteq C_k\}, \qquad C \equiv \begin{pmatrix} c_1^\top \\ \vdots \\ c_L^\top \end{pmatrix} \in \mathbb{R}^{L \times K}.$$

For any $u \in B_K^\infty = [-1, 1]^K$, the induced coefficient on refined cell ℓ is

$$\widetilde{u}_\ell \equiv c_\ell^\top u, \qquad \widetilde{u} = Cu, \qquad u^\top g(x) = \widetilde{u}_\ell \quad (x \in C_\ell^*). \tag{21}$$

Crucially, the set of permissible refined coefficients is $C[-1, 1]^K$, not an independently constrained box in $\mathbb{R}^L$. One may therefore eliminate $\widetilde{u}$ from the optimization, as in (10), or retain it together with the equality $\widetilde{u} = Cu$.

For each observation, intersect its feasible rectangle with each refined cell *before* taking a closure:

$$E_{i\ell} \equiv D_i \cap (\mathbb{R} \times C_\ell^*), \qquad S_i \equiv \{\ell : E_{i\ell} \neq \varnothing\}, \qquad R_{i\ell} \equiv \mathrm{cl}(E_{i\ell}), \quad \ell \in S_i. \tag{22}$$

Each $R_{i\ell}$ is a compact rectangle. Its coordinate-q endpoints, for $\ell \in S_i$, are

$$x_{iq\ell}^L = \max\{x_{iqL}, \xi_{q,\ell_q - 1}\}, \qquad x_{iq\ell}^U = \min\{x_{iqU}, \xi_{q,\ell_q}\},$$

with the y endpoints unchanged. These formulas describe the closure of the intersection only after nonemptiness of $E_{i\ell}$ has been checked using the half-open boundary convention. For example, if $d_x = 1$, D_i restricts x to the singleton $\{b\}$, and a refined cell is $(b, c]$, then that cell does not belong to S_i, although $[b, c]$ intersects $\{b\}$. Closing first would therefore add an incorrect cell and could change the support function.

For fixed θ, assume that $m(\cdot, \cdot, \theta)$ is continuous and coordinatewise monotone on every nonempty $R_{i\ell}$. The direction of monotonicity in each coordinate can depend on θ and on the rectangle but is fixed within that rectangle. Maximizing or minimizing one coordinate at a time shows that the extrema occur at vertices. Write

$$m^-_{i\ell}(\theta) = \min_{z \in \mathrm{vert}(R_{i\ell})} m(z, \theta), \qquad\qquad m^+_{i\ell}(\theta) = \max_{z \in \mathrm{vert}(R_{i\ell})} m(z, \theta).$$

There are at most 2^{1+d_x} vertices of each rectangle; in degenerate rectangles some of them coincide. Since $E_{i\ell}$ is dense in $R_{i\ell}$ and m is continuous, for every real scalar $\widetilde{u}_\ell$,

$$\sup_{(y,x) \in E_{i\ell}} \widetilde{u}_\ell m(y, x, \theta) = \max\big\{\widetilde{u}_\ell m^-_{i\ell}(\theta), \widetilde{u}_\ell m^+_{i\ell}(\theta)\big\}. \tag{23}$$

The upper corner is relevant for $\widetilde{u}_\ell \geq 0$ and the lower corner for $\widetilde{u}_\ell < 0$. A maximizing vertex may lie on an excluded boundary of C^*_ℓ; its contribution is understood as a limit from within $E_{i\ell}$, with the coefficient $\widetilde{u}_\ell$ held fixed. It must not be evaluated with the instrument vector assigned to an adjacent cell.

By definition, $Q_{\theta,K}(W^O_i) = \mathrm{cl}_{\mathbb{R}^K}\{m(y, x, \theta)g(x) : (y, x) \in D_i\}$. A continuous linear functional has the same supremum over a set and its closure. The refined cells partition the covariate

domain, so using (21) and (23) gives

$$\begin{aligned} h(Q_{\theta,K}(W_i^O), u) &= \sup_{(y,x)\in D_i} m(y,x,\theta) u^\top g(x) \\ &= \max_{\ell \in S_i} \sup_{(y,x)\in E_{i\ell}} \widetilde{u}_\ell m(y,x,\theta) \\ &= \max_{\ell \in S_i} \max_{s\in\{-,+\}} \left\{ m_{i\ell}^s(\theta) c_\ell^\top u \right\}. \end{aligned}$$

This proves (9). In particular, there is no numerical approximation to the finite-truncation support function: its value is the maximum of finitely many linear functions of the original direction u.

Recall from Section 3.1 that

$$-\widehat{T}_{n,K}(\theta) = \inf_{u\in[-1,1]^K} \frac{1}{n} \sum_{i=1}^n h(Q_{\theta,K}(W_i^O), u).$$

For any feasible pair (u, t) in (10), its constraints and (9) imply

$$t_i \geq h(Q_{\theta,K}(W_i^O), u) \qquad (i = 1, \ldots, n).$$

Thus its objective is at least the support minimum. Conversely, for any $u \in [-1, 1]^K$, taking $t_i = h(Q_{\theta,K}(W_i^O), u)$ satisfies every epigraph constraint and gives exactly the sample support average. Minimizing over u proves that the optimal value of (10) is $-\widehat{T}_{n,K}(\theta)$.

The sample support average is continuous in u, since each observation-level support function is a finite maximum of linear functions. Compactness of $[-1, 1]^K$ therefore ensures that the minimum is attained. The direction $u = 0$ and $t = 0$ is feasible, so the optimal value is nonpositive and $\widehat{T}_{n,K}(\theta) \geq 0$. The epigraph variables t_i must be unrestricted in sign; imposing $t_i \geq 0$ would generally change the criterion.

The reduced LP has $K + n$ variables, $2K$ box inequalities and $2\sum_{i=1}^n |S_i|$ epigraph inequalities. If the induced coefficients $\widetilde{u}$ are explicitly introduced, there are L additional

variables and L equalities $\widetilde{u} = Cu$, with no change in the optimal value. Because the sample objective is continuous and convex in u, a unique minimum-Euclidean-norm optimizer can be selected from its compact argmin set if a deterministic tie-breaking rule is needed for sample-dependent direction selection.

For a scalar index residual $m(y, x, \theta) = y - F(\theta_0 + x^\top \theta_1)$, where F is continuous and nondecreasing, the moment is nondecreasing in y and monotone in each x coordinate (with direction determined by the corresponding coefficient). This includes linear, probit and logit residuals on bounded rectangular domains. More generally, the argument only needs the cellwise extrema to be attained among finitely many known points. For vector-valued moments, coordinatewise monotonicity of the individual components is not sufficient by itself: a direction-dependent linear combination need not have its extrema at a common finite set of corners. Such models require a separate finite-support argument before an exact LP can be asserted.

F Relation to Alternative Outer Sets

Aucejo et al. (2017) are concerned with the same conditional moment model defined by equation (1) and Assumption 1 above. However, their focus is on the missing data problem, i.e. $[Y_l, Y_u] = \mathcal{Y}$ and $[X_l, X_u] = \mathcal{X}$. Nevertheless, comparing our results with their construction develops further intuition for the identification problem; thus, in this section we discuss both approaches and their relation in detail. For simplicity, let us focus on the one moment case, thus $d_m = 1$, although both their paper and the present work can deal with the more general case. Similarly, let us only cover the missing covariate case and abstract from fully observed additional covariates. Define $Z = 1$ if the covariate is missing and $Z = 0$ otherwise.

As mentioned in the main text, Aucejo et al. (2017) note that the sharp identified set, Θ_I^*, is generally too complicated an object to compute and thus they resort to constructing outer

sets, i.e. supersets of the sharp identified set. The first step in Aucejo et al. (2017) transforms the conditional moment conditions into unconditional moments via instrument functions as we propose for our problem in Lemma 1. They use the class of boxes $\mathcal{G}_{\text{box}} \equiv \{\mathbb{I}[X \in G(x,v)] : \forall (x,v) \in \mathcal{X} \times \mathbb{R}^{d_x}_{++}\}$ with $G(x,v) \equiv \{\tilde{x} \in \mathcal{X} : \{x_j - v_j \leq \tilde{x}_j \leq x_j + v_j\}_{j=1}^{d_x}\}$ as instrument functions.[14] Let us denote the collection of boxes by $\{G_k\}_{k\geq 0} \equiv \{G(x,v) \quad \forall (x,v) \in \mathcal{X} \times \mathbb{R}^{d_x}_{++}\}$. Note that the main point of the following conceptual comparison does not rely on the specific instrument class. They propose two outer sets.

The first outer set in Aucejo et al. (2017) uses the following vector-valued auxiliary functions for each member of $\{G_k\}_{k\geq 0}$:

$$M_{k,1}(W^O) \equiv \begin{pmatrix} -\left(\inf_{y\in\mathcal{Y}, x\in G_k} m(y,x,\theta)\mathbb{I}[Z=1] + m(Y,X,\theta)\mathbb{I}[Z=0, X\in G_k]\right) \\ +\left(\sup_{y\in\mathcal{Y}, x\in G_k} m(y,x,\theta)\mathbb{I}[Z=1] + m(Y,X,\theta)\mathbb{I}[Z=0, X\in G_k]\right) \end{pmatrix}$$

They proceed by defining their outer set as $\Theta_{I,1}^{ABH} \equiv \{\theta \in \Theta : \ \mathbf{E}[M_{k,1}(W^O)] \geq 0 \quad \forall k \geq 0\}$. This solves the moment problem for each box, or in general, each subspace defined by the instrument functions, but not simultaneously for all of them. As Aucejo et al. (2017) mention, the difficult part of characterizing the sharp identified set is to ensure that the moment conditions hold *simultaneously* for all boxes and not only individually; thus, $\Theta_{I,1}^{ABH}$ is an outer set.

The second outer set defined in Aucejo et al. (2017) is based on moments integrating out the missing covariate. Their auxiliary moment conditions do not depend on k in the case of no additional fully observed covariates and are given by:

$$M_{k,2}(W^O) \equiv \begin{pmatrix} -\left(\inf_{x\in\mathcal{X}} m(Y,x,\theta)\mathbb{I}[Z=1] + m(Y,X,\theta)\mathbb{I}[Z=0]\right) \\ +\left(\sup_{x\in\mathcal{X}} m(Y,x,\theta)\mathbb{I}[Z=1] + m(Y,X,\theta)\mathbb{I}[Z=0]\right) \end{pmatrix}$$

[14]Note that in the definition of $G(x,v)$ we use $\tilde{x} \in \mathcal{X}$ directly, thus $G(x,v) \cap \mathcal{X} \neq \emptyset$ is always true and we can avoid carrying the intersection in the definition of M_k below.

The second outer identified set is the collection $\Theta_{I,2}^{ABH} \equiv \{\theta \in \Theta : \ \mathbf{E}[M_{k,2}(W^O)] \geq 0\}$.

Let us now show the relation between $\Theta_{I,1}^{ABH}$, $\Theta_{I,2}^{ABH}$ and our sharp identified set Θ_I. First note that the class of boxes is comprehensively revealing (see Andrews and Shi (2013)), i.e., it fulfills Assumption 2(i),[15] so our main identification result applies. Further, we know that any subset $\mathcal{U} \subseteq B_1^\infty$ defines an outer set. In particular, choose $\mathcal{U}_{point} := \{\pm e_j : j \geq 1\}$, where $e_j := (0, \dots, 0, 1, 0, \dots)$ is the jth basis vector. The outer set implied by this choice is

$$\Theta_O(\mathcal{U}_{point}) = \{\theta \in \Theta : \mathbf{E}[h(Q_{\theta,0}, u)\mathbb{I}[Z = 0]] + \mathbf{E}[h(Q_{\theta,1}, u)\mathbb{I}[Z = 1]] \geq 0 \quad \forall u \in \mathcal{U}_{point}\}$$

where we use the law of total expectation and divide our theorem into fully observed and missing covariates and denote with a slight abuse of notation the respective random sets as $Q_{\theta,z}$.

Further, considering the class of boxes, the support function for the missing covariate case only is given by $h(Q_{\theta,z}, u) = \sup_{x \in [X_l, X_u]} \sum_{k \geq 0} u_k m(Y, x, \theta)\mathbb{I}[x \in G_k]$. Any choice of $\tilde{u} \in \mathcal{U}_{point}$ simplifies the support function considerably. In particular, choosing $\tilde{u} = e_k$ yields $h(Q_{\theta,z}, \tilde{u}) = \sup_{x \in [X_l, X_u]} m(Y, x, \theta)\mathbb{I}[x \in G_k]$. For the cases for which we fully observe the covariates, $Z = 0$, this further collapses to $h(Q_{\theta,0}, \tilde{u}) = m(Y, X, \theta)\mathbb{I}[X \in G_k]$ with $X = X_l = X_u$. Thus, the condition summed over missingness patterns for the specific choice e_k is

$$\begin{aligned} &\mathbf{E}[\sup_{x \in [X_l, X_u]} m(Y, x, \theta)\mathbb{I}[x \in G_k]\mathbb{I}[Z = 1]] + \mathbf{E}[m(Y, X, \theta)\mathbb{I}[X \in G_k]\mathbb{I}[Z = 0]] \geq 0 \\ \iff &\mathbf{E}\left[\sup_{x \in \mathcal{X}} m(Y, x, \theta)\mathbb{I}[x \in G_k]\mathbb{I}[Z = 1] + m(Y, X, \theta)\mathbb{I}[Z = 0, X \in G_k]\right] \geq 0 \end{aligned}$$

where we used $[X_l, X_u] = \mathcal{X}$ for the missing data case. Simultaneously, using $\sup_{a \in A} f(a) =$

[15] Assumption 2(ii) is not fulfilled by the class of boxes alone. Instead we would need to augment them with positive and decaying weights. For the identification argument here, the choice of such weights is immaterial so we abstract from them.

$-\inf_{a\in A}(-f(a))$, and considering $\tilde{u} = -e_k$ we obtain:

$$\mathbf{E}\left[-\left(\inf_{x\in\mathcal{X}} m(Y,x,\theta)\mathbb{I}[x\in G_k]\mathbb{I}[Z=1] + m(Y,X,\theta)\mathbb{I}[Z=0, X\in G_k]\right)\right] \geq 0$$

These expressions are almost equivalent to the elements of the vector $M_{k,1}$ defined above; however, whereas we take the supremum over $x \in \mathcal{X}$, Aucejo et al. (2017) are considering $y \in \mathcal{Y}, x \in G_k$. Additionally, our result includes the indicator $\mathbb{I}[x \in G_k]$ in the objective for the missing observations. Notice that taking the supremum over the entire support of Y even for observed values is not using the entire information provided from the data.

The second difference is that instead of maximizing over the full support as in Theorem 1 above, Aucejo et al. (2017) maximize over the individual boxes. In combination with also varying Y for the missing X observations this still yields an outer set. However, conceptually it is assuming that for each box, the value of the missing covariate must lie within the specific box. This is stronger than required by the conditional moment restriction. The conditional moment model is only implying that the moment condition has to hold over all subspaces, not that it must hold under all subspaces assuming that all realized values are within that space. Instead, our construction maximizes $m(Y,x,\theta)\mathbb{I}[x \in G_k]$ over $x \in \mathcal{X}$, thus it allows for a specific observation not contributing to G_k.[16]

Taking all of the above into account, the preceding comparison does not by itself establish a general ranking between $\Theta_{I,1}^{ABH}$ and $\Theta_O(\mathcal{U}_{point})$. The main takeaway is that our identification result also considers combinations of boxes. It is the requirement that the moment conditions hold simultaneously across boxes that makes the sharp identified set difficult to compute. Adding directions to our support-function approach yields tighter outer sets which, under

[16] Another way to see this point is to consider a simpler setting. Suppose $m(y,x,\theta) = x$ with $\mathcal{X} = [0,1]$, thus the moment condition boils down to $\mathbf{E}[X|X] = 0$. If X is missing for all observations we cannot rule out that $P_x = \delta_0$, a mass point at zero. The latter is the only distribution of X fulfilling the moment condition. For any box in $[0,1]$ and u vector a feasible solution to our support function approach is zero as we can either choose $x = 0$ or another value with associated $u = 0$. Instead, considering any box $[a,b]$ with $a \neq 0$, the first moment of the first outer set in Aucejo et al. (2017) is negative, thus violated. We thank Federico Bugni for bringing up this example.

Theorem 1, converge to the sharp identified set. Since the sharp set is contained in $\Theta_{I,1}^{ABH}$ under the same maintained assumptions, the limiting set is contained in $\Theta_{I,1}^{ABH}$, although the inclusion need not be strict.

For $\Theta_{I,2}^{ABH}$, an equivalence follows from a single box. Because $\mathcal{X}$ is compact, we can include a box $G_{k_0} = \mathcal{X}$ in our instrument class. The directions e_{k_0} and $-e_{k_0}$ then yield, after dividing by the positive instrument weight,

$$\begin{aligned}
\mathbf{E}\left[\sup_{x\in\mathcal{X}} m(Y,x,\theta)\mathbb{I}[Z=1] + m(Y,X,\theta)\mathbb{I}[Z=0]\right] &\geq 0,\\
\mathbf{E}\left[-\inf_{x\in\mathcal{X}} m(Y,x,\theta)\mathbb{I}[Z=1] - m(Y,X,\theta)\mathbb{I}[Z=0]\right] &\geq 0.
\end{aligned}$$

These are precisely the two inequalities defining $\Theta_{I,2}^{ABH}$ in the absence of additional fully observed covariates. Thus $\Theta_O(\{e_{k_0}, -e_{k_0}\}) = \Theta_{I,2}^{ABH}$ and, in particular, $\Theta_O(\mathcal{U}_{point}) \subseteq \Theta_{I,2}^{ABH}$. We note here that the outer sets proposed by Aucejo et al. (2017) can nevertheless be quite informative in certain settings. In particular, in the point-identified design of Section 5, the power against fixed local alternatives of the confidence sets based on Aucejo et al. (2017), as implemented via the bootstrap of Andrews and Shi (2017), compares favorably to our method in finite samples. It seems that in cases in which some parts of the support of the regressor are not affected by censoring, checking the moment condition instrument by instrument or integrating out the unobserved component does not yield a significant loss of information. Figure 4 shows the comparison of our adaptive procedure versus Aucejo et al. (2017), similar to the main text Figure 2.

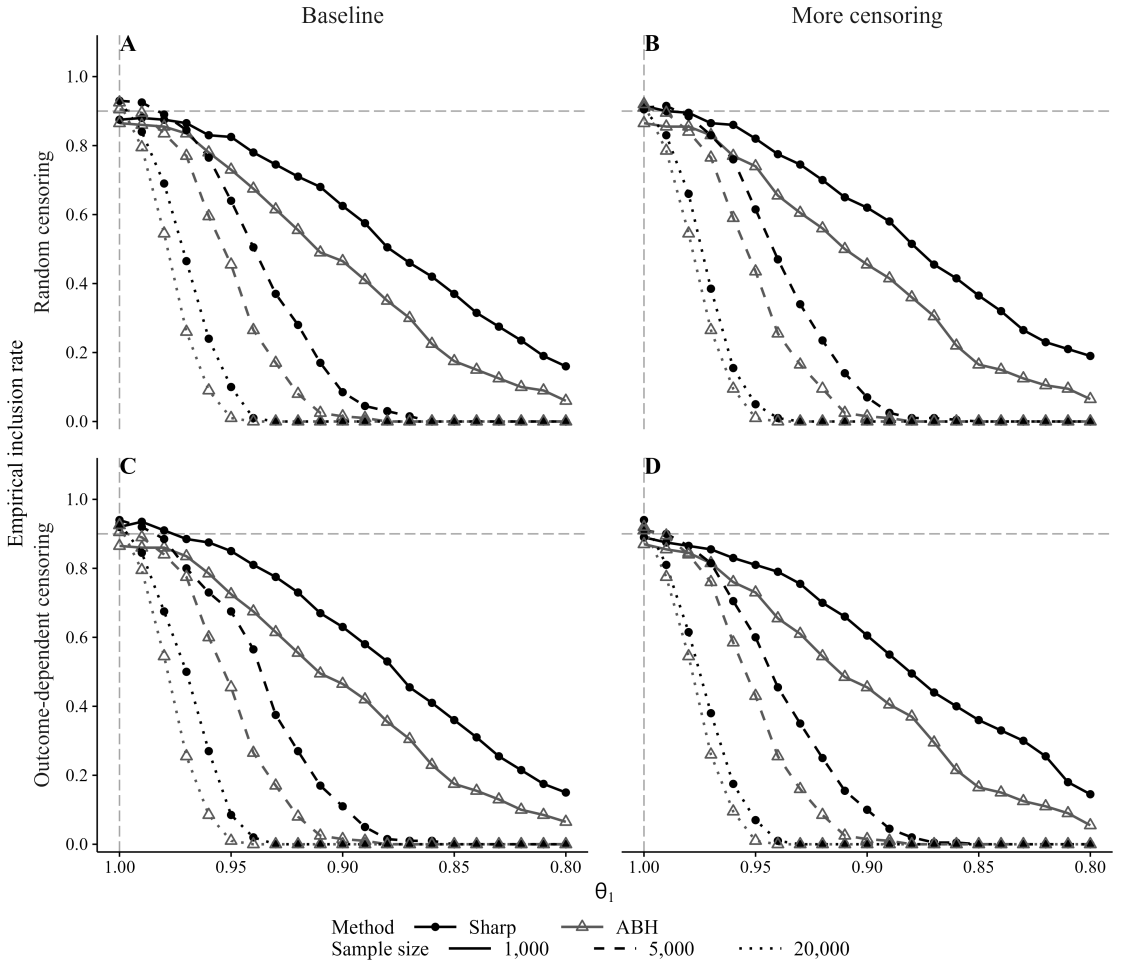


Figure 4: Empirical Inclusion Frequencies along the Slope Path

Note: Share of 200 Monte Carlo replications in which $(0, \theta_1)$ lies in the 90% confidence set, for the adaptive procedure (*Sharp*) and the ABH confidence set. The true parameter is $(0, 1)$, so the inclusion rate at $\theta_1 = 1$ is the coverage and the rates at $\theta_1 < 1$ are one minus the power. The horizontal dashed line marks the nominal level of 0.90. Baseline: $p = 0.05$; more censoring: $p = 0.25$.